Part IV

Principles of Palliative Medicine and Ethics

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Chapter 67

Palliative Care and Special Management Issues

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INTRODUCTION

Palliative medicine is an essential skill set for health care professionals caring for older adults. Supporting the needs and goals of older adults as they age is a complex process that often involves balancing: goals of life prolongation, functional status preservation and restoration, risk and harm mitigation, and symptom relief. Palliative care aims to ensure that these goals match the individualized care plans through interprofessional care and careful communication and goal setting.

Palliative care for older adults poses distinct challenges. For older patients with diminished reserve to respond to stressors, the potential for harm of interventions can be substantially greater than younger populations. Cognitive impairment may limit the participation of patients at times when the most difficult decisions need to be made. The interaction of multiple illnesses in multimorbidity may reduce the potential benefit of interventions, which are otherwise therapeutic in single disease state.

DEFINING PALLIATIVE CARE

Palliative care aims to improve the quality of life for both the patient with serious illness and the family. Palliative care focuses on providing patients with relief from the symptoms, pain, and stress of a serious illness. It is appropriate for any type of diagnosis, any stage of a serious illness, and importantly, can be provided together with curative treatment.

Specialty palliative care consists of care by an interprofessional team of physicians, nurses, and social workers, chaplains and other individuals with expertise in palliative medicine, who work with patients' other health care professionals to provide care that matches patients' goals. In addition to palliative care specialists, the core principles of palliative care can be implemented by all health care professionals (often described as *primary palliative care*). The key components of primary palliative care include symptom management, coordination of care, communication about goals of care and advance care planning (ACP), caregiver support, and when needed, referral to specialist palliative care teams.

Hospice Care

Hospice care is a specialized form of palliative care for patients with limited life expectancy. Hospice care provides medical, psychosocial, and spiritual support to the patient. Hospice also supports family members coping with the complex consequences of illness, disability, and functional decline as death nears and includes bereavement support after death of the patient. Hospice care does not aim to shorten or prolong life, but rather provides comfort and support services to help people live out the time they have remaining to the fullest extent possible.

The Medicare hospice benefit employs an interdisciplinary team of professionals including doctors, nurses,

Learning Objectives

- Identify the palliative needs of an aging population and recognize the potential benefits palliative care delivers in collaboration with or as part of geriatric medicine practice.
- Describe the unique palliative care needs of older adults including patients with dementia, frailty, or multimorbidity, as well as diverse populations.
- Distinguish the various advance care planning (ACP) formats and tools and adapt a three-step process to prognostication in older adults as part of ACP.

Key Clinical Points

1. **Multiple randomized trials show palliative care improves outcomes including quality of life, satisfaction with care, reduced family distress, and increased hospice utilization.**
2. **For older adults with multiple chronic illness, a five-step approach to palliative care includes the following:**
   a. **Determine patient preferences.**
   b. **Interpret the evidence for treatment with recognition of the limitations applying it to an older adult population.**
   c. **Let prognosis frame clinical management decisions.**
   d. **Consider treatment complexity and feasibility as part of management decisions.**
   e. **Optimize therapies and care plans.**

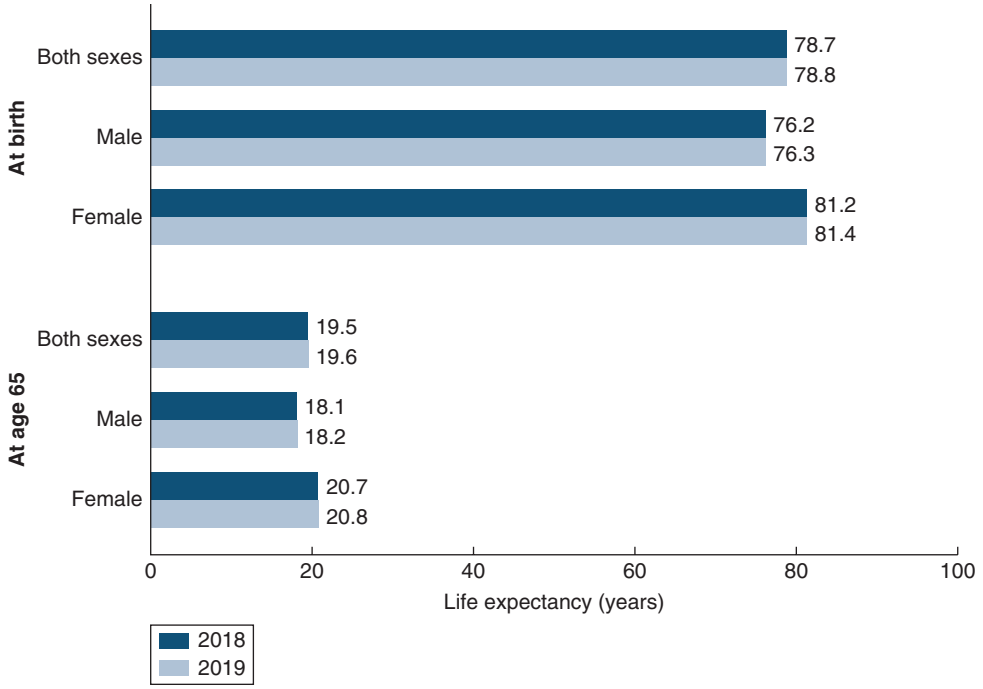

FIGURE 67-1. Life expectancy at birth and age 65, by sex: United States, 2018 and 2019. (Data from National Center for Health Statistics, National Vital Statistics System, Mortality.)

aides, social workers, and spiritual care coordinators to address the patient's physical, emotional, social, and spiritual needs. Under the Medicare hospice benefit, a patient is eligible for hospice care when the patient's attending physician and the hospice medical director certify that she or he is terminally ill with an expected prognosis of less than 6 months "if the disease runs its usual and expected course." The Centers for Medicare and Medicaid Services (CMS) has guidelines entitled local coverage determinations (LCDs) to help physicians determine a prognosis of 6 months or less. These LCDs are available on the CMS website and are *guidelines* to help determine prognosis, they *are not rules or requirements* for hospice admission. Once a patient is admitted, the hospice agency is required to provide services that are reasonable and necessary for the palliation and management of a patient's terminal illness and related conditions, including doctor and nursing visits, hospice aide and counseling services, medications, medical equipment and supplies, and short-term respite and inpatient stays to manage pain and symptoms.

PALLIATIVE CARE NEEDS OF AN AGING POPULATION

Health care in the United States has made tremendous advances which have allowed for an aging population. The typical death has been transformed from the nineteenth-century picture of an acute illness over a short time from infectious disease into a process of decline over years with increasing morbidity and functional dependency from chronic disease. The average age at the time of death in the United States per 2019 data is now 79 (81 years for females and 73 for males) and survivors to age 65 live on average another 20 years (**Figure 67-1**).

While chronic illness of older adults is increasingly complicated by multiple illnesses, in 2018 two major diseases, heart disease and cancer, still accounted for nearly half of all 2018 deaths. Apart from acute respiratory illness, accidents, and suicide, the majority of deaths of older adults are associated with a period of chronic disease. In 2018, two-thirds of people aged 65 and older had multiple chronic conditions per the Centers for Disease Control and Prevention (CDC).

The exponential increase in COVID-19 illness in the United States has led to a significant number of cases and mortality particularly in older adults in 2020 to 2021, becoming deadlier than heart disease and cancer. COVID-19's impact on older adults perfectly demonstrates the unique needs of the older adult population. Geriatric palliative care during the COVID-19 pandemic and future pandemics can work within systems to balance potentially burdensome interventions with complex goals and to deal with the common sequela of decline in performance status after infectious illness.

As the patient's number of comorbid conditions rises, the complexities of delivering care have had profound consequences. Patients with multimorbidity are at increased risk of disability and institutionalization, poorer quality of life, and higher risk of harm from medical treatments. Over time, the care that these patients get often becomes

more fragmented. Among Medicare fee-for-service decedents in 2015, deaths in the home or community setting occurred in 40% and a change in the setting of care in the last 3 days of life occurred among 10% of decedents.

THE BENEFIT FOR PALLIATIVE MEDICINE

Palliative medicine is now well established as a discipline, and as the field matures, there is a growing body of literature showing its benefit.

The Evidence for Palliative Care Systems

The evidence for specialist palliative care interventions includes multiple randomized trials showing palliative care improves outcomes including quality of life, satisfaction with care, reduced family distress, and increased hospice utilization. The landmark Temel study demonstrated the value of concurrent care, that is, palliative care delivered early and simultaneously with curative, disease-focused care in advanced lung cancer patients. Early palliative care for newly diagnosed metastatic lung cell cancer patients was associated with higher quality of life, less depression, and a statistically significant longer median survival (12 months vs 8.9 months). Subsequent, larger cluster randomized trial of advance cancer patients with good performance status representing a wide variety of cancer types demonstrated improved quality of life with early palliative care. Notably, the statistical significance in many outcomes was not achieved until the fourth month of palliative care, emphasizing the importance of early referral to palliative care to impact quality of life measures beyond pain control. The ENABLE III study showed that initiating concurrent palliative care at the time of a cancer diagnosis had a significant impact on 1-year survival compared to initiating palliative care 3 months after initial cancer diagnosis (63% early vs 48% delayed) and involved multiple types of cancer not only lung cancer.

Extensive evidence is accumulating for palliative care delivery for noncancer diagnosis as well. A 2020 meta-analysis of 28 trials of palliative care in noncancer patients (13,664 patients, mean age 74) including patients with heart failure (10 studies), dementia (4 studies), chronic obstructive lung disease (3 studies), and multimorbidity (11 studies) showed that palliative care is associated with less emergency department use, less hospitalization, and lower symptom burden. Trials to date of palliative care have been least promising in dementia and highlight opportunities to better tailor palliative care to the special needs of persons living with dementia.

The Evidence for Hospice

Hospice has been shown in numerous studies to improve symptom control, increase patient satisfaction and quality of life, and help patients and families prepare for death. The highest-quality outcomes tend to occur for patients who were enrolled in hospice care for longer than 30 days. Since 2014 the CMS has collected quality data via the Hospice Item Set, which has seven key domains of patient quality.

A major benefit of hospice over usual care is that hospice allows patients to die in their own place of residence when that is the patient's wishes. Hospice care adds value by delivering high-quality care while reducing Medicare costs primarily through caring for patients in the home setting. Cost savings increase with greater time spent in hospice care. The maximum reduction in expenditures tends to occur when the hospice length of stay exceeded 50 days.

A specific concern of geriatricians and policy makers is whether hospice adds benefits for patients with dementia in long-term care. Nursing home residents with dementia and cancer who received hospice care in the last 30 days of life were less likely to die in a hospital than those who did not receive hospice, and hospice enrollment increases the likelihood that patients with dementia who have pain receive an opioid.

PALLIATIVE CARE IN SPECIAL POPULATIONS

Palliative care and geriatrics are both designed to serve the needs of the most vulnerable patients with a "common central philosophy of patient-centered commitment to holistic and humane care." Due to the growing geriatric population, there is an unprecedented need to provide excellent palliative care to older adults, especially those with dementia, frailty, and multimorbidity. Partnerships between geriatricians and palliative medicine specialists can be especially useful for management of patients with these conditions, with each specialty working together to ensure the highest level of evidenced-based compassionate care to older adults.

Dementia

Older adults with dementia have unique palliative medicine needs given the progressive nature of their illness with significant, prolonged disability and predictable increases in symptom burden as dementia advances. This population especially benefits from combined geriatric and palliative medicine approaches.

It is imperative that health care professionals provide appropriate management of behavioral and psychological symptoms of dementia. As discussed in detail in the management of agitation in Chapter 60, management should involve (1) identifying the specific problem behavior and its severity, (2) identifying potential triggers for the behavior, (3) removing underlying physiologic and environmental triggers whenever possible, (4) attempting nonpharmacologic interventions to improve behavior, and (5) as a final resort, using recommended targeted pharmacotherapy for behaviors that do not respond to attempts

of nonpharmacologic management in situations that risk patient or caregiver safety. As part of the American Board of Internal Medicine (ABIM) "Choosing Wisely" initiative, the American Geriatrics Society (AGS) recommends against the use of antipsychotics as first-line treatment for behavioral and psychological symptoms of dementia. Notably, a cluster randomized trial of nursing home patients with behavioral problems using an empiric stepwise protocol to treat pain resulted in a decrease in the number of symptoms. Most patients were treated with the first step of the protocol, acetaminophen. Opioids may be tried for nonresponders. The Pain Assessment in Advanced Dementia (PAINAD) Scale may be a useful tool in this situation.

In advanced stages of dementia, weight loss and dysphagia are common with the latter putting patients at risk for aspiration pneumonia. Weight loss may be secondary to decreased appetite, depression, oral problems such as ulcers or ill-fitting dentures, increased wandering, or medication side effects. A reversible etiology for weight loss is not always present. The addition of nutritional supplements or appetite stimulants has not been shown to improve clinically important outcomes. For patients with dysphagia, several organizations, in the ABIM "Choosing Wisely" initiative, recommend *against percutaneous feeding tubes* and recommend *in favor of offering oral assisted feeding*. Studies have shown that feeding tubes in advanced dementia do not increase survival or quality of life, prevent aspiration pneumonia, or improve pressure ulcer healing or nutritional parameters (weight, albumin). In fact, feeding tube usage has been associated with the development of pressure ulcers and the use of pharmacologic and physical restraints. The majority of feeding tubes in patients with advanced dementia are inserted during an acute hospitalization.

Infections, specifically urinary tract infections and pneumonia, are frequently diagnosed in advanced dementia and commonly occur as patients near end of life. In nursing home residents with advanced dementia, observational data show suspected urinary tract infections rarely meet criteria for antibiotics; however, the majority are treated with them. Findings from a prospective cohort study suggest that the treatment of suspected urinary tract infections in this population does not prolong survival. Antibiotic therapy for aspiration pneumonia in advanced dementia has been associated with improved survival but not improved comfort. For aspiration pneumonia, the most aggressive treatment approaches with intravenous therapy and hospitalization have been associated with the greatest discomfort, and the survival benefit associated with antibiotic therapy was similar regardless of administration route. Interventions such as opioids, fans, and acetaminophen can increase comfort. In addition to using the minimal clinical criteria for the initiation of antibiotics, appropriate management of infections in advanced dementia requires an attention to a patient's primary goals (ie, life prolongation, comfort, safety) in order to avoid unnecessary or unwanted treatment burden.

Health care professionals caring for older adults with dementia should engage patients and their families in ACP early and include an assessment of understanding of diagnosis, prognosis, and disease trajectory. The hope is to align treatment preferences with patient values and decrease stress on future caregivers. ACP discussions may result in completion of an advance directive document in which the patient-specifies preferences for future care and/or designates an authorized surrogate decision maker to communicate their preferences in circumstances when they no longer possess medical decision-making capacity (see more details in Chapters 7 and 10). Dementia-specific advance directive documents are now available given dementia's distinct clinical course. While barriers to these conversations exist such as difficulty with prognostication, lack of decision-making capacity, and patient or family reluctance to engage, both geriatricians and palliative medicine specialists are well equipped to conduct these vital discussions. Online resources are available for assistance (**Table 67-1**).

ACP discussions with surrogate decision makers should continue as dementia progresses. Older adults with advanced dementia living in nursing homes who have proxies with an understanding of their terminal prognosis are less likely to receive burdensome interventions. Consideration for hospice care in end-stage dementia is also important as enrollment is associated with greater satisfaction in patient care. Determining a 6-month prognosis in advanced dementia can be challenging. Factors such as irreversible weight loss, low body mass index, functional dependence, and advanced age are associated with limited prognosis. Palliative medicine specialists can also help with prognostication.

Frailty

While the exact definition of frailty remains a subject of debate (see Chapter 42), frail patients are particularly vulnerable to adverse outcomes and have significant palliative care needs. For instance, severely frail older adults have shown poor response to treatment and may not benefit from intensive rehabilitation efforts. Attempting rehabilitation of the most frail may in fact cause harm. The key to

TABLE 67-1 ■ RESOURCES AVAILABLE FOR DEMENTIA-SPECIFIC ADVANCE CARE PLANNING

Dementia-Specific Advance Directive https://dementia-directive.org
Communication Tools https://www.capc.org/training/best-practices-in-dementia-care-and-caregiver-support/
For Caregivers and Surrogate Decision Makers https://theconversationproject.org/wp-content/uploads/2020/12/DementiaGuide.pdf https://www.alz.org https://www.caregiver.org

managing the palliative needs of frailty is determining goals and recognizing the patient with poor potential to improve. When in doubt, a therapeutic trial of improving performance status is always appropriate, but if not achieved should prompt a repeat discussion of goals and prognosis.

Determining the severity of frailty and the potential for reversibility can help clinicians recommend appropriate treatment. Frailty definitions, staging, and recommended treatment options for each stage are described in **Table 67-2**.

Patients with frailty and multimorbidity may still be eligible for hospice even if they do not have a single terminal illness. While CMS has stated that frailty, debility, or adult failure to thrive are not accepted as the principal hospice diagnosis reported on the Medicare hospice claims form, this does not mean that frail patients are not eligible for the Medicare hospice benefit. A patient is considered hospice eligible if the attending physician and hospice medical director have certified the patient to be terminally ill with a prognosis of less than 6 months. This prognosis is determined based on the severity and irreversibility of the patient's frailty as well as their multimorbidity. The hospice Medical Director could utilize the hospice LCDs available on the CMS website to support their determination that the patient has a prognosis of less than 6 months.

Multimorbidity

Providing optimal care for older adults with multiple chronic conditions, or multimorbidity, presents one of the greatest challenges in geriatrics. Patients with multimorbidity have increased use of health care resources, poorer quality of life, higher rates of institutionalization, disability, and death. To address the challenge of providing patient-centered care to older adults with multimorbidity, a stepwise approach can integrate patient care with the key principles of geriatric and palliative medicine in five domains: (1) patient preferences, (2) interpreting the evidence, (3) prognosis, (4) clinical feasibility, and (5) optimizing therapies and care plans. The American Geriatrics Society's "Patient-Centered Care for Older Adults with Multiple Chronic Conditions: A Stepwise Approach: Expert Panel on the Care of Older Adults with Multimorbidity" is an excellent online resource: https://geriatricscareonline.org/ProductAbstract/Framework-for-Decision-making-for-Older-Adults/CL026 [geriatricscareonline.org].

Considerations for Providing Inclusive Care

The US population is ethnically diverse and there may be differences on perceptions and readiness for palliative care based on race and socioeconomic factors. Ornstein's 2020

TABLE 67-2 ■ A CLINICAL MANAGEMENT APPROACH TO FRAILTY

STAGE	CLINICAL	MANAGEMENT
Latent stage or "pre-frail" (1-2 frailty criteria met)	Not clinically apparent in the absence of stressors	Likely most responsive to **prevention:** • Minimize and/or treat precipitants. • Prevent and minimize immobility, and maintain physical activity. • Maintain muscle mass and strength through resistance exercises; this can be supplemented by aerobic and balance training. • Prevent nutritional inadequacy (including vitamin D deficiency). • Treat depression. • Reduce polypharmacy.
Early stage or "frail" (3 frailty criteria met)	Clinically apparent; earliest presentations tend to be weakness, slowed walking speed, and/or decreased physical activity	Likely most responsive to **intervention:** • Implement preventive measures (as above). • Encourage resistance and strength exercise. Evidence is substantial that resistance or strengthening exercise is effective in increasing muscle mass, strength, and walking speed in frail older adults. Other forms of exercise, including stretching, Tai chi, and aerobic exercise, are also helpful. • Prevent nutritional inadequacy; nutritional supplementation appears to be effective only when added to resistance exercise. • Offer "prehabilitation" before surgery.
End-stage or "severe frailty" (4-5 frailty criteria met and low cholestrol and albumin levels)	Severely frail older adults appear to be in an **irreversible**, pre-death phase with high mortality over 6-12 months; associated with high short-term mortality rates and suggest a poor response to treatment	Consider **palliative** approaches for these patients: • Focus on optimizing abilities needed to reach individual patient goals. • Compensate for diminished abilities by modification of living environment and/or increased support from caregivers.

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analysis of racial disparities in the use of hospice and end of life treatments found that Black individuals were significantly less likely to use hospice and more likely to have multiple emergency department visits and hospitalizations and undergo intensive treatment in the last 6 months of life compared with White individuals regardless of cause of death. Barzargan examined the awareness of palliative, hospice care and advance directives in a large sample of an ethnically diverse population. Hispanic and non-Hispanic Black participants are far less likely to report that they have heard about palliative and hospice care and advance directives than their non-Hispanic White counterparts. In this study, 75%, 74%, and 49% of Hispanics, non-Hispanic Blacks, and non-Hispanic White participants, respectively, claimed that they had never heard about palliative care.

Patients who are unable to speak the language of the care team are at risk of receiving care that may not match their wishes. Use of family members as interpreters risks information being withheld or incomplete translation. A trained medical interpreter at bedside or a telephone translator service is recommended.

The National Academy of Medicine (NAM) report on health care disparities and unequal treatment determined that patients' attitudes toward health care and treatment preferences were not the major source of disparities. At the level of clinical encounter, factors that lead to disparities include bias or prejudice, stereotypes (beliefs held by the provider about the behavior or health of minorities), and uncertainty about preferences. The key systems factors in disparities per the NAM study were factors such as cultural and linguistic barriers, fragmentation of health care systems, and the types of incentives in place.

While all persons with serious illness endure difficulties, those patients who are lesbian, gay, bisexual, and transgender/transsexual (LGBT) may face unique issues distinct from the heterosexual population. LGBT older adults have a history of having experienced stigmatization, discrimination, victimization, or violence during their lifespan by their families, school, workplaces, and even health care providers. LGBT persons may present at more advanced stages of illness because they are more reluctant to seek health care, or they lack health insurance. LGBT persons may need to rely on close friends rather than on family for caregiving. Nonfamily caregiving may be due to estrangement from the person's biological family or because LGBT persons are less likely to have children. Informal communities of support are also known as "lavender families" or "families of choice" and should be respected and included as family as identified by patients. The support from the "lavender family" may become essential in caring for an LGBT individual at the end of life. It may be challenging for LGBT persons who need custodial care to find a long-term care facility that is "LGBT friendly." LGBT patients may hide their sexual identities to be able to access a long-term care facility also known as "going back in the closet."

Transgender patients may face several unique stressors. Having a gender-variant body and requiring assistance for basic needs of daily living like bathing, dressing, or feeding puts the transgender individual in a vulnerable position and at risk of being physically abused.

It is of extreme importance that the LGBT person takes steps to legally document their preferences for health care and other legal matters. Individuals that may be closer to the LGBT person may not be identified as the default surrogate by law should the LGBT person no be able to make his or her own decision; and it is possible that the default decision maker may not be respectful of the individual's chosen gender identity.

LGBT patients should designate a Health Care Power of Attorney (HCPOA) and additionally explicitly give that person the power to direct health care professionals with the preferences of name, pronoun of choice, and appearance consistent with their gender identity. Given that the HCPOA and advance directives are no longer effective as soon as the person dies, it is important for LGBT persons to also put in place a Disposition of Bodily Human Remains (DBHR) document so that he or she can name who is the person to have authority over the remains of a person once deceased. For example, a same-sex partner who had HCPOA would not be able to claim the deceased body of his or her loved one unless he or she had been designated to do so in a DBHR document.

LGBT persons also may face challenges during the bereavement process, particularly if in the case of a same-sex relationship, the couple had not been openly acknowledged. The surviving individual may experience "silent mourning" and not have access to traditional grief support. **Table 67-3** offers open-ended inclusive questions to use to better understand patients' needs and to interact with them in a way that is aligned with their values and identity.

PROGNOSTICATION IN OLDER ADULTS

Prognostication is a vital aspect of decision making as it provides patients and families with information to determine realistic and achievable goals of care, is used in determining eligibility for benefits such as hospice, and helps in targeting interventions to those likely to live long enough to benefit from a proposed intervention. For example, it may take many years for the benefits to accrue for preventative interventions such as cancer screening and tight glycemic control. For frail older adults whose life expectancy is less than this time horizon to benefit, they are exposed to potential immediate harms of these preventative interventions without the possibility of receiving the benefits.

Prognostication can be broken into three general steps: (1) estimating the probability of an individual developing a particular outcome over a specific period of time, (2) the act of communicating the prognosis with the patient and/or family, and (3) the interpretation of the prognosis by the patient and/or family.

TABLE 67-3 ■ GETTING TO KNOW THE PATIENT IN AN INCLUSIVE WAY

QUESTION	RELEVANCE
What name do you use? How would you like me to address you?	• A person may have a legal name but may have a name of their choice they like to go by that may be different. • Listening to the patient pronounce their name gives you an opportunity to learn the pronunciation if it is a name you are not familiar with pronouncing.
What gender pronouns do you use?	• It can be awkward to ask a person their pronouns, and a way to achieve this is by introducing yourself and the pronouns you use. "My name is … and the pronouns I use are… [she/her, he/him, they/them]. What pronouns do you use?" • You don't have to assume based on the appearance of the person or their name what pronouns are preferred. • The person may not want to correct you.
To whom do you turn for support? Who are those you consider close to you that are either your family of origin or choice?	• Some persons may be estranged from the biological family and have informal communities of support. • Some patients may not be open about their sexual identity and may identify a significant other as a friend. • Asking broadly gives you the opportunity to gather information on who is important to the patient. • After a person dies, those important to the patient, whether they are the family of origin or choice or both, will benefit from bereavement care.
Do you have any groups or organizations that you belong to that are important to you?	• When asked broadly, this opens the opportunity for the patient to share what groups or organizations are meaningful to them. • The answers may be diverse and may include religious affiliation or groups like the Veterans of Foreign Wars (VFW) or the AA (Alcoholics Anonymous).
I'd like to learn about the things that give you meaning or strength. In times when you face challenges, what are some ways you cope and find strength?	• A question like this gives an opportunity for a wide range of answers to assess the spiritual or religious beliefs or existential concerns.
If you are unable to make your own decisions, who would make decisions for you? Have you formally named a medical decision maker?	• Do not assume that because a person refers to someone as their spouse that they are legally married. This is particularly relevant for unmarried LGBT couples, who may need to specify their preference in a formal document because the person the patient may choose may not be in the traditional surrogacy ranking. • Ask not only who their legal decision maker is (eg, the person they have appointed to serve as their durable power of attorney for health care or health care proxy) but also about who they would have included—and excluded—from discussions surrounding decisions about their care.

Step 1: Estimating Prognosis

Estimating prognosis in geriatric populations is more complicated than in younger populations, as older adults are more likely to have more than one chronic progressive illness that impacts life expectancy. Estimation of prognosis in multimorbid older adults requires clinicians to account for interaction of their medical problems, clinical and laboratory findings, and functional and cognitive status. While prognostication based on clinician judgment is correlated with actual survival, it is subject to numerous biases, most importantly that clinicians tend to overestimate patient survival by a factor of 3 to 5. Prognostic indices that incorporate age and clinical characteristics such as multimorbidity and functional status may improve the accuracy of prognostication compared to clinician judgment alone. A helpful repository of published geriatric prognostic indices can be found at www.ePrognosis.org.

Step 2: Communicating Prognosis

Numerous studies have shown that most individuals and their families want prognostic information, yet they are often not given the opportunity to discuss it with health care providers. This is despite evidence that effective communication around end-of-life issues can provide benefits to patients and families, without worsening of anxiety, hopelessness, or depression.

When delivering prognosis, it is first helpful to ask about the patient's understanding of their illness and perception of what the future may have in store for them. It is also helpful to assess the readiness of the patient and/or family member to have a discussion of prognosis by asking them permission to talk about it. Based on this information, the clinician should deliver prognostic information tailored to the patient's current desire and level of understanding.

When delivering prognostic estimates, clinicians should acknowledge the inherent uncertainty in most prognostic estimates. One way to do so is by giving ranges in prognostic estimates (ie, days to weeks, weeks to months, months to years to live) or by giving the best case, worst case, and most common scenarios. In addition, delivering information on prognosis is likely to bring emotional reactions in patients and their family members. Use of empathic statements, including naming the emotion, and the use of therapeutic silence can be helpful in supporting patients and their surrogate decision makers.

Step 3: Interpreting Prognosis

Patients' and surrogates' personal estimates of prognosis are often different, and generally more optimistic than what is communicated to them by health care providers. Furthermore, few surrogates solely rely on prognostication information delivered by physicians to determine their own prognostic estimates. Rather, their interpretation of prognosis is influenced by many factors, including perceptions of the patient's strength, will to live, unique history, individual observations of physical appearance, and the surrogate's presence, optimism, intuition, and faith. Given this information, it is helpful when delivering prognosis to include the factors that influence the clinician's prognostic estimate, as it may also influence the patient's or surrogate's interpretation of prognosis. It is also valuable to enquire how the clinician's prognostic estimate influenced how the patient or surrogate is currently thinking about the prognosis.

ADVANCE CARE PLANNING AND ADVANCE DIRECTIVES

ACP is a process that supports adults at any age or stage of health in understanding and sharing their personal values, life goals, and preferences regarding future medical care. ACP should be ongoing as the values or priorities of people may change as they go through life and should include continued support and preparation for future medical decision making. The focus of ACP is on meaningful conversations in order to create a care plan that aligns recommended treatments with patient preferences. There is evidence that ACP increases patient and surrogate satisfaction with both communication and medical care and decreases surrogate distress. ACP does not increase anxiety or lead to loss of hope. To effectively deliver ACP, health disparities, systemic racism, and cultural differences need to be addressed. These conversations are appropriate at any time in an older adult's life, but clinicians should especially consider initiating goals of care discussions if they answer "no" to the question "Would you be surprised if this patient dies in the next year?" Documentation of these discussions may include advance directives or physician orders further described in **Table 67-4**. **Advance directives** are written documents that state a patient's preferences for addressing *future* medical decisions. Advance directives can be executed at any time by a competent adult person and go into effect when the person is no longer capable of making health care decisions.

TABLE 67-4 ■ ADVANCE DIRECTIVES AND PHYSICIAN ORDERS

ADVANCE DIRECTIVES (FUTURE CARE)
Durable Power of Attorney for Health Care (DPOA-HC) or Health Care Proxy 1. Allows a patient to designate a person to make health care decisions on behalf of the patient should they lose decision-making capacity 2. Does not specify wishes for medical care 3. Ideal proxy is someone who the patient knows well, is a competent adult, and is willing/able to assist in health care decision making respecting the preferences of the patient should the need arise 4. Particularly important when the patient is estranged from their family or if the patient wants to name a friend who would not be in the hierarchy of surrogate decision makers
Living Will 1. Allows a patient to specify in writing wishes for future medical care in the hypothetical situation where they lose decision-making capacity 2. Difficult to cover all future possibilities making the health care proxy choice important
PHYSICIAN ORDERS (CURRENT CARE)
Physician Orders for Life-Sustaining Treatment (POLST) 1. Standing medical order specifying decisions regarding *current* medical care for patients with progressive chronic illnesses 2. Translates patient preferences for resuscitative efforts, artificial nutrition, and other life-sustaining treatment measures into an effective state-authorized medical order 3. Order accompanies patients in different settings (home, hospital, nursing home)

Priorities in ACP discussions vary depending on the setting. In a *relatively healthy* person, the focus may be on making goals of care choices based on hypothetical situations or choosing a health care proxy. If a health care proxy is not chosen, the surrogate decision maker authorized to make health care decisions for an incapacitated patient is based on a hierarchy of closeness to the patient (eg, spouse, adult children, parents, etc), which may vary based on the state. In someone with a *predictable progressive condition* who is likely to lose functional or cognitive abilities, such as ALS or dementia, the conversation should include information on treatment options with significant long-term impact, such as tracheostomy or hemodialysis, as well as prognosis. In *a seriously ill* person, the choices are less hypothetical and an actual decision regarding treatment may be needed. In this circumstance, it is important to inform patients (or their proxies) of the risks, benefits, and alternative treatments available. Consideration should

also be given to deactivating devices such as implantable cardioverter-defibrillators in patients close to death.

ACP is complex, and communication techniques differ based on the clinical situation. Chapter 71 expands on useful communication strategies. The Gerontological Society of America also has resources available to assist clinicians with communication with patients with cognitive and sensory impairments. A variety of resources to improve communication effectiveness in ACP are available online and include vitaltalk.org and capc.org. ICARE, a communication model used within the Veterans Health Administration, is provided as an example of discussing "**Do not resuscitate orders**" in **Table 67-5**. When having conversations about cardiopulmonary resuscitation (CPR), it is important to discuss the likelihood of this intervention meeting a patient's goals. A recommendation for or against CPR may differ, for example, for a patient who prioritizes maintaining independence and one who prioritizes living as long as possible. In adults older than 65 years, 17% of patients who undergo in hospital CPR survive to hospital discharge and 7% are discharged home. The presence of serious conditions such as metastatic cancer, sepsis, or multiple organ failure are associated with worse CPR outcomes.

In 2020 as the COVID-19 pandemic spread throughout the world, ACP required immediate attention and innovation. For older adults at high risk for serious disease, readdressing care plans and eliciting preferences specifically pertinent to COVID-19, such as hospitalization, intubation, and CPR, were urgently needed. Telehealth increased and remains a powerful tool for patient outreach. When using telemedicine for ACP, it is important to create a welcoming environment with the electronic device situated at eye level with adequate sound to promote patient engagement, to invite patients to include those they would want involved in the discussion, to provide an overview of a telehealth visit, and to monitor for nonverbal and environmental cues. Sending an electronic summary of the visit can also be helpful.

Addressing Spiritual Needs

Comprehensive palliative care attends to the whole needs of the patient. The founder of modern palliative care, Dr. Cicely Saunders, coined the term Total Pain, meaning that palliative care attended to the pain that was physical, psychological, social, and spiritual. An international consensus project defines spirituality as "a dynamic and intrinsic aspect of humanity through which persons seek ultimate meaning, purpose, and transcendence, and experience relationship to self, family, others, community, society, nature, and the significant or sacred. Spirituality is expressed through beliefs, values, traditions, and practices."

While patients with serious illness wish to have spiritual needs addressed as part of their care, most physicians report discomfort doing so. Screening for spiritual needs allows for referral for support, which can be in the form of pastoral support of a health care team or within the patient's own spiritual support network. A screening tool that has been well validated is FICA Spiritual History Tool. FICA asks questions about Faith, Importance, Community, and how to Address these issues. Additional questions for communication within the FICA framework can be found at https://smhs.gwu.edu/spirituality-health/.

TABLE 67-5 ■ ICARE GOALS OF CARE FOR CODE STATUS

Introduction	Orient patient or surrogate to the conversation	It is really important for us to honor our patient's choices. I'd like to spend a little time learning about things that are important to you and about the kind of medical treatments you would and would not want in an emergency. Does this sound OK to you?
Care goals	Patient's goals are the foundation of your recommendation	What matters most to you? What does a good day look like for you? As you think about your medical care, are there any medical treatments or procedures that aren't acceptable to you?
Alignment	Repeat what the patient said to ensure you got it right	As I listen, it sounds like what's most important to you is … [reflect back the patient's stated care goals].
Recommendation	Ask permission and make a recommendation aligned with the patient's goals	**If the outcomes of CPR do not align with the patient's goals:** Would it be OK if I offered a recommendation? Given what you have told me is most important, I'm concerned CPR won't help you live the life you want [repeat patient's goals]. For these reasons, I recommend we allow for a natural death instead of attempting CPR. **If the outcomes of CPR align with the patient's goals:** Would it be OK if I offered a recommendation? Given what you have told me is most important, it sounds like you would want to pursue any treatment that would give you a chance of living longer, even if that means your life might be supported by machines. For these reasons, I recommend we attempt CPR.
Exploration	Explore further if decision regarding code status seems inconsistent with stated goals	I worry that CPR won't help you reach your goals. Can you tell me more about what you are hoping for with CPR? Is there a situation you could imagine when you wouldn't want CPR?

CONCLUSION

An essential component of geriatric medicine is the routine delivery of effective primary palliative care that includes attention to symptom management, prognostication, communication, ACP, and individualized care plans. Appropriate referrals to specialty-level palliative care should also be considered when palliative needs are great, given the increasingly robust evidence base of benefit for both palliative care and hospice.

FURTHER READING

Acquaviva, KD. *LGBTQ-Inclusive Hospice and Palliative Care: A Practical Guide to Transforming Professional Practice*. New York, NY: Harrington Park Press; 2017.

American Geriatrics Society Expert Panel on the Care of Older Adults with Multimorbidity. Patient-centered care for older adults with multiple chronic conditions: a stepwise approach from the American Geriatrics Society: American Geriatrics Society Expert Panel on the Care of Older Adults with Multimorbidity. *J Am Geriatr Soc*. 2012;60:1957–1968.

American Geriatrics Society. Ten things physicians and patients should question, 2013. https://www.choosingwisely.org/societies/american-geriatrics-society/ [choosingwisely.org]. Accessed February 18, 2022.

Bakitas M, Tosteson T, Lyons K, et al. Early versus delayed initiation of concurrent palliative oncology care: patient outcomes in the ENABLE III randomized controlled trial. *J Clin Oncol*. 2015;33:1438–1445.

Bernacki RE, Block SD. American College of Physicians High Value Care Task Force. Communication about serious illness care goals: a review and synthesis of best practices. *JAMA Intern Med*. 2014;174:1994–2003.

Fried LP, Tangen CM, Walston J, et al. Frailty in older adults: evidence for a phenotype. *J Gerontol A Biol Sci Med Sci*. 2001;56:M146–M156.

Gerontological Society of America (GSA). *Communicating With Older Adults: An Evidence-Based Review of What Really Works*. Washington, DC: Gerontological Society of America. 2012.

Hanson LC, Ersek M, Gilliam R, Carey TS. Oral feeding options for people with dementia: a systematic review. *J Am Geriatr Soc*. 2011;59:463–472.

Husebo BS, Ballard C, Sandvik R, Nilsen OB, Aarsland D. Efficacy of treating pain to reduce behavioural disturbances in residents of nursing homes with dementia: cluster randomised clinical trial. *BMJ*. 2011;343:d4065.

IOM (Institute of Medicine). *Dying in America: Improving Quality and Honoring Individual Preferences Near the End of Life*. Washington, DC: The National Academies Press; 2014. https://www.nap.edu/catalog/18748/dying-in-america-improving-quality-and-honoring-individual-preferences-near [nap.edu]. Accessed February 18, 2022.

IOM (Institute of Medicine). *Unequal Treatment: Confronting Racial and Ethnic Disparities in Health Care*, 2002. http://nationalacademies.org/hmd/reports/2002/unequal-treatment-confronting-racial-and-ethnic-disparities-in-health-care.aspx. Accessed May 10, 2016.

IOM (Institute of Medicine). (US) Committee on Lesbian, Gay, Bisexual, and Transgender Health Issues and Research Gaps and Opportunities. *The Health of Lesbian, Gay, Bisexual, and Transgender People: Building a Foundation for Better Understanding*. Washington, DC: The National Academies Press; 2011. https://nationalacademies.org/hmd/reports/2011/the-health-of-lesbian-gay-bisexual-and-transgender-people.aspx. Accessed May 10, 2016.

Johnson KS. Racial and ethnic disparities in palliative care. *J Palliat Med*. 2013;16:1329–1334.

Mitchell SL, Teno JM, Kiely DK, et al. The clinical course of advanced dementia. *N Engl J Med*. 2009;361:1529–1538.

National Hospice and Palliative Care Organization. NHPCOs Facts and Figures. Hospice Care in America. http://www.nhpco.org/hospice-statistics-research-press-room/facts-hospice-and-palliative-care. Accessed May 10, 2021.

Quinn KL, Shurrab M, Gitau K, et al. Association of receipt of palliative care interventions with health care use, quality of life, and symptom burden among adults with chronic noncancer illness: a systematic review and meta-analysis. *JAMA*. 2020;324:1439–1450.

Sampson EL, Candy B, Jones L. Enteral tube feeding for older people with advanced dementia. *Cochrane Database Syst Rev*. 2009;2:CD007209.

Sidebottom AC, Jorgenson A, Richards H, Kirven J, Sillah A. Inpatient palliative care for patients with acute heart failure: outcomes from a randomized trial. *J Palliat Med*. 2015;18:134–142.

Singer AE, Meeker D, Teno JM, Lynn J, Lunney JR, Lorenz KA. Symptom trends in the last year of life from 1998 to 2010: a cohort study. *Ann Intern Med*. 2015;162:175–183.

Temel JS, Greer JA, Muzikansky A, et al. Early palliative care for patients with metastatic non-small-cell lung cancer. *N Engl J Med*. 2010;363:733–742.

Teno JM, Gozalo PL, Mor V, et al. Site of death, place of care, and health care transitions among US Medicare beneficiaries, 2000-2015. *JAMA*. 2018;320:264–271.

Zimmermann C, Swami N, Krzyzanowska M, et al. Early palliative care for patients with advanced cancer: a cluster-randomised controlled trial. *Lancet*. 2014;383:1721–1730.

Chapter 68

Pain Management

Roxanne Bavarian, Amber K. Brooks

OVERVIEW

Pain is one of the most common reasons adults seek medical care. Pain prevalence increases with age and is associated with wide-ranging adverse health outcomes including increased risk of falls, increased functional disability, declining mobility, cognitive decline, and decreased quality of life. The most common pain conditions in older adults include osteoarthritis, chronic neuropathic pain (eg, diabetes, herpes zoster), vertebral compression fractures (VCFs), pain associated with cancer and its treatments, and pain associated with other chronic illnesses. The prevalence of chronic (persistent) pain among older adults is estimated at 25% to 75% and higher in residential care settings. Chronic pain tends to be more complex in older adults with 60% to 70% of them describing multisite pain and more than 60% describing multiple types of pains. Diagnosing and treating chronic pain in older adults is further complicated by age-related changes in pathophysiology, such as cognitive impairment and organ system dysfunction; coexisting chronic medical conditions; and complex treatment considerations, such as polypharmacy and increased susceptibility to side effects, among other factors (**Table 68-1**).

Learning Objectives

- Classify different types of pain.
- Describe pathophysiological findings in chronic pain.
- Review key aspects of pain evaluation and management.

Key Clinical Points

1. **Pain in older adults is often underdiagnosed and undertreated.**
2. **Careful evaluation of pain requires a thorough history, physical examination, and use of validated pain measures.**
3. **Comprehensive pain management in older adults requires a patient-centered, multidimensional pain management strategy that recognizes the biological and psychosocial complexities and utilizes a combination of medication and nonmedication therapies.**

CLASSIFICATION OF PAIN

Pain is defined as an unpleasant sensory and emotional experience associated with actual or potential tissue damage.

Acute Versus Chronic Pain

Acute pain is defined by its sudden onset, association with a noxious stimuli, and short duration of tissue healing (approximately ≤ 3 months). Examples of acute pain include trauma and postoperative pain. Poorly controlled acute postoperative pain is a particularly vulnerable period for older adults and is associated with increased stress response (ie, increased heart rate, blood pressure, and respiratory rate), limited mobility, prolonged hospital stays, and increased risk of developing chronic pain. Conversely, acute pain that is appropriately managed during the perioperative period may help prevent the development of chronic pain.

Chronic Pain

Chronic (persistent) pain is defined as pain that persists or recurs for more than 3 months. In 2019, the International Association for the Study of Pain (IASP) updated its chronic pain classifications for the International Classifications of Diseases (ICD-11) (**Table 68-2**). Pain that is deemed a disease in its own right, such as fibromyalgia or nonspecific low back pain, is called "chronic primary pain." In the other six subgroups, pain is secondary to an underlying disease: chronic secondary musculoskeletal pain, chronic neuropathic pain, chronic secondary headache or orofacial pain, chronic secondary visceral pain, chronic cancer-related pain, and chronic postsurgical or posttraumatic pain.

High-impact chronic pain is defined as chronic pain that limits life or work activities on most days or every day during the past 6 months. In 2016, an estimated 8% of US adults (20 million) had high-impact pain, with higher

TABLE 68-1 ■ COMPLEXITIES OF PAIN AND AGING

Pathophysiology of Aging Cognitive impairment Gastrointestinal system dysfunction Hepatic system dysfunction Renal system dysfunction Pulmonary system dysfunction	**Comorbidities** Chronic complex medical conditions
Treatment Considerations Polypharmacy Side effects Caregivers Multiple specialists Compliance	**Diagnosis** Disparities in care (underdiagnosed, underreported, undertreated) Multifactorial/multisite pain Psychosocial considerations

prevalence among older adults (11% for 65 to 84-year-old adults and 16% for ≥85-year-old adults).

Pathophysiology of Chronic Pain

In a normal response to tissue injury with a noxious stimulus, the body will respond by activating immune cells, such as macrophages, leukocytes, and mast cells, that release proinflammatory mediators. These proinflammatory mediators include bradykinin, histamine, tumor necrosis factor, interleukin-1β, and interleukin-6, all of which promote the release of substance P and calcitonin gene-related peptide from nerve endings to ultimately activate spinal pathways that cause pain. These pain signals occur in response to the initial inflammation to help "teach" a person to avoid further injuring the damaged tissue during the healing process. In acute pain, healing occurs over the ensuing days to weeks, leading to resolution of inflammation and pain signals in the body. Contrarily, in patients with chronic pain, the nervous system will continue to send signals for pain even after the initial injury has subsided. The pathophysiology of chronic pain revolves around this concept of central plasticity of the brain ("neuroplasticity") in which neural connections are rewired and sensitivity to stimuli changes in response to an initial injury.

Neurological changes associated with chronic pain include hyperalgesia, allodynia, and the spread of pain. Hyperalgesia is defined as an increased sensitivity to painful stimuli, such as an exaggerated pain response to a gentle pinprick, scratch, or heat. Hyperalgesia occurs when the threshold of local nociceptors in the tissue is lowered, which increases the excitability of nociceptor neurons and pain pathways. Similarly, the receptive field of activation of these nociceptors can spread, leading to the spread of pain to adjacent, noninjured areas. Hyperalgesia and spread of pain is caused by a process known as peripheral sensitization, in which inflammation causes release of chemical mediators like histamine and bradykinin that influence the threshold and field of activation of nociceptors in the peripheral tissues. Central sensitization refers to the maladaptive plasticity within the central nervous system. This maladaptivity leads to an increase in the synaptic strength of pain pathways, with a reduction in pain inhibitory pathways, which ultimately lead to increased pain processing in the brain. Subsequently, allodynia may ensue: nonpainful touch stimuli activate mechanoreceptor neurons that have been rewired to stimulate pain pathways. Patients may also report their pain being triggered by a cool breeze against the painful area or the weight of their clothing or bedsheets brushing against the area.

Although the pathophysiological concepts behind chronic pain, such as hyperalgesia, allodynia, and spread of pain, have been verified in animal models of chronic pain, understanding why some patients develop chronic pain is a subject that warrants further research.

TABLE 68-2 ■ UPDATED CHRONIC PAIN CLASSIFICATIONS

CHRONIC PRIMARY PAIN	CHRONIC SECONDARY PAIN
MUSCULOSKELETAL PAIN	
Chronic primary musculoskeletal pain Chronic widespread pain	Chronic secondary musculoskeletal pain
NEUROPATHIC PAIN	
Complex regional pain syndrome	Chronic neuropathic pain
HEADACHE/FACIAL PAIN	
Chronic primary headache or orofacial pain	Chronic secondary headache or orofacial pain
VISCERAL PAIN	
Chronic primary visceral pain	Chronic secondary visceral pain
OTHER PAIN	
	Chronic cancer-related pain Chronic postsurgical or posttraumatic pain

Nociceptive Versus Neuropathic Pain

For treatment purposes, it is important to distinguish the underlying mechanism of pain in order to best tailor the treatment plan, particularly in regard to medications so as to minimize the risk of polypharmacy. Pain that results from the stimulation of pain receptors is called nociceptive pain. Nociceptive pain may arise from tissue injury, inflammation, or mechanical malformation. Examples include trauma, burns, infection, arthritis, ischemia, and tissue distortion. Pain from nociception usually responds well to analgesic medications.

Neuropathic pain, on the other hand, is caused by a lesion or disease of the somatosensory system, including peripheral nerves and central neurons, and affects 7% to 10% of the population. Neuropathic pain conditions commonly seen in older adults include diabetic peripheral neuropathy, postherpetic neuralgia, and

posttraumatic neuralgia (postamputation or phantom limb pain). Neuropathic pain conditions are often persistent and difficult to treat in older adults, as they are more susceptible to the side effects of commonly used neuropathic pain medications such as oversedation and falls. However, it is important to begin treatment as early as possible with a regimen associated with the least amount of harm in order to prevent long-term complications of persistent pain such as physical and psychological disability.

Aging and Its Effect on Pain Perception

Age-related changes in the nervous system may alter pain perception. A decrease in the number of pain receptors in the skin and other organs, altered nerve conduction, and even central nervous system changes have been linked to altered sensory processing in older adults. Similarly, studies of experimental pain have also been linked to age-related changes in perception. These experimental pain studies used a heat probe, electrical stimulation, or other methods to induce pain in volunteers in an effort to identify a pain threshold or pain tolerance. These studies reveal that aging may decrease sensitivity for pain of low intensity; reduced sensitivity is especially apparent for heat pain; and aging has no strong effect on pain tolerance. It is believed by most researchers that age-associated changes in pain perception are subtle, and their clinical relevance may be minimal. Conversely, older adults may have sensory impairments (visual or auditory), cognitive impairment, or sensory neuropathies that may interfere with the ability to appropriately communicate or even recognize pain symptoms.

Biopsychosocial Model for Older Adults With Pain

Chronic pain requires a multidimensional approach to treatment that incorporates the biological, psychological, and social factors that modulate a person's pain. The biopsychosocial model (BPS) of chronic (persistent) pain describes the intricate interplay between these factors. The original BPS model has since been adapted for older adults (**Figure 68-1**). A number of biological factors are associated with chronic pain in older adults, including age (usually described as ≥ 65 years of age), sex, multiple coexisting chronic medical conditions, genetic factors, common

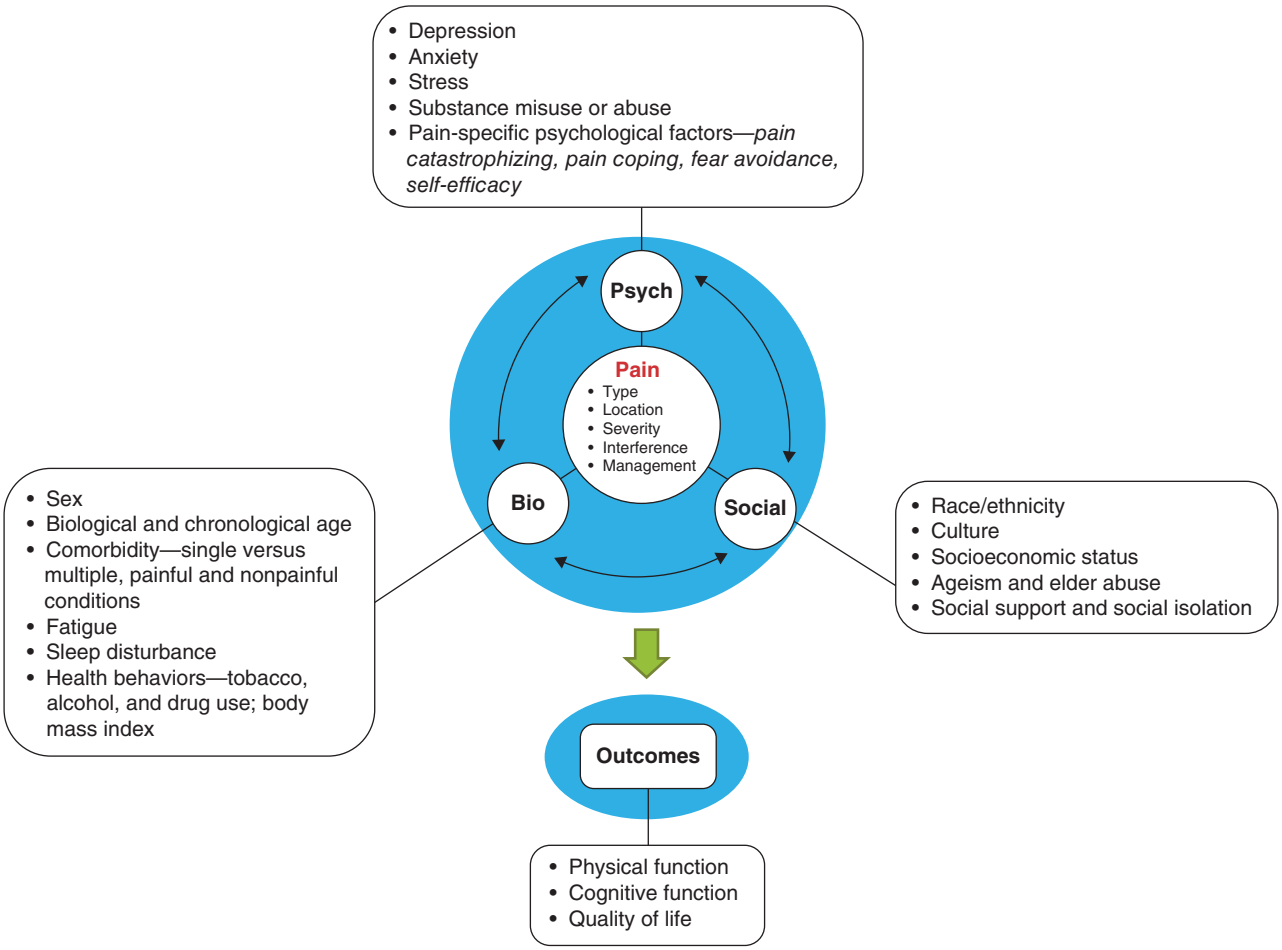

FIGURE 68-1. Biopsychosocial model of pain for older adults. (Reproduced with permission from Miaskowski C, Blyth F, Nicosia F, et al. A biopsychosocial model of chronic pain for older adults. *Pain Med*. 2020;21[9]:1793–1805.)

coexisting symptoms (fatigue and sleep disturbance), and a variety of health-related behaviors (smoking, alcohol use, and illicit drug use). Psychological considerations include depression, anxiety, stress, substance misuse or abuse, pain-specific psychological factors (pain catastrophizing, pain coping, fear avoidance, and self-efficacy). Social influences include race/ethnicity, culture, socioeconomic status, ageism and elder abuse, social support, and social isolation. Ultimately, it is the combination of these biopsychosocial factors that determine outcomes like physical function, cognitive function, and quality of life.

Diagnosis of Pain in Older Adults

The diagnosis of pain relies heavily on self-report from the patient, which is considered the gold standard in measuring pain in both clinical practice and research. A thorough assessment of pain should include a series of questions regarding the location of pain, along with its onset, frequency, intensity, quality of pain, and any modifying factors. When inquiring about pain intensity, it may be helpful to ask the patient about a range of their pain intensity over the past week, rather than solely in the present moment. The assessment of the pain characteristics should be done regularly, both at the initial visit to accurately diagnose a patient and the source of pain (ie, musculoskeletal, visceral, neuropathic, or a combination of the above) and also at follow-up visits to assess the benefit of any treatment rendered. As discussed in the previous sections, the longer pain goes untreated, the more likely for the pain to spread to neighboring areas of the body and become more difficult to diagnose.

Other pertinent questions in the assessment of pain include any history of trauma (eg, falls or injuries) or indirect trauma (eg, whiplash), especially given the risk for falls and occult fractures in older adults. Sleep quality should also be assessed for any indicators of poor sleep quality (eg, insomnia, frequent arousals, daytime hypersomnolence), which could impact the patient's perception of pain. With the patient's permission, a discussion of any recent personal stressors, such as the loss of a loved one, that may have contributed to the onset and/or intensity of pain may be relevant, particularly in neuropathic pain conditions where psychological stress is a known trigger.

In patients with pain, a review of systems should include whether the patient reports generalized muscle pain, generalized joint pain, swelling or erythema, or neurologic symptoms like numbness, tingling, or autonomic symptoms. The medical history should also be reviewed for any other chronic pain conditions and their history of management, which can help clinicians assess what treatment modalities have alleviated pain in the past. Other conditions associated with pain include psychiatric conditions, such as anxiety, depression, and posttraumatic stress disorder, which can potentially contribute or exacerbate symptoms of pain and may indicate a need for interdisciplinary treatment with a pain psychologist. Understanding the patient's lifestyle and social history can also be informative, including their average daily activities, their stress levels, and their social support system. In addition, any history of drug addiction or alcoholism either in the patient or their family should be documented to assess their risk for opioid use disorder.

In older patients, obtaining an accurate history of their pain symptoms and review of systems may be difficult due to cognitive limitations and inability to respond to questions. In older adults, the line of questioning should be simple and transparent. For example, asking "Does this hurt?" may be more informative than "How are you doing?" For patients with difficulty responding to pain questions, a nonverbal observational pain scale may be indicated to better assess their pain. In patients with cognitive impairment or dementia, family or caregivers should also be asked to provide information regarding the patient's pain.

Various verbal and nonverbal pain assessment scales have been reviewed in the evaluation of pain in older patients. These scales are generally classified as unidimensional or multidimensional. A unidimensional pain scale focuses on a single factor, such as the severity of pain or impact of pain on function. In patients with chronic pain, the impact of the pain on physical and mental function, such as their mobility, mood, and fatigue, may be more meaningful than the pain intensity itself. **Table 68-3** provides a review of unidimensional pain scales. Multidimensional scales, on the other hand, include questions to assess pain intensity, interference with enjoyment of life, or history of pain treatments and relief. **Table 68-4** shows a review of multidimensional pain scales that can be useful in assessing pain in older adults.

Following the initial intake, a thorough physical examination should be completed. The affected location of pain should be evaluated for erythema or edema, which could be a marker of inflammation or infection. The musculoskeletal system can be evaluated by palpating for tender trigger points that duplicate the patient's chief complaint, as well as assessing range of motion, posture, and gait. A neurologic examination for signs of focal muscle weakness, atrophy, or sensory impairments such as numbness or tingling should also be performed to rule out any peripheral or central neuropathic condition.

TREATMENT

Physiologic Considerations and Medication Management in Older Adults

Age-related organ dysfunction poses significant challenges for nonopioid and opioid pain medication management in older adults. The gastrointestinal, hepatic, renal, and respiratory systems are particularly important to consider when initiating and dosing pain medications in this population. There is a plethora of age-related pharmacokinetic changes that clinicians should consider before prescribing pain medications (**Table 68-5**). Decreases in gastric secretion

TABLE 68-3 ■ UNIDIMENSIONAL SCALES FOR PAIN MEASUREMENT

SCALE	ORIENTATION OF SCALE	DESCRIPTION	RECOMMENDATIONS FOR OLDER ADULTS
Numeric Rating Scale (NRS)	Horizontal or vertical	Numeric range 0–5, 0–10, 0–20, or 0–100, where 0 represents no pain and 5, 10, 20, or 100 represent extreme pain	• May be difficult for older adults with cognitive impairment • Vertical orientation preferred over horizontal
Visual Analog Scale (VAS)	Horizontal or vertical	100 mm line, with one end representing no pain and the other end representing pain at its worst	• May be difficult for older adults to understand • High rates of failure and low reliability • Avoid in older adults
Colored Analog Scale (CAS)	Horizontal or vertical	Uses colors to represent increasing pain, with white representing no pain and increasing intensity of red representing extreme pain	• Can be used in older adults with and without cognitive impairment • May not be appropriate for all cultures • Requires visual acuity
Iowa Pain Thermometers (IPT)	Vertical	Utilizes a picture of a thermometer with use of colors to represent increasing pain, with verbal descriptors and numbers	• Allows for numeric rating of intensity and description of pain • Recommended in older adults, including those with cognitive impairment, as well as ethnic minorities • Requires visual acuity
Faces Pain Scale-revised (FPS-R)	Horizontal	Utilizes a series of faces expressing pain, where the first face (face 1) is zero pain and the last face (face 6) represents maximum pain	• Recommended in older adults and ethnic minorities • Requires visual acuity

and intestinal motility lead to altered absorption of certain medications. Aging is also associated with increased body fat and decreased lean body mass, total body water, and serum albumin—all of which affect the distribution of drugs throughout the body. Circulating albumin and other proteins bind analgesic medications, such as nonsteroidal anti-inflammatory drugs (NSAIDs) and tricyclic antidepressants (TCAs). More unbound drug can lead to increased toxicity and drug–drug interactions.

In addition, alterations in hepatic and renal function in older adults can affect drug metabolism and elimination. Hepatic blood volume and blood flow decrease with age. Similarly, increasing age is associated with decreased renal blood flow and glomerular filtration rate, which may lead to increased serum concentrations of renally cleared medications and their metabolites. The resultant decline in hepatic and renal function may lead to increased risk for adverse medication events and drug–drug interactions secondary to elevated serum parent drug and metabolite concentrations.

Lastly, changes in the pulmonary systems such as decreased elasticity of the lung and increased chest wall

TABLE 68-4 ■ MULTIDIMENSIONAL PAIN SCALES FOR PAIN MEASUREMENT

SCALE	DESCRIPTION	COMMENTS
McGill Pain Questionnaire	78 words grouped into 20 categories that are related to pain	• Extensively studied; may discriminate between types of pain • Long, difficult to score
Brief Pain Inventory-Short Form (BPI-SF)	15 questions that assess pain intensity from 0 to 100, location, and interference with life, as well as any current treatments or medications	• Useful for gauging impact of pain on life, including activities of daily living • Helpful for assessing fall risk
Geriatric Pain Measure-Short Form	12 questions ask about pain intensity and interference with ambulation, social engagement, ability to accomplish tasks, and sleep	• Useful for gauging impact of pain on life, including activities of daily living • Involves more simple yes or no questions
Pain Intensity, Enjoyment in Life, and General Activity (PEG scale)	3 questions that assess pain intensity, interference on enjoyment in life, and general activity	• Can be given regularly to assess for benefits of treatment and level of pain control
Western Ontario and McMaster Universities Arthritis Index (WOMAC)	41 items in five domains: pain, stiffness, physical function, social function, and emotional function	• Specific for arthritis • Difficult to use

TABLE 68-5 ■ PHYSIOLOGICAL CHANGES ASSOCIATED WITH AGING

ORGAN SYSTEM	PHARMACOKINETIC CHANGES
Gastrointestinal system	Decreased gastric secretion Decreased intestinal motility Decreased absorption
Hepatic system	Decreased hepatic volume Decreased hepatic blood flow
Renal system	Decreased renal blood flow Decreased glomerular filtration rate
Pulmonary system	Decreased lung elasticity Increased chest wall rigidity

rigidity may lead to an increase in respiratory complications such as respiratory depression, which is of paramount importance when prescribing and dosing opioid medications.

Opioid Medication and Prescribing Practices for Older Adults

The first wave of the opioid epidemic began with the increased prescribing of opioids in the 1990s, with overdoses increasing since at least 1999. Furthermore, an analysis of Medicare beneficiaries' prescription opioid use in 2016 showed one in three Medicare Part D beneficiaries received a prescription for an opioid. Unfortunately, older adults are not immune from the detrimental consequences of opioid misuse disorder, including overdoses and deaths. The Centers for Disease Control and Prevention (CDC) reported an increase in drug overdoses and opioid deaths in the aging population, with a 7.7% increase in deaths related to opioid overdose in persons older than 65 years from 2013 to 2014. Older adults with opioid use disorder appear to be at higher risk of death compared to younger adults with the disorder. This disparity in death may be related to altered age-related organ function and accidental medication-related deaths.

In 2016, the CDC released its landmark guideline for prescribing opioids for chronic pain, which includes special considerations for prescribing practices in older adults. First, given reduced renal function and medication clearance even without renal disease, older adults may be more susceptible to the accumulation of opioids and, therefore, identifying the therapeutic window between safe dosages and higher dosages associated with respiratory depression and overdose may be extremely challenging. Cognitive impairment may increase the risk for medication-related errors and opioid-induced confusion. Prescribing providers should pay particular attention to prescribing immediate-release opioids (recommendation #4); prescribing the lowest effective dose possible (recommendation #5); and maintaining close follow-up within 1 to 4 weeks of starting opioid therapy, with subsequent follow-up at least every 3 months or more frequently (recommendation #7). Clinicians should also consider instituting bowel regimens to prevent constipation, performing risk assessments for falls, monitoring for cognitive impairment, and performing random urine drug screen monitoring, especially when there are other caregivers involved in the patient's care. In response to the opioid epidemic in the United States, the CDC has also recommended that prescribers utilize state-wide prescription drug monitoring programs (PDMPs) prior to prescribing any opioid or other controlled substance to monitor a patient's use of controlled substances and minimize the risk for abuse, overdose, and diversion.

Commonly Used Opioid Medications

For patients with moderate acute or chronic pain that has failed to respond to conservative management with nonopioid medications or nonpharmacological therapies, weak opioids such as hydrocodone, codeine, and tramadol may be considered after a thorough discussion with the patient about potential risks versus benefit. In patients with severe acute or persistent pain, more potent opioids are available, including morphine, methadone, fentanyl, oxycodone, buprenorphine, hydromorphone, and oxymorphone. Of note, in opioids combined with nonopioid analgesics, the maximum dose is typically dictated by the maximum dose of acetaminophen, NSAID, or aspirin. A table of commonly used opioid medications with recommended starting doses is featured in **Table 68-6**.

Despite the number of different opioid medications, they generally possess similar mechanisms of action and pharmacokinetics. The mechanism of opioid medications in alleviating pain is mediated by their binding to opioid receptors in the central nervous system to inhibit the ascending pain pathway. In terms of pharmacokinetics, opioids are rapidly absorbed in the gut and then undergo a high rate of first pass metabolism in the liver, where they are conjugated and form metabolites. Opioids then vary in their distribution due to their differing protein affinity, and lastly are excreted via bile to feces or via the kidneys. As discussed above, older adults may be prone to side effects of opioids due to slower gastrointestinal transit time associated with aging and risk of increased gastric pH due to concurrent use of proton pump inhibitors or antacids. In addition, the increased adipose tissue and decreased lean body mass and total body water leads to changes in drug distribution and a longer time for elimination. Thus, with any opioid medication, older patients are at risk for side effects such as central nervous system depression, sedation, respiratory depression, and constipation. Because of the increased risk of side effects in older adults, it is recommended to start opioids at a lower dose, about 25% to 50% of the dose given to younger patients. The CDC recommends carefully evaluating a patient for benefits and risks of opioid doses beyond more than 50 morphine milligram equivalents per day (MME/day) and avoiding doses more than 90 MME/day.

TABLE 68-6 ■ COMMONLY PRESCRIBED ORAL OPIOID PAIN MEDICATIONS

DRUG	INITIAL DOSING	COMMENTS
Morphine, immediate release	10 mg every 4–6 hours	Titrate to comfort: continuous use for constant pain and intermittent use for episodic pain
Morphine, extended release	Tablet: 15 mg BID-TID Capsule: 30 mg QD vs 15 mg BID	Should be avoided in opioid-naive patients
Oxymorphone, immediate release (Opana)	5–10 mg every 4–6 hours	Titrate to comfort: continuous use for constant pain and intermittent use for episodic pain
Hydromorphone (Dilaudid, Exalgo)	1–2 mg every 3–4 hours	Titrate to comfort: continuous use for constant pain and intermittent use for episodic pain
Oxycodone, immediate release (Roxicodone)	5–10 mg every 4–6 hours	Titrate to comfort: continuous use for constant pain and intermittent use for episodic pain
Oxycodone, extended release (OxyContin)	Tablet: 10 mg BID Capsule: 9 mg BID	Should be avoided in opioid-naive patients
Hydrocodone-acetaminophen (Norco)	2.5–10 mg every 4–6 hours	Maximum dose dictated by acetaminophen 4 g total daily dose
Codeine (codeine, Tylenol 3)	15–60 mg every 4 hours as needed; maximum total daily dose: 360 mg/day	Variability in effectiveness between individuals due to drug metabolism into its seven active metabolites
Tramadol (Ultram)	25–50 mg every 4–6 hours; maximum total daily dose: 300 mg	• Acts both as an opioid receptor agonist and a serotonin-reuptake inhibitor • Avoid in patients with other serotonergic drugs or history of seizures
Methadone (Dolophine)	2.5–10 mg QD	Long half-life and risk of accumulation; difficult to titrate
Fentanyl, transdermal (Duragesic)	25-mcg patch (every-72-hour dosing)	• Should be avoided in opioid-naive patients • Dose dependent on previous 24-hour total morphine milligram equivalent dose
Tapentadol (Nucynta)	50 mg every q 4–6 hours moderate to severe pain	• Partial opioid and norepinephrine reuptake receptor activity • May be helpful in treating patients with neuropathic pain
Buprenorphine	Belbuca (SL): 75–150 mcg BID, max dose 900 mcg BID Butrans (transdermal): 5 mcg/h every 7 days, max dose 20 mcg/h Buprenex (IV/IM): 0.3 mg every 6 hours	• FDA-approved for the treatment of pain • Buprenorphine is a partial agonist at the mu opioid receptor with a high binding affinity to the mu receptor and slow dissociation from the receptor • May act as an antidepressant at the kappa receptor

A consideration in the use of opioids in managing chronic pain is the risk for development of opioid-induced hyperalgesia. Opioid-induced hyperalgesia refers to a state of nociceptive sensitization in which patients taking opioids for the treatment of pain develop hyperalgesia, or increased sensitivity to painful stimuli. The mechanism of opioid-induced hyperalgesia is not entirely clear, but is thought to be due to neuroplastic changes in the peripheral and central nervous system that lead to sensitization of pronociceptive pathways. While all patients receiving opioids are at risk for opioid-induced hyperalgesia, groups with increased risk include patients with a history of opioid use disorder, chronic pain patients receiving opioids, and patients on high doses of potent opioids. Clinical signs of opioid-induced hyperalgesia include lack of effectiveness of opioids in the absence of disease progression, which may be difficult to distinguish from opioid tolerance. However, in contrast to patients with opioid tolerance, patients with opioid-induced hyperalgesia will often show increased levels of pain with increased doses of opioids. Patients may also report symptoms such as diffuse allodynia beyond the region of the original pain. Management of patients with opioid-induced hyperalgesia includes attempting to gradually reduce or eliminate the opioid and evaluate symptoms. Combination therapy with nonopioid analgesics, opioids with unique properties such as buprenorphine, or NMDA receptor antagonists may prove to be more effective in patients with opioid-induced hyperalgesia.

Management of Acute and Postoperative Pain

Management of acute pain and pain around the time of surgery is challenging. Poorly controlled and undertreated pain is associated with prolonged hospitalization, delayed wound healing, increase in health care utilization and costs, poor patient satisfaction, and adverse psychological consequences. In addition, one of the most significant long-term

consequences of poorly treated acute postsurgical pain is the development of chronic pain. However, advances in modern medicine offer promising new approaches for the management of acute pain, especially postoperative pain.

Enhanced Recovery after Surgery (ERAS) protocols have become more commonplace across the United States, especially in light of the opioid epidemic. The mainstay of ERAS protocols is the judicious use of multimodal analgesia. ERAS protocols attempt to minimize the use of opioids and offer opioid-sparing techniques such as intravenous or oral acetaminophen, intravenous or oral NSAIDs, intravenous lidocaine infusion, intravenous ketamine infusion, liposomal bupivacaine, neuraxial and peripheral regional anesthesia regional techniques for select surgeries, as well as patient-controlled modalities. Opioids still play an important role in the treatment of acute and postoperative pain, but are limited by their side effects and other sequelae, especially in older adults. If using opioid medications for acute pain in the inpatient setting, it is important to consider the following risk factors for opioid-related respiratory depression (and other adverse events), including a history of sleep apnea, obesity, concomitant administration of other respiratory depressant drugs such as benzodiazepines, opioid-naïve patients, opioid-tolerant patients, and chronic obstructive pulmonary disease. While supplemental oxygen is commonly used after surgery, it may produce acceptable oxygen saturation levels via pulse oximetry even with substantial hypoventilation. Thus, patients with risk factors for opioid-related respiratory depression should be considered for perioperative monitoring with continuous end-tidal carbon dioxide monitoring, especially during the first 24 hours after surgery.

Management of Chronic Pain

For patients with mild chronic pain, nonopioid analgesics, such as NSAIDs or acetaminophen, are recommended. These medications are available in tablet, capsule, topical, and injection form with many being available over the counter.

Acetaminophen Acetaminophen (Tylenol) is a widely available analgesic and antipyretic that can be used for mild-to-moderate pain of any etiology. While its mechanism of action is not yet fully understood, it is thought to activate descending serotonergic inhibitory pathways in the central nervous system to reduce pain perception. While acetaminophen is considered relatively safe in its side effect profile when compared to NSAIDs, the main adverse reaction is hepatotoxicity due to acetaminophen overdose. In older adults, initial dosing is recommended at 500 to 1000 mg every 6 hours. The maximum total daily dose of acetaminophen is 4000 mg. In patients with hepatic impairment or a history of alcohol use, a maximum daily dose of 2000 to 3000 mg is recommended. Given the availability of acetaminophen as an over-the-counter analgesic, prescribed medication, as well as combined with other medications, such as opioids, it is important to review all of a patient's current medications to ensure the patient is not taking higher than the recommended dose.

Nonsteroidal anti-inflammatory drugs (NSAIDs) NSAIDs represent another class of widely available medications that has both anti-inflammatory and analgesic effects that alleviate mild-to-moderate pain, and in particular, pain of an inflammatory origin, such as osteoarthritis or rheumatoid arthritis. There are a variety of NSAIDs available either as a prescription or over the counter, each with different dosing regimens and tolerability profiles. Often, if one particular type of NSAID is not effective for a patient, it may be worthwhile trying a different NSAID. **Table 68-7** reviews the most common NSAID medications and their doses.

NSAIDs work to reduce pain by reversibly inhibiting the activity of cyclooxygenase enzymes, COX-1 and COX-2, and consequently preventing the synthesis of prostaglandins, which mediate inflammation as well as the transmission of pain. NSAIDs that act as nonselective COX inhibitors are not recommended for long-term use due to side effects associated with inhibition of COX-1. COX-1 is present throughout the body and plays a role in protecting the gastric mucosal lining and maintaining renal and hepatic function, among other functions. Inhibition of COX-1 is associated with increased risk for gastrointestinal ulcers and bleeding. Thus, patients on NSAIDs for any period of time should be monitored for any symptoms of gastrointestinal pain or bleeding. Caution should be given when prescribing either NSAIDs or aspirin to older adults who are on anticoagulant therapy due to increased risk of bleeding. Other complications of long-term NSAID use include increased risk for cardiovascular events and kidney disease.

COX-2, on the other hand, is present throughout the body in lower concentrations and is expressed in response to injury and inflammation. Utilization of NSAIDs that selectively target and inhibit COX-2 enzymes, such as celecoxib (Celebrex), can effectively reduce inflammation and pain, while sparing patients from the organ toxicity associated with other NSAIDs. However, COX-2-selective NSAIDs are not risk-free and, similar to nonselective COX-inhibiting NSAIDs are associated with increased risk for cardiovascular events.

For older adults at risk for adverse events associated with systemic NSAIDs, topical agents such as diclofenac 1% gel, represent a safe form of an anti-inflammatory treatment that can be massaged onto intact skin overlying painful areas, such as joints affected by osteoarthritis or other inflammatory conditions.

Other Adjunct Medications for Chronic Pain

In addition to opioids analgesics, a number of other medications may be helpful in the management of chronic pain depending on the diagnosis and etiology of the patient's pain. **Table 68-8** reviews common medications used in the

TABLE 68-7 ■ COMMONLY PRESCRIBED NONSTEROIDAL ANTI-INFLAMMATORY DRUGS (NSAIDs)

DRUG	DOSING	MECHANISM OF ACTION	COMMENTS
Celecoxib (Celebrex)	100–200 mg BID	Selective COX-2 inhibition	Reduced GI toxicity and platelet compared to other NSAIDs
Ibuprofen (Motrin, Advil)	2400 mg maximum total daily dose, dosed every 6–8 hours	Reversibly inhibits COX-1 and COX-2 to decrease formation of prostaglandin precursors	• High risk for side effects, particularly peptic ulcers, in older patients • Avoid long-term use due to GI, renal, cardiovascular toxicity • Older adults at risk for CNS side effects (confusion, agitation)
Naproxen (Aleve, Naprosyn)	1000 mg maximum total daily dose, dosed every 8–12 hours	Same as ibuprofen	Same as ibuprofen
Meloxicam (Mobic)	7.5–15 mg QD	Same as ibuprofen	Same as ibuprofen
Diclofenac (Cataflam)	150–200 mg maximum total daily dose, dosed every 6–12 hours	Same as ibuprofen	Same as ibuprofen
Indomethacin (Indocin)	25–50 mg, up to TID	Same as ibuprofen	Beers Criteria (AGS 2019): avoid use due to high risk for GI toxicity and acute kidney injury
Nabumetone (Relafen)	1,000 mg QD-BID	Same as ibuprofen	Same as ibuprofen
Ketorolac (Toradol)	**IM:** 120 per day, with 30–60 mg loading dose followed by half the loading dose **PO:** 60 mg per day, with 6-hour dosing	Same as ibuprofen	Beers Criteria (AGS 2019): avoid use due to high risk for GI toxicity and acute kidney injury
Diclofenac 1% topical gel (Voltaren gel)	2–4 g applied QID PRN	Same as ibuprofen	Topical gel to avoid systemic side effects, can be massaged onto intact skin

TABLE 68-8 ■ NONOPIOID COANALGESIC MEDICATIONS

DRUG	DOSING	MECHANISM OF ACTION	COMMENTS
Serotonin-norepinephrine reuptake inhibitors (SNRIs)			
• Venlafaxine (Effexor) • Duloxetine (Cymbalta)	37.5–75 mg QD 20–60 mg QD	Modulation of pain pathway via serotonin, norepinephrine	• Low sedation and anticholinergic effects compared to TCAs • Beers Criteria (AGS 2019): avoid SNRIs in patients with history of falls or fracture • Avoid abrupt discontinuation, slow taper over 4 weeks to avoid withdrawal symptoms
Tricyclic antidepressants (TCAs)			
• Amitriptyline (Elavil) • Nortriptyline (Pamelor)	10–50 mg QHS 10–50 mg QHS	Increases concentration of serotonin and/or norepinephrine to modulate pain pathways	• Beers Criteria (AGS 2019): avoid due to high anticholinergic activity; avoid in patients with history of falls or fracture • Avoid abrupt discontinuation, slow taper over 4 weeks to avoid withdrawal symptoms
Gabapentinoids			
• Gabapentin (Neurontin) • Pregabalin (Lyrica)	100 mg QHS, up to 900 mg TID PRN 25 mg QHS, up to 75 mg TID PRN	Block voltage-gated calcium channels to reduce neuronal hyperexcitability	• Initiate at lowest dose and gradually titrate • Contraindicated in those with renal impairment • Avoid in patients with concurrent opioid use
Sodium channel blockers			
• Carbamazepine (Tegretol) • Oxcarbazepine (Trileptal)	200–400 mg BID, maximum total daily dose of 1200 mg daily 150–300 mg BID, maximum total daily dose of 2400 mg daily	Block voltage-gated sodium channels to reduce neuronal hyperexcitability	• Consider testing for HLA-B*1502 allele • Risk for hyponatremia—monitor Na^+ and drug levels

(*Continued*)

TABLE 68-8 ■ NONOPIOID COANALGESIC MEDICATIONS (*CONTINUED*)

DRUG	DOSING	MECHANISM OF ACTION	COMMENTS
Benzodiazepines			
• Clonazepam (Klonopin) • Diazepam (Valium)	0.5 mg QHS 2–2.5 mg up to BID PRN	Increases activity of GABA to reduce neuronal excitability	• Initiate with low doses and monitor, particularly if any hepatic or renal impairment • Risk for sedation and falls—avoid in patients with concurrent opioid use
Muscle relaxers			
• Baclofen (Lioresal)	5–10 mg, up to TID	Muscle relaxer; inhibits transmission of monosynaptic and polysynaptic reflexes to reduce muscle spasticity	• Avoid if possible due to high risk of sedation and cholinergic effects • Gradual taper recommended if discontinuing
• Methocarbamol (Robaxin)	500 mg up to TID	Centrally acting skeletal muscle relaxer	Beers Criteria (AGS 2019): avoid use; limited efficacy at doses tolerated by older adults
• Tizanidine (Zanaflex)	2–4 mg up to QID	Alpha$_2$-adrenergic agonist, inhibits transmission of polysynaptic reflexes to reduce muscle spasticity	Caution in renally impaired patients
• Cyclobenzaprine (Flexeril)	5–10 mg up to TID	Centrally acting skeletal muscle relaxer	High risk of sedation and anticholinergic effects
• Metaxalone (Skelaxin)	800 mg up to QID	CNS depressant	Beers Criteria (AGS 2019): avoid use; limited efficacy at doses tolerated by older adults
Topical adjunct treatments			
• Topical lidocaine	Available as topical creams (3–5%), mucosal jelly 2%, mucosal solution 2%, and transdermal patches (1.8–5%)	Decreases neuronal membrane to sodium ions to reduce nerve initiation and	
• Capsaicin 0.075% cream	Apply thin film to affected area up to QID PRN	Binds TRPV-1 receptors on nociceptors, leading to increased pain followed by pain relief mediated by depletion of substance P	

management of chronic pain. These medications are often categorized as being "adjuvant" or "co-analgesics," as their initial indication was for conditions other than pain, such as depression or seizures; however, many of these medications can be used as a first-line treatment option for specific chronic pain conditions.

Tricyclic antidepressants (TCAs) and serotonin-norepinephrine reuptake inhibitors (SNRIs) can be used in the management of chronic pain. By increasing the activity of serotonin and norepinephrine, these medications modulate the pain pathway to inhibit ascending pain pathways and promote descending pain inhibitory pathways. These medications have been shown to be effective in treating neuropathic pain as well as musculoskeletal pain. Despite their efficacy at low doses, TCAs like amitriptyline and nortriptyline possess potent anticholinergic activity, which can lead to adverse effects, particularly in older individuals. These adverse effects can include memory impairment, confusion, and hallucinations. Patients are also at risk for dry mouth, blurred vision, constipation, nausea, urinary retention, impaired sweating, and tachycardia. SNRIs should also be initiated with caution due to their serotonergic activity and risk for seizures. Additionally, Beers Criteria (American Geriatric Society, 2019) recommend avoidance of SNRIs in patients with a history of falls and fractures.

For patients with chronic neuropathic pain conditions, such as diabetic neuropathies, postherpetic neuralgia, and trigeminal neuralgia, medications in the category of anticonvulsants may be helpful. These include gabapentinoids, such as gabapentin and pregabalin, and sodium channel blockers, such as carbamazepine and oxcarbazepine. These medications have a high risk for side effects such as central nervous system depression, fatigue, dizziness, and disorientation. Such disorientation can cause patients to take the

incorrect dosing or precipitate falls. Thus, dosing should always be initiated at a low dose with a gradual increase of the dose as needed to treat symptoms. Noteworthy, precautions include prescribing gabapentinoids with opioids due to the combined effects of central nervous system depression.

Clonazepam is a benzodiazepine that may be indicated for chronic neuropathic pain conditions, such as tinnitus or burning mouth syndrome, as well as movement disorders. However, benzodiazepines must be prescribed with caution in older adults because of the potential risk for sedation, confusion, and falls. For similar reasons, benzodiazepines and opioid medications should not be taken concomitantly. In addition, due to the risk for abuse and addiction, benzodiazepines such as clonazepam are included along with opioids in state-wide prescription drug monitoring programs and measured as lorazepam milligram equivalents (LMEs). Older adults taking daily benzodiazepines for more than 30 days may develop physiologic dependence and should be gradually tapered to avoid risk of withdrawal symptoms.

While sodium channel blockers like carbamazepine and oxcarbazepine represent highly efficacious frontline treatments for neuropathic pain such as trigeminal neuralgia, these medications require caution and close monitoring when prescribing. Due to the side effects of sedation, clinicians should consider advising the patient to have a family member or caregiver monitor the patient when first initiating carbamazepine or oxcarbazepine. Due to these drugs' effect on sodium levels and on the bone marrow, regular monitoring of sodium levels along with serum drug levels, a complete blood count with differential, and liver and renal function tests are recommended when initiating either of these medications. In addition, patients with either a HLA-B*1502 and HLA-A*3101 allele (most common in South Asian ancestry) are at risk for severe skin reactions, such as Stevens–Johnson syndrome, and thus, genotype screening may be indicated. Carbamazepine is also a potent cytochrome P 450 inducer that can interfere with the metabolism of other drugs, particularly those with narrow therapeutic indices like warfarin or lithium.

In patients with moderate-to-severe chronic musculoskeletal pain, muscle relaxants may be indicated to alleviate pain. A variety of muscle relaxants are available, including baclofen, methocarbamol, tizanidine, and cyclobenzaprine. While these medications are typically indicated on an as-needed basis to treat muscle spasm, patients with chronic musculoskeletal pain such as temporomandibular joint and muscle disorders caused by parafunctional habits (ie, jaw clenching and tooth grinding) may benefit from a nightly dose to reduce their pain. However, all muscle relaxants carry potential side effects of sedation and anticholinergic effects. The lowest possible dose is recommended when initiating therapy, with patients gradually increasing the dose either as needed or as tolerated.

Nonpharmacological Treatments

Chronic pain in older adults is often treated with a multimodal pain management approach: a combination of medications, physical therapy, psychological interventions, or interventional pain management. Interventional pain management is defined as the discipline of medicine devoted to the diagnosis and treatment of pain-related disorders, principally with the application of interventional techniques (commonly referred to as injection therapies), independently or in conjunction with other treatment modalities. Interventional pain procedures are performed by a myriad of providers, including, but not limited to, anesthesiologists, physiatrists, neurologists, neurosurgeons, orthopedic surgeons, rheumatologists, and radiologists. Interventional pain procedures have the added benefit of targeting specific nociceptive transmission sites with the goal of minimizing the intake of oral medications and their end organ effects. Thus, interventional pain management techniques offer older adults an alternative treatment pathway with potentially fewer side effects.

Low back pain *Lumbar epidural steroid injection* (ESI) is a commonly used procedure for treating lumbar spinal stenosis, lumbar disc herniation, lumbar degenerative disc disease, and lumbosacral radicular pain. Pertinent imaging such as plain films, magnetic resonance imaging (MRI), and computed tomography (CT) scans should be reviewed before performing an ESI, although physical examination findings and pain symptoms do not always correlate to image findings. ESIs are preferably performed with image guidance (CT or fluoroscopy) for increased accuracy using an interlaminar approach (midline or paramedian) or transforaminal approach. The transforaminal approach (placement of a needle within the neuroforamen) is preferentially used for patients with lumbosacral radicular pain who may also have low back pain. Contraindications to ESIs include coagulopathy, current anticoagulation use, infection (localized near injection site or systemic), uncontrolled diabetes, allergy to medication being injected (contrast, local anesthetic, steroid), anatomic changes that would prevent a safe procedure (congenital or surgical), or immunosuppression. Potential complications include bleeding, infection, neural injury, inadvertent injection of steroid or local anesthetic outside of the epidural space, and an allergic reaction to medications administered.

There is considerable controversy surrounding the efficacy of ESIs. The results of clinical trials are influenced by the type of interventional pain management specialist, injection approach (interlaminar vs transforaminal), pain type, and injectate. Nonetheless, there is general consensus that ESIs provide short-term relief (weeks to months) in well-selected patients. ESIs should be used in combination with other forms of pain management with the goal of reducing pain and improving function.

Lumbar Facet Injections Lumbar facet-mediated pain is a common cause of low back pain in older adults that is often associated with functional limitations. Lumbar facet joints,

or zygapophyseal joints, are synovial joints in the lumbar region that are innervated by the medial branch (MB) nerves of the dorsal primary ramus. Lumbar facet-mediated pain may manifest in the low back and commonly refers pain to the groin, hip, or thighs, but rarely below the knee. Patients commonly report pain with bending over, twisting movements, lateral rotation, and prolonged sitting or standing. Typical examination findings include pain with direct palpation over the facet joints and pain with extension and lateral rotation, also known as facet loading. Diagnostic studies (plain films, CT, or MRI) can be helpful in characterizing the degree of lumbar facet arthropathy (mild, moderate, or severe). Lumbar facet mediated-pain injections can be performed by injecting directly into the joint space (intra-articular) or by aiming the injection at the junction of the superior articular process and transverse process where the MB nerves reside. Facet MB joint diagnostic nerve blocks are most commonly performed due to fair to good evidence regarding efficacy. In addition, an 80% reduction in pain after facet MB joint nerve blocks offers the added advantage of proceeding to MB nerve radiofrequency denervation in an attempt to provide longer lasting relief.

Sacroiliac Joint Injections The sacroiliac joints (SIJs) are small, diarthrodial joints located at the junction of the sacrum and ilium, whose primary purpose is to act as a shock absorber for the spine. The ability for the SIJ to absorb shock from the spine decreases with age and is associated with increased pain. SIJ degenerative changes may be seen on imaging studies (plain films, CT, or MRI). Physical examination findings may include tenderness to palpation directly over the joint space or positive provocative tests (ie, Faber, distraction, compression, Gaenslen's, or thigh thrust). The SIJ can refer pain to the hip, buttocks, sacrum, or thighs, but usually does not extend past the knee. It can be difficult to distinguish SIJ pain from other causes of low back pain like discogenic pain, lumbar myofascial pain, or lumbar facet-mediated pain. Like facet-mediated pain, intra-articular steroid injections or nerve blocks of the L5 dorsal primary rami and the lateral branch (LB) of the S1-3 dorsal rami may be performed. Evidence regarding the efficacy of intra-articular SIJ with steroids is limited. The best recommendation is to use these injections in combination with other multimodal pain management including physical therapy, medication management (muscle relaxants, anti-inflammatories), SIJ stretching exercises, heat, ice, or transcutaneous electrical nerve stimulation unit. If 80% relief from the nerve blocks is appreciated, the patient may proceed to radiofrequency denervation of the L5 dorsal primary rami and the lateral branch (LB) of the S1-3 dorsal rami for more prolonged relief.

Percutaneous Vertebral Augmentation for Compression Fractures VCFs are the most common fragility fracture reported in the literature. VCFs, by definition, compromise the anterior half of the vertebral body and the anterior longitudinal ligament leading to a characteristic wedge-shaped deformity. In older adults, the most common risk factor for VCFs is osteoporosis. The prevalence is highest among women older than 50 years, affecting an estimated 25% of postmenopausal women in the United States. An estimated 40% to 50% of people older than 80 years have sustained a VCF. Furthermore, people who have had one osteoporotic VCF are at five times the risk of sustaining a second VCF. Only a quarter of VCFs result from falls; most are precipitated by routine daily activities such as bending or lifting. VCFs most frequently occur at the thoracolumbar junction (ie, the segment from T12 to L3) and can be diagnosed with plain films, CT scan, or MRI. An MRI may provide more additional information regarding the acuity of the VCF, regarding whether it is, acute, subacute, or chronic. Patients afflicted with VCFs report wide-ranging symptoms including no pain, while others report severe pain. Symptomatic patients may benefit from conservative treatment including pain medications, immobilization via bracing, and physical therapy, while others fail to respond.

Two minimally invasive procedures, vertebroplasty (VP) and kyphoplasty (KP), are commonly used to treat persistent, acutely painful VCFs. In VP, cement is injected percutaneously into the fractured vertebral body via a cannula. KP is similarly performed percutaneously, but involves injection of the cement into an inflated balloon that creates a cavity that encapsulates the cement. Potential complications of both VP and KP include cement leakage into the spinal canal with resultant neurologic deficits and cement leakage into surrounding vascular structures with subsequent risk of pulmonary embolism. Contraindications to VP and KP include existing coagulopathy or use of blood thinners, burst fractures with retropulsed bone, and vertebral height loss greater than 66%.

Osteoarthritic joint pain and the use of interventional pain management Osteoarthritis (OA) is highly prevalent in older adults, with 60% of people 65 years and older diagnosed with arthritis or chronic joint pain. OA is a leading cause of disability, economic burden, and functional decline in older adults. OA is characterized by an active, dynamic process arising from an imbalance between the repair and destruction of joint tissues. Structural alterations in articular cartilage, adjacent bone, ligaments, synovium, and capsule have all been described in OA. Interventional pain management for knee and hip OA offers older adults nonsurgical and relatively low risk pain management therapy options. Interventional pain management for knee and hip OA is most often performed using fluoroscopy, MRI, CT scan, or ultrasonography (US) to help guide needle placement and increase accuracy of the procedure.

Intra-Articular Injections In general, three types of intra-articular injections may be performed including corticosteroid, hyaluronic acid (HA) also known as viscosupplementation, and platelet-rich plasma (PRP). These injections are commonly offered to patients with moderate-to-severe

symptomatic OA that have responded unfavorably to physical therapy or nonsteroidal anti-inflammatory medications. Corticosteroid medications usually contain 40 to 80 mg of a depot steroid preparation such as triamcinolone acetate or methylprednisolone acetate in combination with a local anesthetic such as bupivacaine or lidocaine.

HA is naturally found in the knee and acts as a shock absorber and lubricant. Decreasing levels of HA are associated with OA. There are several commercially available HA preparations in the United States including Supartz, Synvisc, and Orthovisc. These formulations vary in origin (rooster combs vs bacterial), molecular weight, and the number of recommended injections (1–5).

PRP is the injection of autologous plasma with a high concentration of platelets into an affected joint or other area of interest. PRP injections are thought to relieve OA-related symptoms via the interaction of platelets, intrinsic joint tissues, and the release of growth factors and cytokines by activated platelets, which ultimately act to reduce inflammation and promote local healing.

In general, all three types of injections are safe and have relatively few side effects. Decisions about frequency, type, and imaging guidance for injection therapies are highly variable, but should be tailored to the patient.

Knee Pain Injection Treatments The knee is the largest joint in the body and the most common site of OA. Its prevalence is two times higher in men than women. Frequently reported symptoms are pain with ambulation, crepitus (cracking or popping sensation), warmth with palpation of the joint, or effusion (ie, swelling). Patients are good candidates for knee injections if they have radiographic evidence of moderate or severe OA. Corticosteroid, viscosupplementation, and PRP may be injected into the knee joint. In the short term (< 3 months), intra-articular steroid injections of corticosteroids and local anesthetic mixtures are more effective than placebo. However, long-term benefits are not well established. PRP injections for knee OA, especially end-stage OA, show promising efficacy (up to 12 months in some studies). However, there remains high variability in the administration of PRP including injection techniques, injection schedules, and number of centrifugations of autologous plasma, which can drastically alter the concentration of platelets injected.

Hip Pain Injection Treatments The hip joint is a ball and socket joint consisting of the articulation of the femoral head with the acetabulum. OA-related hip pain commonly travels to the groin but can also travel to the low back or buttock area. Patients report increased pain with standing and walking. Pain with internal rotation of the hip is a hallmark physical examination finding. Corticosteroid in combination with a local anesthetic is the most commonly performed intra-articular hip injection, which is done under the guidance of imaging (fluoroscopy, CT, or ultrasound). Additional clinical studies are needed to evaluate the short- and long-term efficacy of HA and PRP injections for hip pain.

Neuropathic pain conditions and the use of interventional pain management Neuropathic pain conditions are more common in older adults, as the risk for conditions that cause neuropathic pain increase with age. For example, conditions like diabetes, herpes zoster, stroke, as well as cancer treatment can all result in neuropathic pain. While most patients respond to standard medications like anticonvulsants and gabapentinoids, older adults may not tolerate frontline treatment with anticonvulsants or gabapentinoids due to side effects of sedation and impaired cognition. Although relatively rare, neuropathic pain affecting the orofacial pain region, such as trigeminal neuralgia and postherpetic neuralgia, can be particularly difficult to manage due to its impact on speaking, chewing, and swallowing. Nerve blocks represent a safe, minimally invasive treatment option that can be both diagnostic, allowing clinicians to localize the source of pain, and therapeutic, in reducing the intensity or frequency of pain. For patients with suspected neuropathic pain of the facial region, an MRI of the brain with trigeminal protocol is recommended to rule out any mass lesion or vascular compression of the trigeminal nerve. Nerve blocks frequently performed in the orofacial region include inferior alveolar nerve blocks for trigeminal neuralgia of the V3 distribution, posterior superior alveolar nerve blocks for trigeminal neuralgia of the V2 distribution, occipital nerve blocks for occipital neuralgia and cervicogenic headaches, and sphenopalatine ganglion blocks for cluster headaches. While the duration of relief with a nerve block can be variable, for some patients, periodic nerve blocks can be helpful for treating flares of pain while avoiding polypharmacy with multiple analgesics and anticonvulsant medications. For older adults with persistent, refractory orofacial pain who cannot tolerate frontline treatments such as anticonvulsant medications, referral to neurosurgery is recommended for consideration of microvascular decompression of the trigeminal nerve, rhizotomy, or gamma knife radiosurgery targeting the trigeminal nerve.

Behavioral Therapies for the Treatment of Pain

In addition to the aforementioned pharmacological and nonpharmacological interventions, behavioral therapy plays a central role in the management of pain, particularly in older patients who are prone to the side effects of medications or may not be candidates for interventional treatment. Behavioral therapies represent safe and noninvasive techniques targeted to reduce pain, restore function and mobility, and improve overall physical and mental health. In fact, many patients themselves often directly state to their doctors that they are eager to try a more conservative treatment rather than initiate a new medication or undergo an invasive procedure.

For patients with chronic musculoskeletal pain, such as osteoarthritis or myofascial pain, physical therapy represents a noninvasive treatment to improve mobility,

alleviate pain, and ultimately reduce disability and improve function for patients. For patients who do not have access to physical therapy, home care with stretching exercises and use of moist heat or ice as needed can help improve pain. In chronic pain conditions such as fibromyalgia, physical activity is recommended to reduce symptoms of pain and stiffness with the ultimate goal of improving daily function. While a combination of aerobic, resistance, and flexibility training is ideal, each regimen of exercise should be tailored to the individual and their functional goals. For example, in older patients with chronic pain and stiffness, a low impact alternative such as aquatic-based physical therapy or aquatic-based exercise may be indicated to maintain their mobility and reduce their pain.

Patients with chronic pain frequently experience significant emotional limitations due to their condition. In addition to their persistent, distressing pain, they may experience difficulties in obtaining a diagnosis and effective management, leading to feelings of discouragement and depressed mood. Some patients may develop disability due to their pain and isolate themselves from others. Psychological interventions have been well studied in the treatment of chronic pain conditions such as fibromyalgia, chronic low back pain, and rheumatoid arthritis. The goal of this treatment is to help patients cope with their pain and minimize any disability and distress. Evidence-based psychological interventions for patients with chronic pain include both talk-based and behavioral therapy, such as cognitive behavioral therapy, acceptance and commitment therapy (ACT), and biofeedback. For patients who either lack access to a psychologist trained in cognitive behavioral therapy or are otherwise resistant to initiate treatment with a psychologist, mindfulness-based stress reduction techniques may be helpful as another mind-body approach to improve a patient's physical and mental health. Examples of stress reduction techniques include meditation, yoga, deep breathing exercises, and body scans. Such techniques are evidence-based to improve chronic pain such as low back pain, with the underlying goal of helping patients increase their awareness from moment to moment and allow them to accept any uncomfortable emotions and physical discomfort.

There are various complementary and alternative therapies that can potentially reduce pain, including acupuncture, massage therapy, Reiki healing, and dietary modifications or supplements, which many patients may seek out as an alternative to conventional Western medicine. As a clinician, it is important to support the patient as they develop their own approach to adapt to their chronic pain and maintain functionality and discuss these treatment options. Dietary modifications and supplements should also be reviewed with the patient to ensure no harmful side effects or interactions with their current medication regimen.

FURTHER READING

Booker SQ, Herr KA. Assessment and measurement of pain in adults in later life. *Clin Geriatr Med*. 2016; 32(4):677–692.

Brooks AK, Udoji MA. Interventional techniques for management of pain in older adults. *Clin Geriatr Med*. 2016;32(4):773–785.

By the 2019 American Geriatrics Society Beers Criteria® Update Expert Panel. American Geriatrics Society 2019 Updated AGS Beers Criteria® for potentially inappropriate medication use in older adults. *J Am Geriatr Soc*. 2019;67(4):674–694.

Centers for Disease Control and Prevention. (2016, January 1). Increases in drug and opioid overdose deaths—United States, 2000–2014. *Morbidity and Mortality Weekly Report*. https://www.cdc.gov/mmwr/preview/mmwrhtml/mm6450a3.htm?s_cid=mm6450a3_w. Accessed May 29, 2021.

Chau DL, Walker V, Pai L, Cho LM. Opiates and elderly: use and side effects. *Clin Interv Aging*. 2008;3(2):273–278.

Chiu IM, von Hehn CA, Woolf CJ. Neurogenic inflammation and the peripheral nervous system in host defense and immunopathology. *Nat Neurosci*. 2012;15(8):1063–1067.

Dahlhamer J, Lucas J, Zelaya, C, et al. Prevalence of Chronic Pain and High-Impact Chronic Pain Among Adults — United States, 2016. *MMWR Morb Mortal Wkly Rep* 2018;67:1001–1006.

Department of Health and Human Services. (2016). Opioids in Medicare Part D: Concerns about extreme use and questionable prescribing. *Office of the Inspector General*. https://oig.hhs.gov/oei/reports/oei-02-17-00250.asp. Accessed May 29, 2021.

Dowell D, Haegerich TM, Chou R. CDC guideline for prescribing opioids for chronic pain—United States, 2016. *MMWR Recomm Rep*. 2016;65(No. RR-1):1–49.

Lautenbacher S, Peters JH, Heesen M, Scheel J, Kunz M. Age changes in pain perception: a systematic-review and meta-analysis of age effects on pain and tolerance thresholds. *Neurosci Biobehav Rev*. 2017;75: 104–113.

Lee M, Silverman SM, Hansen H, Patel VB, Manchikanti L. A comprehensive review of opioid-induced hyperalgesia. *Pain Physician*. 2011;14(2):145–161.

Mitra S, Carlyle D, Kodumudi G, Kodumudi V, Vadivelu N. New advances in acute postoperative pain management. *Curr Pain Headache Rep*. 2018;22(5):35.

Pergolizzi J, Böger RH, Budd K, et al. Opioids and the management of chronic severe pain in the elderly: consensus statement of an International Expert Panel with focus on

the six clinically most often used World Health Organization Step III opioids (buprenorphine, fentanyl, hydromorphone, methadone, morphine, oxycodone). *Pain Pract*. 2008;8(4):287–313.

Treede RD, Rief W, Barke A, et al. Chronic pain as a symptom or a disease: the IASP Classification of Chronic Pain for the International Classification of Diseases (ICD-11). *Pain*. 2019;160(1):19–27.

Williams ACC, Fisher E, Hearn L, Eccleston C. Psychological therapies for the management of chronic pain (excluding headache) in adults. *Cochrane Database Syst Rev*. 2020;8(8):CD007407.

Chapter 69

Management of Common Nonpain Symptoms

Christine S. Ritchie, Alexander Smith, Christine Miaskowski

OVERVIEW: COMMON NONPAIN SYMPTOMS IN OLDER ADULTS

Advances in medicine have led to our ability to stave off many acute life-threatening events and have contributed to the development of numerous co-occurring chronic conditions defined as multimorbidity. The experience of multiple conditions is often characterized by an array of symptoms associated with these conditions and/or their treatments. The presence of bothersome symptoms in older adults may contribute to illness burden in ways that may not be predictable on the basis of any one diagnosed disorder. However, these symptoms have a negative influence on older adults' function and quality of life (QoL).

Many symptoms are addressed throughout this textbook: anorexia is covered in Chapter 30, sleep disorders in Chapter 44, dizziness in Chapter 45, pain in Chapter 68, depressed mood and anxiety in Chapters 65 and 66, and constipation in Chapter 87. The focus of this chapter is on overall symptom assessment, evaluation, and management of fatigue and shortness of breath, especially for those with advanced illness, and special considerations in the context of multimorbidity.

Multiple symptoms occur frequently in chronically ill older adults. Because many individual conditions, such as cancer, heart failure, and chronic obstructive pulmonary disease (COPD), are associated with high symptom burden, their accumulation contributes to a large and complex array of symptoms (**Figure 69-1**). In a nationally representative study of older adults that assessed for pain, fatigue, breathing difficulty, sleeping difficulty, depressed mood, and anxiety, 28% of the population had three or more symptoms. Of note, reduced physical function and falls were more common in those with higher symptom burden. In a population-based sample of older adults using a 10-item symptom tool, the average number of symptoms was 3.7 and one-third of the population had five or more. In this cohort, the most common nonpain symptoms were fatigue (48%), weakness (39%), constipation (36%), anxiety (36%), anhedonia or depression (37%), shortness of breath (35%), and poor appetite (17%). In a study of 318 adults followed by a housecalls program, 43% reported severe burden from one or more symptoms. The symptoms with the highest severity ratings were depression, pain, loss of appetite, and shortness of breath. Few longitudinal studies of the general older adult population have assessed changes in the symptom experience over time. Among 754 older adults in their last year of life, the monthly occurrence of one or more "restricting symptoms" was fairly constant in the first half of the year prior to death (20%). At approximately 5 months prior to death, the rate increased rapidly from 27% to 57% in the month prior to death.

Learning Objectives

- Review the prevalence of and assessment strategies used to evaluate common nonpain symptoms in older adults.
- Address the evaluation and management of fatigue and dyspnea.
- Describe symptom management considerations in the context of multimorbidity.

Key Clinical Points

1. **Many older adults experience a higher symptom burden associated with multiple co-occurring conditions. Screening for common symptoms should be a routine component of a comprehensive geriatric assessment (CGA).**
2. **Multicomponent interventions that rely on nonpharmacologic treatments are most effective for the management of fatigue and breathlessness.**
3. **Because the evidence base for symptom management in older adults is sparse, especially for those with multiple chronic conditions, a thoughtful approach that is informed by patient preferences is necessary to optimize benefit and minimize adverse effects.**

	Cancer	CHF	COPD	CKD	Dementia	AIDS
1.	Pain (84%)	Breathlessness (88%)	Breathlessness (97%)	Fatigue (88%)	Breathlessness (46%)	Fatigue (73%)
2.	Fatigue (69%)	Fatigue (88%)	Fatigue (88%)	Sleep disturbance (82%)	Constipation (40%)	Pain (69%)
3.	Anorexia (66%)	Pain (75%)	Dry mouth (85%)	Anorexia (55%)	Pain (39%)	Dry mouth (54%)

FIGURE 69-1. Pain and nonpain symptoms in common chronic conditions. Top three most common (in rank order) symptoms by condition. AIDS, acquired immunodeficiency syndrome; CHF, congestive heart failure; CKD, chronic kidney disease; COPD, chronic obstructive pulmonary disease.

OVERALL EVALUATION OF SYMPTOMS

A CGA, historically did not include an assessment of pain or nonpain symptoms. More recent CGA versions have included systematic assessments of these symptoms.

Many symptom measures were developed initially for cancer patients and only evaluate symptoms over short time intervals. Other symptom measures focus on a particular disease without capturing the impact of nonindex conditions on symptoms. With few exceptions, nondisease-specific symptom inventories are not available for older adults with multiple conditions. One screening tool developed from a community-dwelling population of 1000 older adults, the Brief Symptom Inventory, assesses 10 common symptoms in older adults: shortness of breath, feeling tired or fatigued, problems with balance or dizziness, weakness, daily pain, stiffness, constipation, poor appetite, anxiety, and anhedonia. The tool shows discriminant and convergent validity. Another recently developed tool in Denmark captured 36 symptoms in a representative sample of 100,000 Danish people 20 years and older. The average number of symptoms reported by the population was 5.4. For each additional comorbid condition, one new symptom was added.

Many comprehensive symptom assessments were borrowed from oncology. The Memorial Symptom Assessment Scale (MSAS) measures the occurrence, severity, and distress of 32 symptoms and the Edmonton Symptom Assessment Scale measures 9 common symptoms on a 0 to 10 numerical rating scale. These instruments have been used to assess symptoms and multiple dimensions of the symptom experience in both geriatric and palliative care noncancer patients. The large number of items in the MSAS can be a practical barrier to its use in the clinical setting.

EVALUATION AND MANAGEMENT OF FATIGUE AND DYSPNEA

Evaluation of Fatigue

The prevalence of generalized fatigue in older adults varies from 5% to almost 70% depending on the measure used, the characteristics of the older adult population, the time of day fatigue is assessed, and the cut points used to determine fatigue. Regardless of the assessment tool, fatigue is associated with decreased function and increased mortality. In a study of 492 older primary care patients, the question "Do you feel tired most of the time?" identified older adults with a one and a half to almost twofold increased risk of mortality over 10 years. Other assessments for fatigue tend to be longer and assess multiple dimensions of fatigue, again, predominantly derived from oncology. The Brief Fatigue Inventory, a nine-item instrument that assesses severity of fatigue and the impact of fatigue on daily functioning in the past 24 hours, has good psychometric properties but has been predominantly used in cancer populations. More recently, the concept of fatigability developed as a complement to the symptom of fatigue. Fatigability measures how fatigued an individual feels in relation to defined activities (eg, how fatigued a person feels after a 5-minute treadmill test). It offers

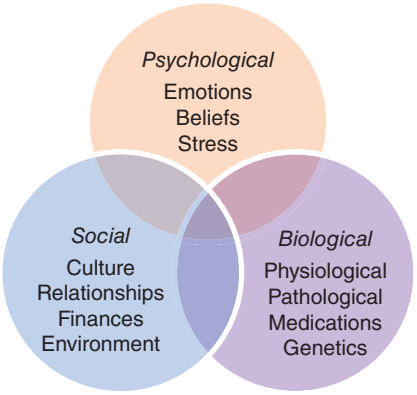

FIGURE 69-2. Biopsychosocial model.

a more standardized way to measure fatigue but is more difficult to integrate into standard practice.

A more comprehensive assessment of fatigue should address (1) when the fatigue began, whether the onset was sudden or more gradual; (2) the pattern of fatigue over the course of a day; (3) what exacerbates or improves the fatigue; and (4) the impact that fatigue has on daily function and relationships. Sleep characteristics and medications should be reviewed. Because psychological factors are often associated with fatigue, systematic assessment for depression is warranted.

The biopsychosocial model, commonly used to characterize pain, applies well to other symptoms, including fatigue. The biopsychosocial model posits that psychological, social, spiritual, and physical factors influence an individual's symptoms (**Figure 69-2**). In the case of fatigue, other factors such as obesity, depression, social isolation, loneliness, and spiritual distress may influence the experience of fatigue. A number of physical conditions are characterized by fatigue including electrolyte disturbances, occult malignancy, polymyalgia rheumatica, occult hepatitis, or HIV. Fatigue may be a silent harbinger of anemia or infection. In older adults, where atypical presentations of acute conditions are common, new-onset fatigue may indicate a recent myocardial infarction (MI) or heart failure. Hypogonadism is a less common cause of fatigue. Findings from the history and physical examination should dictate further diagnostic evaluation.

Management of Fatigue

Very few studies have addressed the management of fatigue in older adults; more trials have focused on chronic fatigue syndrome in younger adults. A few studies have evaluated fatiguability in context of frailty. Approaches that have not shown promise include protein supplementation, thyroid supplementation (in subclinical hypothyroidism), or testosterone (in men). Counseling, vitamin D3 (in those with vitamin D deficiency), and functional training have shown some promise. Improvement of older adults' sleep hygiene through both nonpharmacologic and pharmacologic strategies (see Chapter 44) may reduce fatigue. While antidepressants may be beneficial, antidepressants with strong anticholinergic effects should be avoided (see Chapter 65). In the palliative care setting for older adults with advanced or life-limiting illnesses, psychostimulants such as methylphenidate or modafinil may be considered.

In a study of predominantly older prostate cancer patients (ages 52–94), methylphenidate reduced fatigue severity compared to control. However, a number of patients had to discontinue treatment due to increased blood pressure and tachycardia. A recent within-person crossover trial cast doubt on methylphenidate treatment of fatigue in advanced cancer. Among 43 participants who alternated between methylphenidate and placebo three times over a 9-day period, no improvement in fatigue during days with methylphenidate was observed. In older adults with known cardiovascular disease or arrhythmias, psychostimulants should not be used. While anecdotal reports touted the benefit of donepezil in treatment of opioid-induced fatigue in cancer patients, a subsequent clinical trial did not support these claims. In summary, fatigue is a common condition in older adults with likely multiple factors contributing to its prevalence.

Evaluation of Dyspnea (Breathlessness)

Due to the high rates of COPD and heart failure in older adults, along with an array of other comorbid conditions, shortness of breath (dyspnea) is very common in older adults. In an Australian primary care practice, over half of those who presented with shortness of breath were older than the age of 65. The prevalence of dyspnea in community-dwelling older adults ranges between 17% and 62%, depending on the population studied and cut point used to define "dyspnea." Dyspnea will likely occur at some point during a number of serious illnesses experienced by older adults (eg, cancer, heart failure, advanced lung disease), and at the end of life. The biopsychosocial model applies to dyspnea as well as it does to fatigue. Anxiety, disappointment, financial stressors, and questions about meaning often contribute significantly to the experience of dyspnea. Dyspnea, in turn, serves as a source of patient and caregiver distress and is associated with decreased QoL, decreased function, and increased health care utilization.

Physiologic mechanisms of dyspnea often fall under three categories: increased respiratory effort due to obstruction (eg, COPD, asthma, masses) or restriction (eg, obesity, pleural effusion); weakness (eg, multiple sclerosis, amyotrophic lateral sclerosis); or ventilation/perfusion mismatch (eg, anemia, pulmonary embolism, heart failure). More subtle systemic changes can contribute to the occurrence of dyspnea, especially at the end of life. For example, in the National Hospice Survey, 24% of patients with no known cardiopulmonary disease experienced dyspnea. The most appropriate measure of shortness of breath is the older adult's self-report. Dyspnea does not always correlate with hypoxia, hypercarbia, or the presence of tachypnea. Most self-report

TABLE 69-1 ■ MEDICAL RESEARCH COUNCIL BREATHLESSNESS SCALE

GRADE	DEGREE OF BREATHLESSNESS
1	Not troubled by breathlessness except on strenuous exercise
2	Short of breath when hurrying on level ground or walking up a slight hill
3	Walks slower than most people on level ground, stops after a mile or so, or stops after 15 min walking at own pace
4	Stops for breath after walking about 100 yd or after a few minutes on level ground
5	Too breathless to leave the house, or breathless when undressing

tools to assess dyspnea come from the obstructive lung disease literature. One of the most common assessment tools is the Medical Research Council (MRC) breathlessness scale. First published in the 1950s, the MRC scale characterizes, through five statements, a range of disability caused by level of breathlessness (**Table 69-1**). It correlates well with other dyspnea scales and with other direct measures of function, such as walking speed.

Management of Shortness of Breath

Nonpharmacologic approaches to breathlessness have few side effects, unlike morphine for example, and avoid polypharmacy, a major concern in the older adults. They should be considered alongside pharmacologic approaches as first-line therapy. Nonpharmacologic approaches that can be helpful in the treatment of dyspnea include the use of a fan, breathing techniques, mindfulness and relaxation, anxiety management, and energy conservation. Breathing techniques that can reduce the sensation of dyspnea include pursed lip breathing, prolonged exhalation, and posture modification. In addition to mindfulness and relaxation, guided imagery and distraction strategies (eg, music, TV, reading by self or caregiver) were shown to reduce the sensation of breathlessness.

In patients with COPD and hypoxia, long-term oxygen therapy increases QoL and prolongs survival. Likewise, in a subset of patients with COPD, noninvasive ventilatory support such as with bilevel positive airway pressure (BiPaP) can improve QoL and prolong survival. In patients with advanced cancer and other nonpulmonary causes of dyspnea, the value of oxygen supplementation in improving outcomes is less clear. These results may be in part due to the fact that less than half of advanced cancer patients with dyspnea are hypoxic. In a study of nasal cannula–delivered air versus nasal cannula–delivered oxygen (median age 65), no difference was found in dyspnea relief between the two modalities when patients were not hypoxic; in two studies of hypoxic cancer patients, more benefit from oxygen was noted. Recent studies of noninvasive ventilation in advanced cancer patients suggest that it may improve symptoms. In a multisite study of 200 predominantly older adults with advanced cancer (mean age 71), noninvasive ventilation was more effective than oxygen supplementation alone in reducing dyspnea and decreasing the amount of morphine needed to control symptoms. However, more patients in the noninvasive ventilation arm discontinued treatment primarily due to mask intolerance and anxiety.

Pharmacologic treatment for dyspnea should first and foremost be directed to the underlying cause of dyspnea, if known. Treatment may include a β-agonist for COPD or a diuretic in the case of heart failure. In a subset of patients, breathlessness persists at rest or on minimal exertion despite optimal treatment of the underlying chronic condition. These patients are described as having refractory dyspnea. In patients with refractory dyspnea, opioids may be beneficial. A number of studies of opioids for the treatment of refractory dyspnea in the setting of advanced illness demonstrate some benefit. However, the findings of reduced dyspnea (compared to control) are not consistent, and many studies remain underpowered to produce conclusive findings. Unfortunately, many of these studies use morphine, which has active metabolites (eg, morphine-6-glucornide, morphine-3-glucoronide). Morphine-3-glucoronide builds up in renal insufficiency, common in older adults and is responsible for symptoms of neurotoxicity (eg, hyperalgesia, allodynia, myoclonus). In older adults, opioids are associated with decreased mental functioning. For some patients, decreased mental functioning will not be acceptable despite its potentially positive impact on breathing.

IMPACT OF MULTIMORBIDITY ON SYMPTOM MANAGEMENT AND PATIENT DECISION MAKING

For older adults, multimorbidity is the norm not the exception. Over 90% of Americans age 65 and older have two or more chronic conditions. In older adults, symptom burden increases with increasing numbers of co-occurring conditions and is associated with doubled mortality rates and reduced QoL.

Symptom Management in Multimorbidity

Multimorbidity can constrain options for management of symptoms. Many pharmacologic treatments for symptoms have adverse effects that are magnified in patients with multimorbidity. For those with heart failure or hypertension, nonsteroidal anti-inflammatory agents often exacerbate these conditions. For those with cognitive impairment, opioids may make confusion more marked. Well known to those caring for older adults, initiation of one medication for one condition or symptom often leads to side effects

that require use of a second medication to manage the side effects—the "prescribing cascade." Analgesics provide an apt illustration of this phenomenon. The use of a nonsteroidal anti-inflammatory drug (for those in whom it is not contraindicated) often requires additional medications to reduce the risk of potential negative gastrointestinal (GI) (eg, the addition of a proton pump inhibitor to reduce risk of GI bleeding) or cardiovascular (eg, the addition of aspirin to reduce the risk of MI or stroke) outcomes. In the case of opioids, patients must routinely be started on bowel stimulants or other laxatives to avoid the expected side effect of constipation.

Even without taking into account these challenges, many medications used to address symptoms have not been evaluated in older adults or in adults with multimorbidity. Among the symptom-focused studies done in older adults with advanced illness or multimorbidity, the most evidence exists for management of pain, and to a much lesser degree, dyspnea in cancer. There is a dearth of evidence for how to manage symptoms in noncancer illnesses, let alone in patients with several diseases. Therefore, the true benefits and the true risks are largely unknown. For this reason, pharmacologic management of symptoms warrants thoughtful discussions with the patient and/or family caregivers, as well as cautious initiation and careful monitoring of both the positive and negative effects of the treatment. For many older adults, symptom management involves both relief of discomfort and development of new drug-induced challenges. Anticipatory and ongoing discussion of the benefits and burdens of pharmacologic symptom management in the context of the patient's goals and preferences offers an ongoing person-centered approach to improving the older person's QoL.

The Role of Patient Preference in Symptom Management

Given the lack of evidence for optimal symptom management for older adults in general and in particular for older adults with multimorbidity, treatment strategies for symptom management are preference sensitive—that is there is often more than one reasonable treatment option, or a particular option offers uncertain benefit. For preference-sensitive decisions, the clinician must understand what is most important to the patient to determine what might be the best treatment option. A starting point may involve asking patients to prioritize a set of universal health outcomes that can be applied across individual diseases. Typical outcomes would include living as long as possible, maintaining function, staying cognitively intact, and alleviating pain and other symptoms. Other outcomes may include staying out of the hospital or dying at home. While not always mutually exclusive, understanding how patients prioritize these various outcomes can help guide decisions about symptom management. This broad-based approach provides a more unified starting point, rather than asking patients or caregivers to make specific treatment decisions.

Optimal decisions regarding symptom management require that the patient is adequately informed about the expected benefits and harms of different treatment options for their symptoms, recognizing that in many instances optimal information is lacking. Actual decision-making preferences vary widely by patients and family caregivers. Some individuals prefer to make the decision themselves, while others prefer that the decision for a specific treatment be made by the clinician. In either instance, most individuals want their opinion used to guide the decision-making process. Then, the clinician can develop a management plan and evaluate at regular intervals whether the management plan remains concordant with patient's wishes. Reevaluation of treatment plans is critical as studies suggest that older adults with multimorbidity engage in dynamic reassessments of their conditions, as they shift between experiencing disruption from the condition and finding the ability to adapt to the condition's challenges.

The following case illustrates these decision-making issues. Consider an 86-year-old man with advanced COPD who presents to the emergency department acutely dyspneic. He wants to live a few more days to say goodbye to family members and friends, even if it means living in the hospital. He does not want to be intubated or sent to the intensive care unit (ICU). He agrees to try BiPAP while initiating diureses. He expresses immediate profound relief of dyspnea upon securing the mask. However, the next morning, he asks that the mask be removed as it seems suffocating. He has had the opportunity to talk to his family and the distress associated with the BiPAP mask no longer makes it worthwhile. The patient's clarity around what is important to him can guide the clinician in these preference-sensitive treatment decisions.

SUMMARY

Experiencing multiple symptoms is common in older adults and particularly common in older adults with multimorbidity. For most symptoms, pharmacologic and nonpharmacologic approaches have some evidence base. Unfortunately, the evidence base is very limited for older adults, especially for those with multiple chronic conditions. Because the evidence base for symptom management is so sparse, treatments are preference sensitive and should take into account the values and preferences of older patients and their caregivers. Universal outcomes around function, cognition, comfort, and survival may be good starting points regarding which treatment options make the most sense. Ultimately more research will be needed to ascertain which treatments are the most beneficial for older adults and those with multimorbidity.

FURTHER READING

Eldadah BA. Fatigue and fatigability in older adults. *PMR*. 2010;2(5):406–413.

Elnegaard S, Andersen RS, Pedersen AF, et al. Self-reported symptoms and healthcare seeking in the general population—exploring "The Symptom Iceberg". *BMC Public Health*. 2015;15:685.

Glynn NW, Santanasto AJ, Simonsick EM, et al. The Pittsburgh Fatigability Scale for older adults: development and validation. *J Am Geriatr Soc*. 2015;63(1):130–135.

Hardy SE, Studenski SA. Fatigue predicts mortality among older adults. *J Am Geriatr Soc*. 2008;56(10):1910–1914.

Jones PW, Harding G, Berry P, Wiklund I, Chen W-H, Leidy NK. Development and first validation of the COPD assessment test. *Eur Respir J*. 2009;34:648–654.

Jones PW, Quirk FH, Baveystock CM. The St George's respiratory questionnaire. *Respir Med*. 1991;85(suppl B): 25–31.

King DE, Xiang J, Pilkerton CS. Multimorbidity trends in United States adults, 1988-2014. *J Am Board Fam Med*. 2018;31(4):503–513

Lehti TE, Öhman H, Knuutila M, et al. Symptom burden is associated with psychological wellbeing and mortality in older adults. *J Nutr Health Aging*. 2021;25(3):330–334.

Mahmoud AM, Biello F, Maggiora PM, et al. A randomized clinical study on the impact of Comprehensive Geriatric Assessment (CGA) based interventions on the quality of life of elderly, frail, onco-hematologic patients candidate to anticancer therapy: protocol of the ONCO-Aging study. *BMC Geriatr*. 2021;21(1):320.

Martinez-Amezcua P, Simonsick EM, Wanigatunga AA, et al. Association between adiposity and perceived physical fatigability in mid- to late life. *Obesity (Silver Spring)*. 2019;27(7):1177–1183.

Mendoza TR, Wang XS, Cleeland CS, et al. The rapid assessment of fatigue severity in cancer patients—use of the Brief Fatigue Inventory. *Cancer*. 1999;85:1186–1196.

Mitchell GK, Hardy JR, Nikles CJ, et al. The effect of methylphenidate on fatigue in advanced cancer: an aggregated N-of-1 trial. *J Pain Symptom Manage*. 2015;50(3): 289–296.

Morris RL, Sanders C, Kennedy AP, Rogers A. Shifting priorities in multimorbidity: a longitudinal qualitative study of patient's prioritization of multiple conditions. *Chronic Illn*. 2011;7(2):147–161.

Nava S, Ferrer M, Esquinas A, et al. Palliative use of non-invasive ventilation in end-of-life patients with solid tumours: a randomised feasibility trial. *Lancet Oncol*. 2013;14(3):219–227.

Patel KV, Guralnik JM, Phelan EA, et al. Symptom burden among community-dwelling older adults in the United States. *J Am Geriatr Soc*. 2019;67(2):223–231.

Portenoy RK, Thaler HT, Kornblith AB, et al. The Memorial Symptom Assessment Scale: an instrument for the evaluation of symptom prevalence, characteristics and distress. *Eur J Cancer*. 1994;30A:1326–1336.

Ritchie CS, Hearld KR, Gross A, et al. Measuring symptoms in community-dwelling older adults: the psychometric properties of a brief symptom screen. *Med Care*. 2013;51(10):949–955.

Smith MEB, Nelson HD, Haney E, et al. *Diagnosis and Treatment of Myalgic Encephalomyelitis/Chronic Fatigue Syndrome. Evidence Report/Technology Assessment No. 219. AHRQ Publication No. 15-E001-EF*. Rockville, MD: Agency for Healthcare Research and Quality; December 2014. Addendum July 2016. www.effectivehealthcare.ahrq.gov/reports/final.cfm. Accessed January 3, 2022.

Chapter 70

Palliative Care Across Care Settings

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INTRODUCTION

Consider Mrs. M, an 85-year old woman with congestive heart failure, mild cognitive impairment, and multijoint osteoarthritis, who lives alone in a second-floor, walk-up apartment. Mrs. M retired from her part-time secretarial work at the age of 70 to care for her husband with Lewy body dementia and has been widowed for the past 5 years. Mrs. M recently agreed to try ambulating with a walker after her third trip to the emergency room for falls but frequently forgets to use it. Her daughter, an only child, lives in the same city and helps her mother shop and clean. In the past, she accompanied her mother to medical appointments but has been unable to do so regularly in the past 2 years due to her work schedule and helping care for her grandchildren. Although Mrs. M never missed medical appointments in the past, she now has "no-showed" to most visits including for follow-up after being evaluated in the emergency room. Her daughter receives a call from a concerned neighbor who reports that her mother only rarely comes out of her apartment, and when she does, seems confused and unsteady. Her daughter has had similar concerns, and additionally worries that her mom appears to be in pain most days, depressed, and losing weight. She takes a day off from work to bring her mom to see her primary care doctor; but, prior to the appointment, she gets a call from the emergency room that her mother fell again resulting in a broken hip requiring surgery. Mrs. M's postoperative course is complicated by delirium and pain. She is transferred to a skilled nursing facility for rehabilitation where the team raises the concern that Mrs. M now has moderate dementia, frailty, and significant gait impairment. They recommend a more supportive living environment. Mrs. M moves in with her daughter and enrolls in a home-based primary care program. Initially, Mrs. M does well, with notable improvements in her mood, weight, walking, pain control, and cognitive health. However, over the next 2 years, her dementia progresses with increasingly severe behavioral and psychological symptoms and rising care needs. She becomes nearly wheelchair bound due to pain when walking. With limited support for her mom at home and now facing her own medical issues, her daughter admits Mrs. M to a nursing home. Mrs. M is hospitalized three times over the next year. She enrolls in hospice after the third hospitalization and survives another year.

Learning Objectives

- Conceptualize the ideal model of palliative care provision for the aging population facing multimorbidity and progressive functional impairment across care settings.
- Identify existing deficiencies and barriers to the adequate delivery of palliative care for older adults across the care continuum.
- Illustrate how effective collaboration between geriatrics and palliative care services with integration of palliative care interventions into existing models of care can help bridge these gaps.

Key Clinical Points

1. **The palliative care needs of older adults differ from younger populations due to differences in illness trajectories, treatment preferences, and patterns of health care utilization.**
2. **Although the highest proportion of palliative care needs for older adults exists in the community setting, access to palliative care is concentrated in acute care settings and through hospice utilization.**
3. **Older adults often experience advancing frailty and multiple complex conditions, necessitating an integrated approach between geriatrics and palliative care specialties to meet the needs of this population across their trajectory of functional decline and increasing needs for care and support.**

The case of Mrs M illustrates the need for a truly integrated system of palliative and geriatric care that meets the medical and social needs of older adults across the care continuum. Palliative care aims to improve the quality of life for persons with serious and advanced illness by decreasing symptom burden, addressing psychological and spiritual distress, and promoting well-being of patients and

their families. The types of care that would help support Mrs. M's quality of life vary as her health circumstances and care settings change. These include, but are by no means limited to, (1) advance care planning conversations to delineate the patient's goals, values, and preferences and ensure goal-concordant care; (2) assessment and treatment of postoperative pain and delirium with heightened monitoring for postoperative cognitive and functional decline; (3) an interprofessional team that can provide longitudinal care for the patient across various settings; (4) caregiver training and support for the patient's daughter; and (5) functional assessment and home adjustments to address needs for adaptive devices and therapy services.

Unfortunately, in reality, our current health care system is fragmented with a disproportionate focus on acute care and immediate post-acute care, often leading to a lack of emphasis on matching treatments to patient goals. In this chapter, we will describe the ideal of a palliative care system for older adults that delivers integrated care across the care continuum, including hospice. We will contrast this ideal with the barriers to and lack of an integrated system for delivery of palliative care across the continuum. We will identify opportunities for improvement. We begin by discussing the tremendous overlap between the fields of geriatrics and palliative care.

GERIATRICS AND PALLIATIVE CARE ARE BETTER TOGETHER

The case of Mrs. M illustrates the blurred boundaries between geriatrics and palliative care (**Figure 70-1**). Geriatrics and palliative care both emphasize improving quality of life in late life. Both fields require excellence in management of persons with multimorbidity, dementia, and disability. Both fields emphasize attention not just to the patient, but also the patient's caregiver, and the situation of the patient within the context of family and community. Both fields are directed primarily toward the sickest and most frail older adults, those that account for about half of health care spending in the United States.

By 2030, the number of Americans aged 65 or older is predicted to approach 73 million, representing 21% of the total US population. Eighty percent of all older adults have at least one chronic condition and 50% have a least two or more chronic conditions. Notably among these, the number of Americans living with Alzheimer disease is anticipated to double from 5.8 million, currently, to 13.8 million by 2050. It is also increasingly recognized that a substantial proportion of community-dwelling older adults are frail (~ 15%) and prefrail (45%). Like dementia, frailty is even more common at older ages, among women and racial and ethnic minorities, and with additional variability in prevalence based on geographic region, location of residence, and socioeconomic status.

In the setting of frailty and multiple chronic illnesses, older adults commonly experience long trajectories of functional decline and episodic fluctuations in health status. They frequently face repeated, nonlinear, and often difficult transitions across care settings, including the community, hospital, rehabilitation, and nursing facilities, and ultimately experience disjointed care. Older adults' family and physicians often have a poor understanding of their goals of care and have rarely engaged in discussions regarding treatment preferences.

Spurred by the needs of the aging population, there is increasing recognition that palliative care does not equate to "comfort care only" for the terminally ill. The recognition that geriatric palliative care approaches can have a major impact on symptom management, communication around goals of care, management of multiple chronic conditions, and caregiver support for all patients with advanced illness and frailty from the time of diagnosis is an invaluable step toward improving the quality of life for older adults.

The fields of geriatrics and palliative care have many common grounds, including providing goal-oriented care,

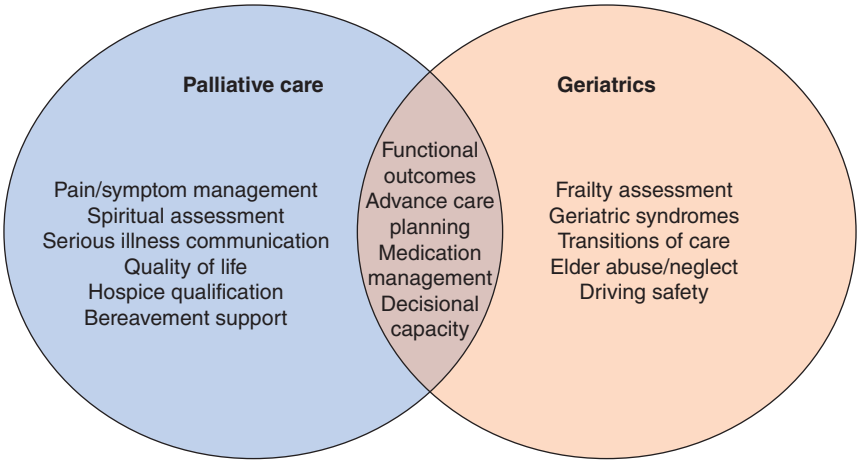

FIGURE 70-1. Intersection of palliative care and geriatrics.

working within an interprofessional team, using multidimensional assessments to identify unmet needs, attending to psychosocial factors, addressing caregivers' needs and including them in care planning and delivering, and adding valuable services to the most vulnerable and frail older adults. On the other hand, these intersections between geriatric medicine and palliative care give rise to unclear boundaries and limited understanding between the two professions. In addition, there are several gaps between the two fields. Palliative care contributes to patient care by offering intensive symptom management and psychosocial support; prognostication; alignment of goals and treatments between patients, families, and clinicians; and support in ethical decision making. Geriatric medicine offers a comprehensive geriatric assessment, including a better understanding of frailty and resilience, functional status, and expected trajectory. As such, the two specialties best serve patients when applied in a blended, interconnected way to deliver integrated patient care rather than in series (geriatrics, then palliative care). These care models can be implemented in different settings and best serve mutual patients as each specialty brings a unique and important aspect to the patient's care, with an ability to intensify or de-intensify a particular service's involvement depending on the patient's health state.

By simultaneously advocating for the holistic care of older adults, recognizing the specialized knowledge base and skills possessed by each discipline, and ensuring competency of future trainees in both geriatrics and palliative care, clinicians from both fields will be able to work together to target the sickest 5% of our population at highest risk, and match these patients with delivery models best suited to address their needs.

THE IDEAL OF A CARE CONTINUUM

The ideal of a continuum of care has been defined as, "a client-oriented system composed of both services and integrating mechanisms that guides and tracks clients over time through a comprehensive array of health, mental health, and social services spanning all levels of intensity of care." The introductory chapter in this section on palliative care introduced the concept that palliative care is appropriately initiated from the time of diagnosis with serious or advanced illness. For many, though not all, serious and advanced illnesses are diagnosed in the outpatient setting. The ideal care continuum would integrate palliative care into the settings where patients spend the most time after diagnosis, that is, in their homes. This would involve access to robust palliative care services along with integration of basic palliative care principles in ambulatory and home-based settings.

As patients develop more serious conditions, such as advancing frailty and dementia in the case of Mrs. M, they may require more care in acute institutional settings like skilled nursing facilities, hospitals, and nursing homes. The ideal continuum of care would track patients as they move between settings, coordinate care across providers, and offer comprehensive services as people move between the community and the inpatient setting. As a patient's goals increasingly shift toward a focus on quality of life, high-quality palliative care services should take on a greater role in care, and be available in all settings.

THE REALITY OF PALLIATIVE CARE FOR OLDER ADULTS

Palliative care has been noted to improve quality of life and symptom management for patients, increase satisfaction of patients and their family members, and reduce health care costs. However, in the current health care system, palliative care is most readily made available to patients during times of crisis (acute inpatient hospitalization) or at the end of life (hospice). The majority of older adults' last months and years are still spent in the community, with institutionalization occurring only in the very late stages of life. There has been less progress noted in the area of nonhospice palliative care in nursing homes or the community setting, including assisted living and home-based programs, suggesting that the current level of palliative care services does not adequately address the needs of older adults.

Palliative care needs for older adults with advancing frailty differ from the needs of younger adults. Older adults tend to have prolonged illness duration with numerous chronic and complex medical conditions. Older patients experience greater functional and cognitive decline, and they also have increased caregiving needs for longer periods of time. Less is known about how older adults and their caregivers cope with the stressors of chronic illness compared to younger adults with sudden changes in health such as a new cancer diagnosis. Additionally, due to exclusion of older adults with multimorbidity from many research studies, there is a paucity of evidence to guide symptom management in this population. While older adults typically die of chronic, slowly progressive illnesses with multiple acute exacerbations, current Medicare coverage is targeted to the acute episodes of illness. This has resulted in palliative care that has been implemented in a reactionary fashion to these acute episodes. This approach insufficiently addresses patients' longer-term needs and creates missed opportunities for palliation along the disease trajectory (**Figure 70-2**).

In addition, frailty increases with age. While there are many different ways to define and capture frailty, they all are based on multidimensional vulnerabilities that cause susceptibility to stressors and reduced adaptive capacity and resilience to these acute events. Since increasing levels of frailty can present with gradual functional and cognitive decline, the trajectory might be less clear than single disease-based clinical evaluation. This also means that palliative care and supportive care might be needed for a longer period of time. Currently, there is no indicator of frailty in hospice-eligible criteria since the removal of

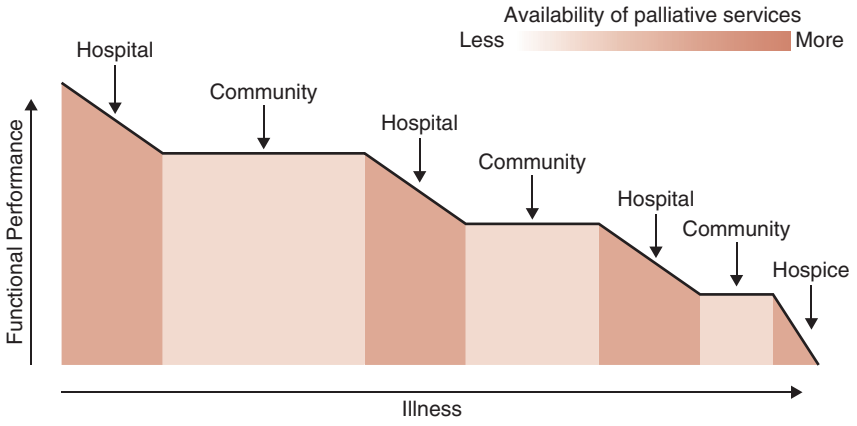

FIGURE 70-2. Gaps in palliative care services for older adults.

"adult failure to thrive" and "debility" as principle diagnoses for hospice eligibility in 2014 despite the Centers of Medicare and Medicaid Services 2009 report indicating a marked increase in the use of these diagnoses.

Better defining frailty and its expected trajectory, together with periodic measures and assessment of increasing health care needs, might assist in engaging palliative care approaches earlier and help in understanding poor outcomes from medical interventions. One area that can be addressed with increasing levels of frailty is polypharmacy and deprescribing, which has been shown to be beneficial in frail older adults, especially when the focus becomes more on quality of life and symptom control and avoiding side effects and medications that are unlikely to contribute any meaningful effect. Palliative care approaches are needed when patients and family are facing advancing frailty; however, due to the complexity of definition and wide range of presentation, these needs are often unmet. A true collaboration between geriatrics and palliative care experts can address this increasing gap.

In order to adequately address the needs of older adults, palliative care must be instituted in a timely manner, earlier in the course of a patient's illness. Upstream involvement of palliative care in the care of older adults, particularly while they are still residing in the community, allows for increased opportunities for shared decision making, constant evaluation of a patient's prognosis and symptom burden, and longitudinal care throughout different care settings. In addition to routine palliative care, clinicians have the opportunity to work with patients' primary care physicians, geriatricians, and specialists to emphasize function, financial planning, and maintenance of social roles within a community. This collaboration is even more integral when caring for older adults who often experience multiple concurrent progressive conditions that may challenge the formulation of prognostic estimates in terms of time and function. Effective palliative care for older adults will require moving away from a focus on single disease trajectory toward understanding and assessment of patient's reserves and vulnerabilities in addition to their values and treatment preferences to guide medical decision making. This foundation will aid in easing transitions to higher levels of care and help patients avoid inappropriate or burdensome treatments.

PALLIATIVE CARE IN THE ACUTE CARE SETTING

Of all settings, palliative care services in the acute inpatient setting have enjoyed the most successful implementation and advancement in the last 20 years. In the inpatient setting, palliative care teams are interprofessional, comprised of a mix of physicians, social workers, nurses, and chaplains. These teams offer expertise in symptom management, support to medical teams as well as patients and families, knowledge about community-based palliative care services and hospice organizations, and assistance in prognostication over a variety of illnesses.

Given that the majority of patients with advanced or serious illness will spend some time in the hospital throughout the course of their illness, and that up to 50% of adult deaths occur in the hospital setting, ensuring adequate palliative care services in the inpatient setting is of particular importance. There is increasing evidence that inpatient palliative care interventions improve the quality of clinical care, increase patient and family satisfaction, improve transitions of patients out of the hospital, and reduce health care costs. Patients who receive inpatient palliative care consultations incur less laboratory and radiology costs, are less likely to be admitted to the intensive care unit (ICU), and spend less days in the ICU once admitted compared to patients receiving usual care, particularly among patients who die during hospitalization.

Older adults receiving palliative care consultation differ in characteristics and interventions compared to younger patients. Referral for palliative care consultation

for older adults is more likely to be requested for disease processes other than cancer and for goals of care discussions rather than symptom management. Older adults tend to be referred to palliative care earlier in their hospital courses and have documented code statuses of "do-not-resuscitate" at time of consult. In the oldest subset of the population (80 and older), patients were more likely to be referred to palliative care for diagnosis of dementia, less likely to be included in discussions around goals of care due to lack of decisional capacity, and more likely to be discharged to nursing homes. In reviewing palliative care interventions offered by age, there are fewer interventions for pain or symptom management offered in older adults and an increased likelihood of limits placed on life-sustaining treatments after consultation. Given these differences, inpatient palliative care teams should be prepared to meet the unique palliative care needs of older adults.

Despite the advancements that have been made in inpatient palliative care, as well as the positive patient and family outcomes that have been observed, challenges remain in the delivery of palliative care in the inpatient setting. Inpatient palliative care teams continue to struggle with late requests for consultation. In addition, constant turnover of inpatient treatment teams, hesitation to move away from a curative focus, and varying levels of knowledge about and alignment with palliative care from referring clinicians pose ongoing barriers to optimal palliative care provision. Due to the acuity and complexity of medical issues present in the hospitalized patient population, it is often difficult for palliative care teams to address all of the needs of these patients during the acute hospitalization, particularly when caring for older adults with complex comorbidities and multifaceted care needs.

In order to address deficiencies that remain in the delivery of palliative care in the inpatient setting, there are opportunities to consider upstream involvement of palliative care as well as integration of palliative care services within existing models of care in the inpatient setting. Rates of presentation to the emergency department (ED) increase with age, and the percentages of patients requiring hospital admission are highest in older adults. As the population continues to age, there will be a rising number of seriously ill older adults who will present to the ED. These needs may be met by identification of ED palliative care "champions" such as nurse managers and care facilitators, routine use of screening tools to evaluate the need for palliative care assessment in this population, development of care pathways as a product of collaboration between ED and palliative care teams, availability of palliative care consultative services to the ED, and familiarity with community palliative care and hospice resources among ED teams.

An organized approach to the identification, assessment, and management of older adults' complex medical, social, psychological, and functional concerns and vulnerabilities is key. For patients admitted to inpatient hospital floors, utilization of such a systematic approach by an interprofessional team will ensure the needs of older adults are met. As previously discussed, older adults receiving palliative care consultation are more likely to have nononcologic diagnoses, which are associated with less predictable hospital courses and illness trajectories. As the hospital course unfolds, communication interventions should target the status of the patient's overall condition, estimated prognosis in terms of functional recovery and life expectancy, and available treatment options in the context of associated risks and likelihood of benefit. Clinicians with an understanding of basic palliative care assessment may be successful in appropriately identifying palliative care needs on day of admission, engaging palliative care teams upstream in the hospital stay of seriously ill older adults, monitoring progress on clinical improvement and achievement of patient's goals, and frequently reevaluating to identify new palliative care needs in the older adult population as they arise throughout the hospitalization (**Figure 70-3A** and 3**B**).

Emerging opportunities for early introduction of palliative care models include programs such as geriatric comanagement with orthopedic surgery and general surgery, with the geriatric-surgery verification program from the American College of Surgeons being adopted by an increasing numbers of medical centers, and with emerging geriatric-ED initiatives. Such programs advocate for early collaboration with geriatric medicine clinicians, who perform an early frailty, cognitive and functional assessment, and assist the team in tailoring treatment plans to specific characteristics and needs of the older patient. Within these models, comanagement with palliative care teams may be considered for patients in instances such as facilitation of complex communication and medical decision making, reevaluation of goals of care based on changes in the patient's clinical trajectory, or easing end of life transitions.

The ICU is another location in the inpatient setting with the most opportunities for successful delivery of palliative care to older adults. Older adults admitted to the ICU have high mortality rates during the course of their hospitalization, and older ICU survivors have elevated palliative care needs for ongoing physical and/or psychological distress. They also remain at increased risk for rehospitalization and 6-month mortality. Studies have demonstrated that patients, families, and clinicians may possess unrealistic expectations about prognosis and effectiveness of ICU interventions. Critically ill patients and their caregivers often seek more honest and open discussions regarding prognosis and appropriate therapies based on patients' values and goals. ICU clinicians should all receive training in core skills of primary palliative care, including serious-illness communication skills and symptom management. Palliative care "bundles" have been suggested to improve communication and promote the comfort of critically ill patients (IPAL-ICU). Key quality measures in these bundles include early identification and documentation of surrogate decision makers and advance directives, regular

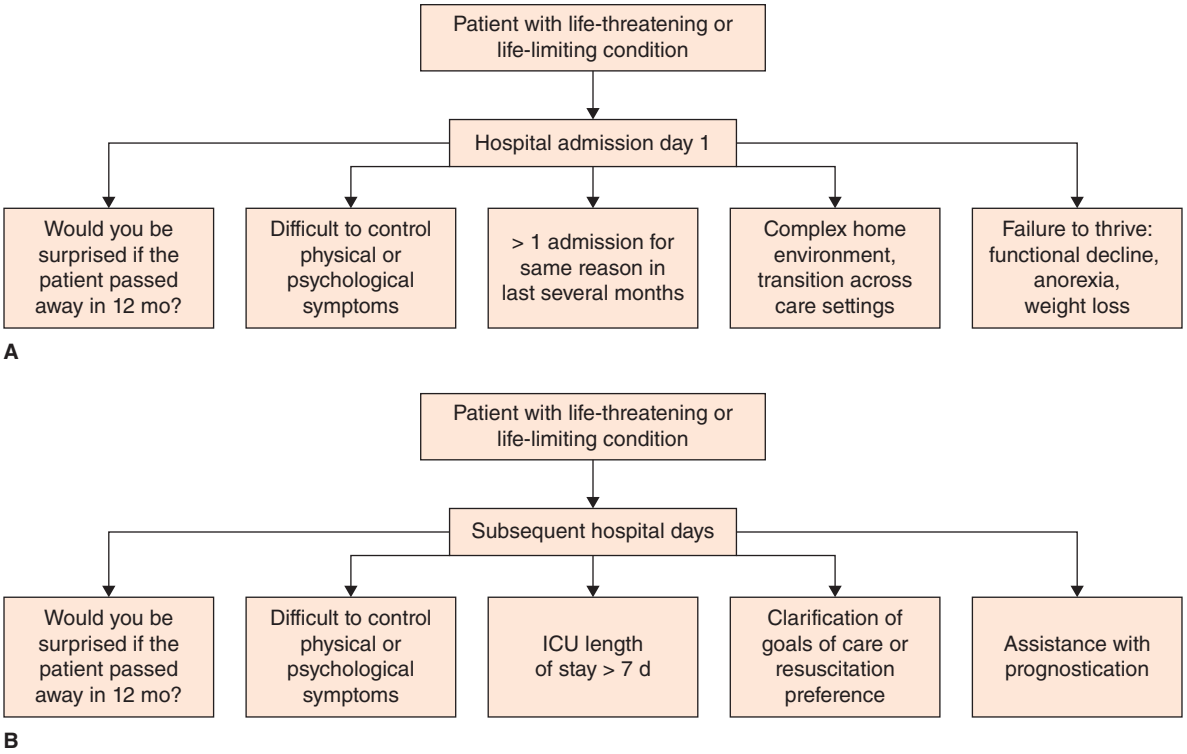

FIGURE 70-3 **A.** Inpatient criteria: guide for primary teams to prompt palliative care consultation on day of admission. **B.** Inpatient criteria: guide for primary teams to prompt palliative care consultation on subsequent days of hospital admission.

symptom assessment, involvement of social work and spiritual care, and regular family meetings to update the patient and family on the status of the patient's illness. By emphasizing compassionate communication, symptom management, and shared decision making, palliative care can be successfully integrated for critically ill patients at all stages of illness.

Across the inpatient setting, collaboration between palliative care teams and patients' longitudinal outpatient clinicians, in particular geriatricians and/or primary care physicians, care managers or care transition specialists can bridge gaps in knowledge of patients' values and treatment preferences, communication, and appropriate identification of community resources to best meet the patients' needs for their current health state. Additionally, novel pathways to meet the unique hospital-based considerations of older adults offer further opportunity for palliative care integration and comanagement, particularly for patients experiencing high symptom burden or for those patients and families benefiting from ongoing goal clarification and assistance in complex medical decision making.

PALLIATIVE CARE IN POST-ACUTE CARE SETTINGS

Many older adults with complex care needs are living and now also dying in the community. Older individuals of advancing age often live alone, particularly women, with estimates of over 50% for women ages 85 and older in 2018. Approximately 21% of these community-dwelling older adults, totaling approximately 7.5 million individuals, are partly or completely homebound, and tend to be older, female, ethnic and cultural minorities, and have lower income. Despite a high burden of chronic conditions, especially dementia, and reports of lower health status, these older adults struggle or are not able to access the ambulatory care needed for their complex medical, psychosocial, and neurologic needs, and have high rates of hospitalization, leading to increased fragmentation of care. Despite the cycle of recurrent fluctuations in health and transitions in care, older adults' family (if present) and physicians often have limited understanding of their goals of care and care needs, prognosis, and have rarely engaged in discussions regarding treatment and care preferences. This is particularly problematic as family caregivers still provide the majority of direct care for US older adults with complex health conditions, and much of this care occurs in the community as opposed to nursing facilities. Additionally, growing evidence suggests that older adults living at home with dementia are more likely to have bothersome pain, dyspnea, and mood symptoms compared with people living in nursing or other residential facilities, with similar patterns seen among other vulnerable older adults in the community. Community-based palliative care services are currently available through both home-based and institutional-based models (**Figure 70-4**).

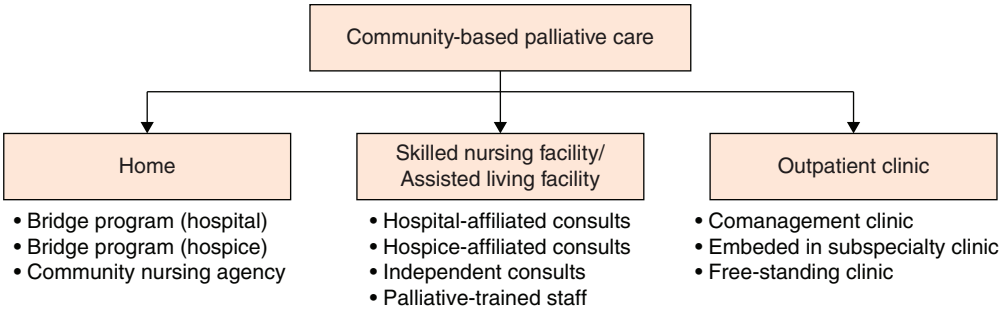

FIGURE 70-4. Palliative care delivery models in the community.

PALLIATIVE CARE IN THE CLINIC

For older adults whose functional status remains preserved or who have sufficient assistance to access care in the ambulatory setting, palliative care clinics facilitate the development of an early relationship between clinicians and patients, improve communication between primary care clinicians and other subspecialty clinicians, and increase satisfaction with care from the perspective of patients and their families. These clinics also serve as an appropriate setting for posthospital discharge follow-up for patients seen by inpatient palliative care teams.

There are a variety of existing models of palliative care clinics in our current health care system (Improving Outpatient Palliative Care [IPAL]). Some clinics are affiliated with particular specialties by which they generate all of their referrals. Other clinics are stand-alone clinics that function independently or share space and staffing with a primary care clinic or specialty clinic in a partially or completely embedded framework. Additionally, clinics vary in the focus of their care, ranging from providing task-specific to symptom-specific palliative care. Palliative care clinicians will often choose between providing care in a consultative or comanagement fashion for the majority of patients, while some patients will be taken care of entirely by the palliative care specialist. Finally, some clinics have the capability to extend their services beyond the clinic and into the community, providing visits to clinic patients who become homebound or transfer to a facility.

Outpatient palliative care services have been associated with improved patient outcomes, such as significant symptom improvement and high satisfaction with services. Care provided in palliative care clinics has also been associated with better maintenance of performance status, increased discussions about advance care planning and end-of-life care, and better long-term outcomes for caregivers. Patients receiving care in palliative care clinics in addition to usual care experience fewer hospital days and skilled nursing facility days, as well as increased hospice use. Clinicians also benefit, with increased rates of satisfaction observed in referring clinicians as well as palliative care clinicians.

Presently, the majority of outpatient palliative care clinics focus on care for patients with cancer diagnoses. As outpatient clinic services expand, particularly to nononcology populations, clinicians may face challenges, including the lack of standardization of the development process, need for adequate marketing, and insufficient staffing to accommodate anticipated growth of the clinics. Promising examples of outpatient palliative care for individuals with serious, life-limiting non–cancer-related diagnoses are emerging. A 2020 pragmatic trial of integrated palliative care at three academic tertiary centers for individuals with Parkinson disease and related disorders demonstrated better quality of life, improved symptom burden and motor symptoms, as well as higher rates and quality of advance care directives in the intervention group compared to those receiving standard of care. Additionally, the study was suggestive of a benefit on caregiver strain by 1 year. Although care at each site was provided in an ambulatory setting and by an interprofessional team using standardized checklists, the specific model of care varied by institution. Encouragingly, the impact on outcomes was equally effective across sites and suggests flexibility may exist in the design of integrated palliative care in the outpatient clinical setting if fidelity to key palliative care quality domains is maintained.

There are several limitations that impede the delivery of ideal palliative care in the clinic setting. Patients with severe and advanced illness, particularly those with chronic illnesses, are always at risk of declining to the point that they are unable to make their outpatient clinic appointments. Unless clinics are associated with bridge programs designed to extend palliative care services to homes and facilities until patients qualify for hospice, provided the latter is an acceptable option to them, chronically ill patients with progressive functional and cognitive decline often experience a gap in the continuation of their palliative care services. In addition, clinics are often limited by their hours of operation and staffing, particularly lacking coverage by a health care provider 24 hours a day to manage any acute crises that may arise. Finally, given the heavy reliance on referrals from the inpatient setting, primary care clinics, and subspecialty clinics, palliative care clinics require a

certain amount of "buy-in" from other disciplines in order to continue to operate. As the population with chronic conditions continues to grow and age, increased attention to availability of palliative care clinics, reimbursement for palliative care services, and relationship building between hospitals, long-term care institutions, and community-based home care organizations is needed to ensure that this population of patients will be able to receive palliative care outside of the hospital setting.

PALLIATIVE CARE IN THE HOME

There are an increasing proportion of older adults living at home with chronic illness with intensive symptom management and advance care planning needs who do not qualify for hospice. As older adults become increasingly homebound, they are at increased risk for symptom and illness burden. Completely homebound patients have a 2-year mortality of approximately 40%, demonstrating a natural intersection of palliative care provision within home-based care. The ideal home-based palliative care model is comprised of an interdisciplinary team of physicians, nurses, social workers, chaplains, home health aides, and pharmacists (similar to the home hospice model) who are able to provide services in coordination with the primary care clinician and specialists. Services would include pain and nonpain management, medication management, discussions about health care decision making and goals of care, caregiver support, attention to spiritual concerns and quality of life, arrangement of respite care, and assistance with the transition to hospice if needed. Specifically tailored to the chronically ill, frail older adult, home-based palliative care would ideally integrate with home-based primary care or other community geriatric care models with an expanded focus that includes education and support for caregivers, timely access to medications, optimal use of applicable community resources, home safety and comfort, and 24/7 access to a health care clinician in order to ensure continuity of care and manage acute crises. A closely integrated model of home-based palliative care with home-based primary/geriatric care is especially beneficial for patients and families who are facing the strain of longer, often unpredictable, but ultimately terminal illness trajectories with challenging, day-to-day prognostication and frequent health exacerbations due to even minor physiologic stressors. This situation is commonly faced when caregivers and clinicians provide longitudinal care for patients with frailty, dementia, and complex multimorbidity. While highly integrated palliative and primary care models existed in limited, most often capitated, health care settings, the most notable and long-standing being the VA Home-Based Primary Care program, new partnerships are now arising between other home-based primary/geriatric and palliative care programs in the context of the United States' evolving alternative payment models and the growth of for-profit practices in home-based medical care. Home-based palliative care organizations would benefit from relationships with home-based primary care as well as institutions such as hospitals, primary care clinics, specialty clinics, nursing homes, and hospice agencies in order to increase the referral base and promote longitudinal care across the care continuum.

Studies on existing models of home-based palliative care have demonstrated beneficial outcomes. The patient populations receiving home-based palliative care include patients with both cancer and noncancer diagnoses (eg, dementia, cardiac disease, pulmonary disease, etc). Benefits noted include improved symptom scores in areas such as depression, dyspnea, and anxiety, decreased overall cost of care, increased completion of advance directives, and increased rates of dying at home. Fewer visits to the ED and lower rates of hospitalization have been observed, without a difference in overall survival. Rates of referral to hospice are higher than among control patients without home-based palliative care. Of the patients who do enroll in hospice, those who have received home-based palliative care prior to this transition are noted to have longer days of enrollment in hospice as compared to those with usual care.

Despite the positive impact home-based palliative care programs have on community dwelling patients with serious and advanced illnesses, the advancement of palliative care in the arena of home-based care has limitations. Home-based palliative care delivery is highly variable and delivered by diverse sources including hospitals and large health systems, community hospice and home health agencies, and individual practices or organizations. Programs face economic barriers such as insufficient reimbursement for services in proportion to time spent, particularly for professions other than physicians and nurses, and a lack of incentive to reduce costs. Organizational barriers exist as well, primarily driven by the lack of adequate staffing to meet the needs of homebound older adults, as well as the lack of recognition and training around quality standards and measurement. At present, there is no standardization of home-based palliative care practices or institutional support, leading to difficulty in translating the beneficial outcomes that have been noted in existing home-based palliative care models to the broader population. While progress has been made in developing a quality framework that adequately represents the needs of the homebound population, there are ongoing efforts to address implementation, measurement, and reimbursement challenges. These standardized quality metrics have the potential to ensure that frail, cognitively impaired and functionally limited people experience improved quality of life while potentially reducing burdensome hospital visits and health care costs.

PALLIATIVE CARE IN LONG-TERM CARE FACILITIES

In adults older than the age of 65, approximately 25% of all deaths will occur in a nursing home, with projections of an increase to up to 40% of deaths of older Americans

occurring in nursing homes by 2030. The majority of residents in long-term care facilities are older adults, experience progressive frailty and functional impairment, and require increased assistance in activities of daily living. For older adults in long-term care facilities, life expectancies are typically shorter in the order of years. With projections suggesting that the number of frail older adults may triple or quadruple in the next 30 years, closer examination is needed on the state of palliative care services in long-term care facilities. For many long-term care residents, hospice is the only form of specialty palliative care available.

What is known about palliative care services in long-term care facilities at present is limited. There are several studies that demonstrate long-term care residents suffer from uncontrolled symptoms, depression, existential suffering, and difficulty adjusting to the long-term care facility as their permanent place of residence. Older adults who do receive palliative care consultation have lower rates of end-of-life hospitalizations, with those who receive consultation further upstream in their illness experiencing lower overall rates of hospitalization and an almost 50% reduction in potentially burdensome transitions such as an emergency room visit within 30 days of death or admission to hospice within 3 days of death.

A number of significant barriers to providing adequate palliative care in long-term care facilities exist. Demographics of patients in long-term care facilities, particularly nursing homes, are changing, with increased numbers of patients with chronic illness and multiple medical problems. A large percentage of these patients have noncancer diagnoses, leading to increased difficulty with prognostication and late referrals to hospice. Nursing home residents are often high utilizers of acute care settings, increasing the risk that patient's values and treatment preferences are unknown or have not sufficiently been elucidated. These issues are further complicated by the fact that there is often high staff turnover at long-term care facilities, making it difficult to sustain palliative care educational efforts and ensure palliative care competency among staff.

With current reimbursement policies focusing on rehabilitation needs and sustenance, a seemingly high proportion of patients undergo aggressive restorative treatment such as intravenous hydration or insertion of feeding tube placement instead of shifting to aggressive palliative care or hospice care. In addition, short-stay residents in nursing homes who are paid for under the Medicare skilled nursing facility benefit following a hospitalization are only able to access the Medicare Hospice benefit if they revoke the skilled nursing facility benefit and pay for room and board out of pocket. These out-of-pocket costs are a major financial disincentive to short-stay residents to exercise the Medicare hospice benefit.

Guidelines for quality palliative care in nursing homes emphasize that palliative care does not require patients to forgo curative treatments nor relinquish subacute rehabilitation or hospitalization (National Consensus Project Guidelines for Quality Palliative Care). Three different models of delivering palliative care services in nursing homes have been proposed: (1) hospice partnerships, (2) external palliative care consultation services, and (3) facility-based palliative care teams. Regardless of the approach taken, sustainability is key. Measures that are critical to sustainability of palliative care interventions in the long-term care facility setting include ongoing education efforts, "buy-in" of palliative care by nursing home staff and clinicians, increased availability of nurses and patient aides, and working relationships with hospitals and community organizations including hospice to ensure continuity of palliative care across all settings.

HOSPICE

Hospice care is the most well-known provider of palliative care services in the community. Hospice care is a philosophy of care in which patients facing a life-limiting illness are approached in a holistic manner, with a focus on medical care and symptom management, as well as an interest in the emotional, spiritual, and future planning needs of the patient and family. There are now more than 4600 hospice organizations nationally, and these organizations serve over 1.5 million people. Of Medicare beneficiaries receiving hospice care in 2018, greater than 60% were older adults aged 65 and older. The greatest increase in hospice utilization was observed in patients aged 85 and older. Research on hospice care has demonstrated increased survival compared to usual care, increased patient and family satisfaction with end-of-life care, increased rates of patients dying in their location of choice, and decreased inappropriate health care resource utilization. Despite tremendous growth in the use of hospice services over the last 30 years, hospice care is still underutilized. A substantial number of patients who would prefer to die at home still die in the hospital. At present, the majority of patients receive hospice care at home. Nationally, hospice is not available in all long-term care facilities, and only 6% of residents in nursing homes elect the hospice benefit annually, far lower than one would expect given the short lengths of stay prior to death for many long-term care facility residents.

The Medicare hospice benefit requires that patients have a life expectancy of 6 months or less. An unfortunate but common misconception is that palliative care and hospice care are equivalent. This leads to missed opportunities for palliative care involvement in patients independent of prognosis, as well as late referrals to hospice care until clinicians are confident the patient is dying. This has translated to a median length of service in a hospice organization of 18 days. Over half of patients are enrolled in hospice for 30 days or less, of which 28% of patients die in the first 7 days. The extent of comprehensive palliative care delivered by hospice organizations also varies, with larger hospice organizations having the ability to provide more intensive services such as 24-hour physician staffing, ethics

committees, and regularly using standardized assessment tools for pain and symptom management.

Cancer remains the most frequent principal hospice diagnosis of beneficiaries of hospice care. As discussed previously, the diseases affecting older adults are largely chronic in nature, with fluctuating trajectories marked by acute exacerbations and difficulties in prognostication, which in turn may reflect the timing with which clinicians refer older adults for hospice. To avoid the continued pattern of late referrals of older adults to hospice until they are in an advanced terminal state, efforts will need to be shifted upstream, with increased emphasis on improved palliative care delivery across care settings, and increased communication and collaboration between primary care and/or specialty clinicians caring for older adults with specialists in palliative care and hospice. Access and utilization of hospice care by older adults will improve by expanding coverage of hospice services to patients with longer life expectancies and eliminating the need for patients to choose between hospice services and other home-care or institutional-level services. Newer programs such as the Medicare Care Choices Model are exploring opportunities to provide eligible beneficiaries access to hospice services while they continue to receive disease-modifying treatment. Thus far, participants in this program have demonstrated increased likelihood of hospice enrollment, decreased reliance on inpatient care, and positive caregiver experiences. By reconceptualizing the population for whom hospice care is offered away from eligibility based on prognosis, and instead, on patients and their caregivers with increasing vulnerabilities and need for care and in-home supports, older adults may be able to benefit from the supportive and holistic care provided by hospice agencies for longer periods of time.

CONCLUSION

The field of palliative care has enjoyed significant advancements over the years. Its rapid growth is evidenced by an increased number of clinicians choosing to train in the field and the increased availability of palliative care teams in a variety of care settings. The more widespread implementation of palliative care has led to improved quality of life and symptom management for patients; improved satisfaction among patients, families, and health care providers; and, decreased health care costs. However, the presence of adequate palliative care for older adults has lagged behind. Older adults have unique palliative care needs that are best met by providing longitudinal palliative care across the care continuum. Implementation of palliative care for older adults at the time of diagnosis of advanced or serious illness or at a time where they are noted to develop progressive frailty or cumulative burden from multiple comorbidities would be ideal. Increased involvement of palliative care with a focus on collaboration and comanagement with geriatricians and other longitudinal clinicians will help older adults reside longer in their communities, maintain their level of function, cope with care transitions, and participate in shared decision making regarding their treatment preferences. The health care system will also need to change in order to train clinicians in primary palliative care skills, expand access to specialty palliative care particularly in the community, and increase coordination of care among institutions and community organizations. Addressing these gaps in the current palliative care models will improve the quality of palliative care delivered to older adults across the care continuum.

FURTHER READING

Bandeen-Roche K, Seplaki CL, Huang J, et al. Frailty in older adults: a nationally representative profile in the United States. *J Gerontol A Biol Sci Med Sci*. 2015;70(11):1427–1434.

Bowman BA, Twohig JS, Meier DE. Overcoming barriers to growth in home-based palliative. *Care. J Palliat Med*. 2019;22(4):408–412.

Bryant EA, Tulebaev S, Castillo-Angeles M, et al. Frailty identification and care pathway: an interdisciplinary approach to care for older trauma patients. *J Am Coll Surg*. 2019;228(6):852–859.e1.

Chen CY, Thorsteinsdottir B, Cha SS, et al. Health care outcomes and advance care planning in older adults who receive home-based palliative care: a pilot cohort study. *J Palliat Med*. 2015;18(1):38–44.

Cooper L, Abbett SK, Feng A, et al. Launching a Geriatric Surgery Center: recommendations from the Society for Perioperative Assessment and Quality Improvement. *J Am Geriatr Soc*. 2020;68(9):1941–1946.

Hamaker ME, van den Bos F, Rostoft S. Frailty and palliative care. *BMJ Support Palliat Care*. 2020;10(3):262–264.

Harrison KL, Ritchie CS, Patel K, et al. Care settings and clinical characteristics of older adults with moderately severe dementia. *J Am Geriatr Soc*. 2019;67:1907–1912.

Kamal AH, Currow DC, Ritchie CS, Bull J, Abernethy AP. Community-based palliative care: the natural evolution for palliative care delivery in the U.S. *J Pain Symptom Manage*. 2013;46(2):254–264.

Kluger BM, Miyaski J, Katz M, et al. Comparison of integrated outpatient palliative care with standard care in patients with Parkinson disease and related disorders: a randomized clinical trial. *JAMA Neurol*. 2020;77(5):551–560.

Leff B, Carlson CM, Sailiba D, Ritchie C. The invisible homebound: setting quality-of-care standards for home-based primary and palliative care. *Health Affairs*. 2015;34(1):21–29.

Miller SC, Lima JC, Intrator O, Martin E, Bull J, Hanson LC. Palliative care consultations in nursing homes and reductions in acute care use and potentially burdensome end-of-life transitions. *J Am Geriatr Soc*. 2016;64(11):2280–2287.

Nelson JE, Curtis JR, Mulkerin C, et al. Choosing and using screening criteria for palliative care consultation in the ICU: a report from the Improving Palliative Care in the ICU (IPAL-ICU) Advisory Board. *Crit Care Med*. 2013;41(10):2318–2327.

Olden AM, Holloway R, Ladwig S, Quill TE, Van Wijngaarden E. Palliative care needs and symptom patterns of hospitalized elders referred for consultation. *J Pain Symptom Manage*. 2011;42(3):410–418.

Ornstein KA, Leff B, Covinsky KE, et al. Epidemiology of the homebound population in the United States. *JAMA Intern Med*. 2015;175(7):1180–1186.

Pacala JT. Is palliative care the "new" geriatrics? Wrong question—we're better together. *J Am Geriatr Soc*. 2014;62(10):1968–1970.

Ritchie CS, Leff B. Population health and tailored medical care in the home: the roles of home-based primary care and home-based palliative care. *J Pain Symptom Manage*. 2018;55(3):1041–1046.

Smith AK, Thai JN, Bakitas MA, et al. The diverse landscape of palliative care clinics. *J Palliat Med*. 2013;16(6):661–668.

Visser R, Borgstrom E, Holti R. The overlap between geriatric medicine and palliative care: a scoping literature review. *J Appl Gerontol*. 2021;40(4):355–364.

Voumard R, Rubli Truchard E, Benaroyo L, Borasio GD, Büla C, Jox RJ. Geriatric palliative care: a view of its concept, challenges and strategies. *BMC Geriatr*. 2018;18(1):220.

Zimbroff RM, Ritchie CS, Leff B, Sheehan OC. Home-based primary and palliative care in the Medicaid program: systematic review of the literature. *J Am Geriatr Soc*. 2021;69(1):245–254.

Chapter 71

Effective Communication Strategies For Patients with Serious Illness

Brook Calton, Matthew L. Russell

INTRODUCTION

Older adults are living longer, and with more chronic diseases, than ever before. The number of Americans ages 65 and older is projected to double from 52 million in 2018 to 95 million by 2060. The average US life expectancy increased from 68 years in 1950 to 79 years in 2017. Approximately 80% of older adults have at least one chronic medical condition; with 77% having at least two. In fact, the majority of hospice diagnoses are now noncancer related, such as congestive heart failure, chronic obstructive pulmonary failure, or dementia. Taken together, these data suggest effective communication around advanced illness is of ever-increasing importance for our older adults facing longer life trajectories, complex chronic illness, and debility and decline. Having achieved longevity, many older adults living with advanced illness prioritize goals such as function, comfort, and family support. Skillful communication is needed to ensure the treatments we offer align with these goals.

Effective communication with patients and families can have a number of benefits. It has been shown to improve diagnostic accuracy, health outcomes, treatment adherence, and patient satisfaction. In addition, effective end-of-life communication has been associated with decreased intensity of care, increased quality of life, and improved quality of dying for patients.

Studies consistently confirm the majority of patients facing advanced illness desire to have honest conversations about their goals, values, and end-of-life care with their providers. A recent systematic review of advance care planning in older adults found between 61% and 91% of older individuals wanted to discuss their end-of-life care with their providers. Benefits cited by patients included assurance their wishes would be respected, an opportunity to address important medical care and treatment issues before cognitive impairment or being physically unwell occurred, and to assist loved ones with decision making. They felt this responsibility to begin the discussion lies with physicians, they wanted to have the conversation in an open and honest manner, and preferred these conversations began early in their illness course.

However, despite the fact that communication around advanced illness is not only beneficial but also desired by most patients, this communication often remains inadequate. In studies, only an estimated 2% to 29% of frail older adults have discussed some form of end-of-life plans with a health care professional. In general, these discussions are often dominated by the clinician—in one study of 60 recorded patient-internist advance directive discussions, physicians spent 70% of the time talking and initiated more detailed discussions about patients values and goals less than 33% of the time. There are a number of challenges to effective communication that may contribute to the gaps described above. Potential barriers and suggested solutions are presented in **Table 71-1**.

In this chapter, we will provide an approach as well as key skills to prepare providers for having important

Learning Objectives

- Describe a suggested six-step process for addressing goals of care and treatment preferences with patients who are seriously ill.
- Identify unique challenges to communicating with seriously ill older adults and strategies to address them.
- Summarize key communication techniques including ASK-TELL-ASK and SPIKES.

Key Clinical Points

1. **Data suggest patients and families generally desire to have open conversations with their medical providers around their goals of care, treatment preferences, and end-of-life wishes.**
2. **Conversations around goals, values, and treatment preferences should be held over time, as a patient's health status and/or perspectives evolve.**
3. **Clarifying patients, and families, understanding as well as effectively addressing emotions around serious illness can help make these conversations easier.**

TABLE 71-1 ■ PROVIDER BARRIERS AND SUGGESTED SOLUTIONS TO ENHANCE COMMUNICATION AROUND ADVANCED ILLNESS

BARRIER	SUGGESTED SOLUTIONS
Provider time	• Reserve specific (and if possible, longer) visits for goals of care, advance care planning conversations • Have conversations over time, in small chunks, rather than in crisis only • Utilize resources (eg, prepareforyourcare.org) that help patients and family to reflect on their wishes before and after visits
Provider training	• Engage in standardized communication training during medical school and residency • Participate in continuing medical education (potential resources include Vital Talk and the American Academy of HealthCare Communication)
Prognostic uncertainty	• Acknowledge uncertainty upfront with patients • Frame discussions within best, worst, and most likely scenarios • Use prognostic calculators (eg, eprognosis.org)
Fear of offending patients	• Routinize goals of care and advance care planning conversations with all patients • Ask patient for permission to have conversation and how much information they want

conversations about goals of care and advance care planning with patients and families. We will also discuss unique challenges and situations specific to communication with older adults.

SUGGESTED APPROACH TO COMMUNICATING WITH PATIENTS WITH ADVANCED ILLNESS

Communication around advanced illness can take many forms. Potential conversations may include, but are not limited to, providing prognostic information, discussing treatment preferences, engaging in advance care planning, and transitioning to comfort care and/or hospice. It is important to remember, particularly in the outpatient setting, that any of these topics should be thought of as a process that occurs over time using open and honest communication that accounts for the patient's personal desire for information.

It is likely that patients' goals and preferences will shift over time as their illness progresses and/or what is important to them changes. Below we provide a suggested six-step approach that can be adapted to any of the conversations described above.

Step 1: Prepare

In the outpatient setting, if possible, it is optimal to ensure an adequate amount of protected clinic time to have the conversation. Encourage the patient to bring family or caregivers with them that they would like to hear and/or participate in the conversation. Clarify medical treatment options and prognostic information with other providers engaged in the patient's care prior to the visit. In the inpatient setting, it is important to arrange for a private place to have the conversation and again, invite key family members, caregivers, and health care providers (eg, consulting MDs, bedside nurses, chaplains, etc) based on the patient's preferences. Prior to the conversation, meet briefly with the participating health care providers (premeeting) to agree upon an agenda, prognosis, suggested treatment course, etc.

Step 2: Clarify Patient's Understanding of Illness and Preferences for Information/Decision Making

Any conversation around advanced illness requires that the provider first assess the patient's understanding of their illness. If the patient is unable to participate in the conversation, the same process should be undertaken with the patient's family and/or designated medical decision maker, depending on the circumstances. This step, using open-ended questions, facilitates the provider's ability to successfully provide education around the patient's illness, discuss treatment options based on a shared understanding of the patient's condition, and avoid potential misunderstandings (**Table 71-2**).

Next, the patient's preferences for information and medical decision making should be clarified. This is critical because while most patients desire information about their illness and want to be involved in decision making, a significant minority do not. Asking these questions (and then proceeding accordingly) will build the patient's trust and empower them in these conversations (**Table 71-3**).

Step 3: Education and Exploration

Step 3 involves patient education and exploration of patients' hopes and values to facilitate medical decision making.

Once you have a sense of the amount of information the patient desires, the foundational communication skill ASK-TELL-ASK can be useful for providing patient education. You've already done the first "ASK" in step 2, by eliciting the patient's understanding of their illness. The provider should next "TELL" the patient in straightforward language the information they are hoping to

TABLE 71-2 ■ SUGGESTED LANGUAGE FOR ELICITING A PATIENT'S UNDERSTANDING OF THEIR ILLNESS

"Tell me how things are going for you."

"Just so we're on the same page, could you tell me what you have heard about your medical condition?"

"What do you understand about your current health situation?"

"What do you think is causing your illness?"

TABLE 71-3 ■ SUGGESTED LANGUAGE FOR ELICITING A PATIENT'S PREFERENCES FOR INFORMATION AND MEDICAL DECISION MAKING

Preferences for Information
"How much do you want to know about your illness?"
"I have information about your condition. Some patients want to know the details, other patients want me to talk to someone else. How do you feel?"
"Are you the type of person who wants detailed information about their health?"
Preferences for Medical Decision Making
"How involved would you like to be in medical decisions regarding your illness?"
"How do you like to make medical decisions with your doctors?"
"Some patients prefer to make medical decisions entirely on their own, others prefer to make the decision jointly with their doctor, and others prefer the doctor to decide for them. Where do you fall?"

convey (eg, help them better understand their illness, treatment preferences, difficult news, etc) in short, digestible chunks. The second "TELL" is an additional check for understanding or perspective. An easy way to ask is, *"To make sure I was explaining things clearly, can you summarize what I just said?"* This ASK-TELL-ASK cycle is often repeated several times during the same conversation, to ensure information is provided in a sensitive manner in digestible pieces.

Understanding a patient's hopes and values can help the provider provide sound medical advice in line with what is most important to the patient. A list of suggested questions is given in **Table 71-4**.

Step 4: Respond to Emotions

Providers often focus on the details of medical care and miss emotional cues. Responding to emotional cues is a key aspect of successful communication, building trust, and supporting patients. Clinicians can express empathy nonverbally and verbally. Eye contact, open body position, leaning in toward the patient, and when appropriate, light touch can be useful tools for showing nonverbal empathy. An invaluable tool for expressing verbal empathy employs the mnemonic "NURSE." Details of the mnemonic and sample language are provided in **Table 71-5**.

TABLE 71-4 ■ SUGGESTED LANGUAGE FOR EXPLORING HOPES AND VALUES

Psychosocial Aspects of Illness
"What is the most difficult part of this illness for you and your family?"
What Gives Life Meaning
"As you think about what lies ahead, what is most important to you?"
"What do you enjoy?"
"How do you weigh quantity vs quality of life? What does quality of life mean to you?"
Hopes and Concerns
"When you think about the future, what do you hope for?"
"What do you fear about your illness?"
"When you think about getting very sick, what worries you the most?"

Step 5: Guide Decision Making

Every conversation is different and in many cases, exploring hopes, values, and emotions can be extremely beneficial, even without a specific medical decision to make. If there are decisions to make, the next step involves providing options to the patient that are consistent with what has been found out is important to them. Providers do not need to identify every possible option, particularly those that are inconsistent with the patient's expressed goals. Information should be given in small chunks and jargon should be avoided. Again, using Ask-Tell-Ask can allow for frequent stops to check for comprehension and provide clarifications or corrections if necessary. In many situations, based on what is understood about the patient's decision-making preferences from step 2, providers can ask for permission to provide a recommendation based on the patient's values and what is known about the medical circumstances. Many patients and families may need time and multiple discussions before they can make big changes in care or decisions, for example, to enroll in hospice or decide whether to go into a nursing home.

TABLE 71-5 ■ NURSE STATEMENTS AND SUGGESTED LANGUAGE

SKILL	TASK	SUGGESTED LANGUAGE
N: NAME	State the patient's emotion	*"It sounds like this experience has been frustrating."*
U: UNDERSTAND	Empathize with and legitimize the emotion	*"I can understand why you're nervous."*
R: RESPECT	Praise the patient for things they are proud of/have done well	*"You've taken incredible care of your mother."* *"I'm impressed by how thoughtfully you've approached this decision."*
S: SUPPORT	Show support	*"No matter what decision we make, I'll be here for you."*
E: EXPLORE	Ask the patient to elaborate on the emotion	*"Tell me more about how you're feeling."*

TABLE 71-6 ■ PROVIDING A RECOMMENDATION, SUGGESTED LANGUAGE

TASK	SUGGESTED LANGUAGE
Restate primary goal(s) and confirm understanding	*"Given what you have told me, it sounds like you want to continue treatments that may extend your life if your quality of life is similar to what it is now. Even if that means coming back to the hospital."* *"What I understand as most important to you is that you remain comfortable and we get you back home. Is that right?"*
Suggest possible plan(s) of care to meet goals	*"Have you heard of hospice? In my experience, it can be one of the best programs for achieving the goals you mentioned for symptom management and care in the home."* *"I'm hearing that you don't want a lot of medical interventions. Given that, I think we should consider focusing on symptom control as opposed to putting you on life support or doing CPR if your breathing gets worse. What do you think?"*

CPR, cardiopulmonary resuscitation.

In practice, this often means not expecting a decision to be made at the end of one discussion. Suggested language is in **Table 71-6.**

Step 6: Summarize and Plan

To close the conversation, the provider should summarize any decisions that were made during the discussion, offer to answer questions, and arrange a plan for follow-up. This is also an opportune time to reaffirm nonabandonment regardless of the outcome of the conversation and the goals of care. In both the outpatient and inpatient setting, this is also an opportune time to thank the patient for having this difficult conversation and remind them you are hoping this process of exploration and decision making will be an ongoing one, as time goes on and their health evolves.

STRATEGIES TO ADDRESS COMMUNICATION CHALLENGES WITH OLDER ADULTS

Communicating with older adults can come with a unique set of challenges. Specific age-related issues (eg, hearing and/or vision loss, decline in memory, slower processing of information) alongside psychosocial factors related to aging (eg, loss of identity, lessening of power/influence of one's life, and separation from family and friends) can impact communication. Studies have found in general, providers spend less time and take more of a paternal approach with older patients. Older patients may withhold information about symptoms or conditions the patient considers "normal for their age," such as pain—potentially hindering successful management of these important issues.

Hearing

Hearing loss is the third most common chronic condition reported by older adults.

Although older adults can compensate to a certain extent by devoting more cognitive effort to comprehension, this practice may limit the patient's ability to encode information into long-term memory, draw interferences, and follow complex conversations. Tips for communicating with patients with hearing loss include the following:

- Increase volume of speech slightly, speak a bit slower, and present information as clearly as possible.
- Avoid shouting as speaking loudly raises the pitch of the voice, making it difficult for patients with hearing impairment to understand.
- Keep good eye contact and sit directly facing the patient so the patient can supplement what they hear with lip reading. Keep lips at the patient's face level.
- Minimize use of the computer if possible.
- Make sure the meeting space is as quiet as possible. Close the door to noisy hallways. Turn off background music and TVs.
- Make use of pocket-talkers and encourage the patient to use their hearing aides, when available.

Memory Loss and Dementia

Working memory declines with age. Working memory is important for processing complex sentences (specifically, sentences with multiple imbedded clauses), which are common when discussing serious news, goals of care, and advance care planning with patients with advanced illness. To enhance comprehension, providers should attempt to break up individual pieces of information into separate sentences that could stand on their own. ASK-TELL-ASK (described earlier in this chapter) is a useful tool for confirming comprehension.

In cases where a patient suffers from dementia advanced enough to impair their ability to make their own medical decisions (or the patient lacks decision-making capacity for other reasons, such as delirium), effective communication with surrogate decision makers is critical. Important medical decisions by surrogates for patients with advanced dementia are common. In a study of 323 health care proxies for patients with advanced dementia, 40% recalled making at least one medical decision—the most common being around feeding (27%) and infections (21%). In a study of nursing home residents living with advanced dementia and in their last 3 months of life, surrogate decision makers who had an understanding of the patient's poor prognosis and the expected clinical complications in advanced dementia were much less likely to elect burdensome interventions than surrogates who lacked

this understanding. This finding demonstrates the need for providers to, over time, educate the family for what to expect in the advance stages of dementia and to foreshadow what decisions may arise in the future.

In situations where a medical decision does need to be made by a surrogate decision maker for a patient with advanced dementia (or a patient who otherwise lacks capacity), the following communication approach is recommended. First, background information should be obtained, including reviewing the patient's advance directive with the surrogate decision maker, if available, and assessing what conversations the surrogate decision maker has had with the patient previously regarding their values and treatment preferences. This information may be sufficient to make some medical decisions, and in most situations, preferences on an advance directive should be honored.

When further discussion is needed, the next decision-making step relies on substituted judgment—asking what the patient would have wanted if he or she could tell us. Time spent with the surrogate decision maker reflecting on the patient's values (eg, what brought the patient pleasure of life, what did quality of life mean to them) can be invaluable in aiding with complex medical decisions. These include deceptively simple questions like, *"Tell us about your mother."* We recommend, for this step, bringing the patient's "voice" into the decision-making process; for example, "*What would your mother choose if she could tell us?"* or *"Knowing your mother, what do you think would be most important to her right now?"* An appeal to substituted judgment may remove some of the burden, by framing the decision as the patient's own choice rather than the surrogate's.

When there is no reasonable basis on which to interpret how a patient would have made a medical decision, decisions should be made that are in the best interest of the patient. That is, the best path to promote the well-being of the patient as a unique person, in the context of their relationships, values, religion and spiritual beliefs, and culture. Providers should encourage surrogates to weigh the harms and benefit of various treatment options, including pain and suffering, the degree of and potential of benefit, and any impairments that will result from the treatment in question. Provider treatment recommendations, based on your understanding of the patient's values, can be very helpful here—and are encouraged. When best interest standards apply, the question becomes, *"What do you think is best for your mother?"* Occasionally, conflicts between what providers and surrogates feel is best to do arise, including questioning whether the surrogate is truly making decisions in the patient's best interest. Providers are encouraged to consult with their medical center's administration regarding next steps in these difficult, ethical situations.

Engaging the Supportive Network

Older patients may have formal or informal caregivers as well as family and friends who form a part of their support network. Engaging patient-identified supports is key to promoting fruitful discussions. Family members and caregivers often accompany older adults to medical appointments. It is particularly important that caregivers (specifically, surrogate decision makers) be present for key conversations regarding an older adult's goals of care so they too can understand the patient's preferences. However, a common occurrence is that caregivers of older adults may unintentionally (or intentionally) fall into a pattern of "speaking for" the older patient. Sometimes, but not always, this is due to difficulties in communication including hearing or vision problems or memory impairment.

Although it is sometimes necessary for a caregiver to provide supplemental information, it should not serve as a substitute for direct communication between the older adult patient and provider. Maintain direct eye contact with the patient, allow extra time for the patient to answer questions, and avoid speaking in the third person about the patient. To help establish the patient's sense of autonomy and participation in their health care, continually attempt to redirect the conversation back to the patient if you find the caregiver is tending to talk over the patient.

Prognostic Uncertainty

As mentioned in the introduction, older adults are living longer, with more chronic conditions that affect their quality of life and prognosis. When it comes to prognosticating, this population can be more challenging to the provider than, for instance, patients with a more predictable, terminal illness such as metastatic cancer. Prognostic uncertainty may heighten providers' hesitancy to discuss prognosis with these complex patients. However we know these conversations are important, and desired by patients. In one study of frail older patients with a mean age of 73, a life-limiting illness, and a need for assistance in at least one instrumental activity of daily living, over half of the patients whose physician had never discussed prognosis reported wanting to discuss it. While providers often consider prognosis conversations as important to medical decision making (including, but not limited to, chronic disease management, cancer screening, initiation of dialysis, and advance care planning), many older adults find this information helpful in determining life choices (eg, financial decisions, long-term care and housing, and quality-of-life considerations like spending time with family). Using prognostic indexes, many of which can be found on the user-friendly E-Prognosis website (www.eprognosis.com), can help inform your estimates alongside the clinical picture for your individual patient. Additional techniques for approaching conversations where prognostic uncertainty is present are summarized in **Table 71-7.**

SPECIAL SITUATIONS

Discussing Difficult News

Difficult news is defined as any news that drastically and negatively alters the patient's view of his/her future. Although

TABLE 71-7 ■ ADDRESSING PROGNOSTIC UNCERTAINTY

TASK	SUGGESTED LANGUAGE
Acknowledge it is difficult to make a 100% certain prediction about the patient's prognosis	*"Like the weather, it is difficult for me to be 100% accurate regarding how long you have to live."*
Use NURSE statements to respond to the patient's emotion around uncertainty	*"I can imagine it's frustrating to not know exactly what will happen."*
Provide prognostic estimates using ranges, eg, days to weeks, weeks to months, months to years	*"My best estimate is that you have weeks to months to live."*
Provide estimates that relate to a population of patients, not just the patient in front of you	*"In patients with medical issues similar to yours, the data suggest patients live an additional 1–2 years."*
Ask open-ended questions that enhance understanding of what is important to your patient	*"How do you view your quality of life currently?"* *"How do you think about balancing quality of life with quantity?"* *"When you think about the future, what do you hope for?"* *"Would there be any circumstances under which you would find life not worth living?"*
After addressing emotion, gently encourage patients and families to refocus on the present, as opposed to fixating on uncertain future	*"What can we do now, given that we are unsure of exactly what the future will bring?"*

we often associate discussing difficult news with the practice of oncology, all providers, regardless of specialty, are called upon to discuss difficult news with patients. Difficult news may relate to a diagnosis (eg, terminal, chronic, cancer, etc), prognosis, test results significant for disease progression, a bad outcome, or even medical mistakes. For older adults, difficult news may come in the form of a loss of independence, such as being deemed unsafe to continue driving a car or, requiring a higher level of care that will force the patient to move from their home of many years. It is important to remember that (1) most patients want to know difficult news (in studies, 75–90%) and (2) how a patient responds to news is influenced by a patient's perspective, life experiences, occupation, personality and coping skills, religion, and social support. It is also important to recognize that a patient's desire for medical information and a family's desire to disclose such information (eg, the difficult news of a new cancer diagnosis) is influenced by a number of factors including, but not limited to, culture, ethnicity, religion, and past experiences with illness. Asking specifically about your patient's preferences regarding how much information they want to know about their health, diagnosis, and treatment options and also how they prefer to make medical decisions is key to providing person-centered medical care.

When difficult news is communicated in an effective manner, it can have an important impact on patient satisfaction and decreasing patient anxiety and depression. Discussing difficult news poorly can not only negatively impact patient outcomes, but can also have short-term (eg, anxiety) and long-term (eg, feelings of failure, regret, identification, higher rate of "burn-out") effects on the provider. Challenges to discussing difficult news include providing information consistent with the patient's understanding of their disease, concerns about taking away patient hope, and addressing emotion. A useful roadmap for discussing difficult news uses the acronym SPIKES (**Table 71-8**). The GUIDE framework from VitalTalk (www.vitaltalk.org) can also be useful.

Tube Feeding in Advanced Dementia

Outcomes including death, aspiration pneumonia, functional status, and patient comfort are similar, if not better in some cases, for careful hand feeding compared to tube feeding in patients with advanced dementia (eg, bed-bound, unable to ambulate, and typically limited, if any, ability to communicate verbally alongside feeding difficulties, such as refusal of food, dysphagia, and/or recurrent aspiration). Additionally, the risk of hospitalization, ER visits, agitation, use of physical and chemical restraints, and incidence of pressure ulcers all appear to be higher with tube feeding as compared to careful hand feeding. Given this data, the American Geriatrics Society, in their "Choosing Wisely Campaign," does not recommend percutaneous feeding tubes in patients with advanced dementia and recommends oral assisted feeding instead.

Despite data that feeding tube placement in advanced dementia may not provide benefit, and may even cause harm, one of the most common communication challenges providers caring for older adults face revolves around these decisions. Insufficient education of patients and surrogates on the potential risks of tube feeding likely contributes to this challenge as does the important social and symbolic role food plays in many cultures and religions.

One of the best strategies for effectively addressing this issue is to start the conversation early. In most cases, outpatient providers will have an opportunity over months to years to observe their patient with advanced dementia's swallowing ability and progressively decreasing oral intake. After ruling out reversible causes, providers should continually educate patients (if able to engage), family members, caregivers, and surrogates that feeding difficulties are a sign of progressive, advanced dementia and you anticipate they will worsen over time. ASK-TELL-ASK, summarized earlier in *Step 3: Education and Exploration*, can be a particularly useful skill here to assess what a patient or family understands and provided targeted education around the natural history of dementia and feeding difficulties.

Discussions regarding preferences for feeding support should begin early in the course of dementia, ideally

TABLE 71-8 ■ SPIKES FOR DISCUSSING DIFFICULT NEWS

S: Setup	• Ensure a private space where everyone can be seated • Obtain collateral information from other providers • Check in with yourself regarding your emotional status and worries regarding the conversation
P: Perception	• Use ASK-TELL-ASK to assess the patient's current understanding of their condition • Clarify misunderstandings
I: Invitation	• Find out how much information the patient wants to know (eg, *"How would you like me to talk with you about your test results?" or "I have some serious news about your health, would you like me to discuss with you or someone else?")*
K: Knowledge	• Convey the difficult news in plain language without jargon (eg, *"The CT scan shows your cancer is not responding to the chemotherapy" or "Unfortunately, your vision has gotten worse to the point I think it is unsafe for you to be driving")* • Pause for 10–15 seconds after you have delivered the difficult news to allow time for processing and emotion
E: Empathize	• Use empathetic statements to respond to patient emotions (draw upon the mnemonic NURSE found in **Table 71-5**) • Resist the temptation to make things better • Consider using "I wish" statements (*eg, "I wish I had better news" or "I wish things had turned out differently"*)
S: Summarize and Strategize	• Summarize the clinical information and plan next steps • Provide reassurance you will not abandon the patient • Ask patient to summarize information in his/her own words to check for understanding

when the patient's cognition is intact enough to express their own preferences. These discussions should not be delayed until a crisis develops. Any decisions made with the patient should be documented in the medical record and an advance directive. This decision-making process includes aligning what patients and families are hoping for, with the realistic expectations and potential outcomes of tube feeding. Questions that can be particularly helpful, both early in the patient's illness and in crisis situations (eg, recurrent hospitalizations for aspiration pneumonia in a patient with advanced dementia) include, *"What do you hope a feeding tube would do for you (or your loved one)?," "What worries do you have about feeding tubes?,"* and *"Have you seen any other patients with severe dementia with a feeding tube? What do you think that experience was like for them?"*

End-of-Life Preferences

As older adults with advanced illness reach the end of their life it is important to be sure that the patients' wishes regarding resuscitation and ICU care are confirmed and documented. A recent study from Teno et al. found an evolving trend toward less in-hospital care, more deaths at home and in assisted living facilities, and fewer transitions of care among Medicare beneficiaries near the end of life. Equally if not more important is ensuring patients' wishes, including desire for hospitalization, hopes for what the end of their life would look like, and preferred place of death are discussed. Patient preferences should be revisited as their health status evolves as patient preferences for medical care change over time. When possible, it is critically important to ensure surrogate decision makers are present for these important conversations so they are aware of the patient's wishes and appreciate nuisances of patient's preferences that may go uncaptured on advance care planning documents. Suggested language for exploring a patient's end-of-life preferences, beyond resuscitation, is given as following:

- *"If you were close to the end of your life, what would be most important to you?"*
- *"Do you have a sense of where you would prefer to be at the end of your life?"*
- *"Some patients feel strongly they'd like to die at home, while others don't have a preference. Do you?"*

CONCLUSION

Effective communication with older adults with advanced illness requires a diverse and flexible set of skills. Techniques including ASK-TELL-ASK, use of NURSE statements to address emotion, and the SPIKES model for delivering difficult news can be invaluable in overcoming some of the challenges to caring for older adults, including sensory deficits, memory problems, and prognostic uncertainty for patients facing multimorbidity and/or frailty. Despite the challenges, having these important conversations can be extremely rewarding. Most patients want to discuss these important issues with their providers. Furthermore, conversations around a patient's quality of life, goals of care, and advance care planning allow clinicians to match our treatment recommendations to patient's ultimate goals and values to provide them with the best care possible.

FURTHER READING

AGS Choosing Wisely Workgroup. American Geriatrics Society identifies five things that healthcare providers and patients should question. *J Am Geriatr Soc*. 2013; 61(4):622–631.

Back AL, Arnold RM, Baile WF, Tulsky JA, Fryer-Edwards K. Approaching difficult communication tasks in oncology. *CA Cancer J Clin*. 2005;55(3):164–177.

Back AL, Arnold RM, Tulsky JA. *Mastering Communication with Seriously Ill Patients: Balancing Honesty with Empathy and Hope*. Cambridge, MN: Cambridge University Press; 2009.

Clayton JM, Hancock KM, Butow PN, et al. Clinical practice guidelines for communicating prognosis and end-of-life issues with adults in the advanced stages of a life-limiting illness, and their caregivers. *Med J Aust* 2007;186:S77, S79, S83–108.

Fischer R, Ury W, Patton B. *Getting to Yes: Negotiating Agreement Without Giving In*. New York, NY: Penguin Books; 1981.

Mitchell SL, Teno JM, Kiely DK, et al. The clinical course of advanced dementia. *N Engl J Med*. 2009;361(16): 1529–1538.

Ritchie CS, Roth DL, Allman RM. Living with an aging parent: "It was a beautiful invitation." *JAMA*. 2011;306(7):746–753.

Smith AK, Williams BA, Lo B. Discussing overall prognosis with the very elderly. *N Engl J Med*. 2011;365(23): 2149–2151.

Sulmasy DP, Snyder L. Substituted interests and best judgments: an integrated model of surrogate decision making. *JAMA*. 2010 3;304(17):1946–1947.

Teno JM, Gozalo P, Trivedi AN, et al. Site of death, place of care, and health care transitions among US Medicare beneficiaries, 2000-2015. *JAMA*. 2018;320(3):264–271.

The Gerontological Society of America. *Communicating with Older Adults, An Evidence-Based Review of What Really Works*. Washington, DC; 2012.

van Vliet LM, Lindenberger E, van Weert JC. Communication with older, seriously ill patients. *Clin Geriatr Med*. 2015;31(2):219–230.

von Gunten CF, Ferris FD, Emanuel LL. The patient-physician relationship. Ensuring competency in end-of-life care: communication and relational skills. *JAMA*. 2000;284(23):3051–3057.

Widera EW, Rosenfeld KE, Fromme EK, Sulmasy DP, Arnold RM. Approaching patients and family members who hope for a miracle. *J Pain Symptom Manage*. 2011;42(1):119–125.

Chapter 72

Ethical Issues

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INTRODUCTION

Health care professionals who work with older individuals in diverse settings such as their homes, assisted living facilities, skilled nursing facilities, outpatient clinics, and hospitals may encounter ethical dilemmas in their work. Ethical dilemmas arise when there is uncertainty or disagreement between stakeholders about the right thing to do in a situation. Ethical dilemmas pertinent to older patients may include questions about the balance of patient privacy versus patient safety, patients' abilities to make their own medical decisions, decisions about preferred treatments at the end of life, and decisions made by surrogate decision-makers.

In this chapter, we will discuss a framework for approaching ethical dilemmas, and provide an overview of the ethics topics especially relevant to the care of older adults. Sections include: an approach to thinking about ethical dilemmas, moral distress in professional caregivers, breaking bad news, surrogate decision-making, ethical decisions at the end of life, dilemmas between honoring self-determination and individual/societal safety, ethical challenges in the nursing home setting, and resource allocation and ageism.

AN APPROACH TO THINKING ABOUT ETHICAL DILEMMAS

Case

Mrs. C is an 88-year-old woman with congestive heart failure, hypertension, atrial fibrillation, and chronic kidney disease. She lives alone in her own home. She was hospitalized last month for delirium secondary to a urinary tract infection. She is admitted again this month after a fall. While in the hospital, she is evaluated by physical therapy and occupational therapy who both recommend a short stay in a nursing home for rehabilitation. Mrs. C refuses to go to a nursing home and demands to return to her home. Her medical team insists that going directly home is not a safe discharge plan, and wants to require her to go to rehab.

Question: What is an ethically appropriate discharge plan for Mrs. C?

Health care professionals working with older patients may encounter ethical dilemmas, regardless of their discipline

Learning Objectives

- Discuss the steps in the CASES approach and how it is used to address ethical dilemmas.
- Define moral distress and identify risk factors for developing it.
- Explain the steps in the SPIKES approach to breaking bad news.
- Understand the need for surrogate decision-making to ensure appropriate care of many older adults.
- Explore the concepts of substituted judgment versus best interest.
- Gain insight into the experience of surrogate decision-makers and potential for surrogate distress after making difficult medical decisions.
- Describe the indications for palliative sedation.
- Recognize the barriers to cessation of driving.
- Discuss examples of ethical dilemmas that arise in nursing homes when the desire to maintain quality of care indicators is in conflict with promoting patient preferences and quality end of life care.
- Recognize ethical principles underlying health care resource allocation under conditions of resource scarcity, including arguments against excluding patients from healthcare resources based on age.

Key Clinical Points

1. **Health care professionals who work with older patients may encounter ethical dilemmas which arise out of conflicts in values about the right thing to do.**
2. **Patients with decisional capacity have the right to make decisions that their medical teams and families don't agree with.**
3. **Moral distress is a condition in which clinicians feel forced to provide care they believe is wrong, which can lead to burnout and to them leaving their jobs and professions.**

(*Continued*)

(Cont.)

4. The majority of people want to be told if they have dementia.
5. Making medical decisions for loved ones can be very stressful for surrogate decision-makers.
6. Through a process of shared decision-making, patients or their surrogates share information about their goals and care preferences with clinicians, and their clinicians share pertinent medical information and make care recommendations based on the patient's values and goals.
7. Palliative sedation is a legal intervention of last resort to promote relief of intractable suffering for terminally ill patients who have comfort as their primary goal of care.
8. Physician aid in dying (PAD), also called physician-assisted suicide, is legal in some US states and countries. Euthanasia is not legal in the United States, but is legal in Belgium and the Netherlands.
9. Patients may raise the topic of PAD as a way to begin a conversation about the end of life. Clinicians should view this as an opportunity to talk about sources of intractable suffering and fears about dying regardless of whether or not they live in a state where the practice is legal.
10. Questions about whether or not to report concerns about an older individual's driving are answered by weighing the clinician's duty to protect the patient's safety, with the duty to maintain confidentiality, and the duty to protect the public. Some US states have mandatory reporting requirements.
11. Ethical issues prevalent in nursing homes include not only those in which the patient and family's care goals are in conflict, but also those in which regulations are in conflict with quality end of life care.
12. Triage committees, and not front-line health care providers, should make decisions about allocation of limited heath care resources under conditions of resource scarcity, and patient age should not factor into these decisions.

or work setting. Ethical dilemmas usually stem from a conflict in values between stakeholders. For example, an older patient with multiple comorbidities, who values being alive regardless of his condition, may disagree with his physician about the best way to manage his advanced kidney disease. To add complexity, older adults are a heterogeneous group with diverse values, preferences, and goals that may or may not be influenced by their race, ethnicity, or religious backgrounds. In terms of their beliefs about end-of-life care, for example, older patients may have more heterogeneous views than their clinicians. Because of this heterogeneity, clinicians should not assume that they know a patient's values and goals without engaging in a conversation with the patient.

Ethical dilemmas may be distressing to health care professionals. Although health care professionals learned about ethics in their training, many are not familiar with a standard approach to thinking about and responding to ethical dilemmas. Without knowledge of such an approach, ethical dilemmas encountered may lead to moral distress and burnout. Familiarity with a framework and using a standard approach to thinking about ethical dilemmas may help health care professionals to find ways to resolve them, while honoring patient preferences, and reducing their own stress.

Different frameworks for approaching ethical dilemmas exist. One approach is the four-box method from Jonsen et al., in which important information about the patient's medical condition, preferences, quality of life, and other contextual features of the case is collected. Another is the CASES approach to ethical dilemmas which has been developed and is used by ethics consultants throughout the Veterans Administration (VA) health care system (see **Table 72-1** and Further Reading). Although this approach was developed for use by ethics consultants, the approach also can be applied by others who are trying to better understand ethical dilemmas. Since the second step of the CASES approach is similar to the four-box method, let us apply the CASES approach to the case of Mrs. C above.

The first step of the CASES approach is to Clarify whether the dilemma is really about ethics or something else, such as a legal issue. This is done by identifying whether there is a conflict in values between two of the involved stakeholders about the right thing to do. To do this, we try to understand the perspectives and values of the involved stakeholders, and identify which values are in conflict. In the case of Mrs. C, the conflict in values is between Mrs. C, who doesn't want to go to a nursing home for rehab, but wants to go home, and her medical team that believes sending her home would be harmful to her. This dilemma also can be viewed as a conflict between ethical principles. In the case of Mrs. C, the conflict in principles is between honoring Mrs. C's autonomy/self-determination and promoting both beneficence and nonmaleficence.

In the second step of the CASES approach, information relevant to the case is Assembled. This information includes medical information, patient preferences, other parties' preferences, and ethics knowledge. In the case of Mrs. C, we already know that she has multiple medical problems, lives alone, is in her second hospitalization, and

TABLE 72-1 ■ THE CASES APPROACH

1. C—Clarify
   a. Does this case involve an ethics concern about the right thing to do?
   b. What is the conflict in values that is leading to the concern?
      i. Values are defined as strongly held beliefs, ideals, principles, or standards that inform ethical decisions or actions.
2. A—Assemble the relevant information
   a. What types of information is needed?
      i. Medical facts
      ii. Patients preferences and interests
      iii. Other parties' preferences and interests
      iv. Ethics knowledge
         1. Codes of ethics, ethics guidelines, and consensus statements
         2. Published literature
         3. Precedent cases
         4. Institutional policy and documents, and law
         5. Outside ethics experts
   b. Which stakeholders need to be interviewed?
3. S—Synthesize the information
   a. Analyze the information
      i. Review the range of ethically justifiable options
      ii. Weigh the different potential outcomes and the impact of these outcomes on the stakeholders
4. E—Explain the synthesis
   a. Communicate the ethically justifiable options to the key stakeholders
5. S—Support the consultation process
   a. Follow up with stakeholders

Data from Berkowitz KA, Chanko BL, Foglia MB, et al. Ethics Consultation: Responding to Ethics Questions in Health Care, 2nd ed. Washington, DC: U.S. Department of Veterans Affairs; 2015.

is staunch in her refusal to go to a nursing home for rehab. Knowing her reason for refusing this would be helpful. For example, she might tell us that her husband died in a nursing home after a prolonged decline from dementia, and that she gets flashbacks and panic attacks whenever she enters a nursing home. Or, she might tell us that she needs to get home to care for her frail neighbor. We would want to hear more from Mrs. C's medical team about their concerns, and what they believe they can and cannot allow their patients to do. In thinking about this case, it also might be helpful to look at the hospital's policies around informed consent and refusal of care, and to search for pertinent literature about this issue (see Further Reading).

Once we have gathered the relevant information about Mrs. C's case, including the stakeholders' perspectives, we would need to begin to consider the different possible options and the impacts of each option on the stakeholders. In the Synthesize step, we would weigh the potential risks and benefits to Mrs. C of honoring her preference and sending her home versus insisting that she be discharged to sub-acute rehab. An important aspect of the case is whether Mrs. C. has decisional capacity to make the "poor" decision to return to her home or not. (Decisional capacity is discussed in Chapter 10.) In assessing her capacity to make this specific decision, it would be important to find out if she understands the risks and benefits of going to a nursing home for rehab versus going directly home.

If Mrs. C has decisional capacity to make the decision to go home, her medical team cannot force her to go to a nursing home against her will. In light of this, her team should consider ways to promote her continued recovery and safety at home. This could include arranging for more resources in her home, such as a visiting nurse, physical therapist, occupational therapist, Meals on Wheels, and a product she could use to quickly call for help in the future. Or it might involve brainstorming about how to get help for her frail neighbor, so she would feel more comfortable going to the nursing home for rehab.

If Mrs. C lacks decisional capacity to make the decision to go home, then the medical team will need to involve her legal surrogate decision-maker to help make this decision. Since some states have laws forbidding placement of individuals in nursing homes against their will, finding a solution could prove to be difficult, and might involve substantial negotiating and developing creative solutions.

When the CASES approach is used by health care professionals to understand the nuances of ethical dilemmas in their work, the last two steps of the approach are less critical than they are for ethics consultants. These last two steps are included for ethics consultants to help ensure that their recommendations are shared with those involved with the case and that appropriate follow-up occurs. However, as a health care professional uses this approach and gains a deeper understanding of a case, he/she may want to share that information with others who are involved. It also may be helpful to identify recurrent ethical dilemmas within an organization, and determine if systems level interventions are needed.

ETHICAL DILEMMAS CAN LEAD TO MORAL DISTRESS IN PROFESSIONAL CAREGIVERS

Case

Mr. G is a 78-year-old skilled nursing facility resident with multiple medical problems, including mild dementia, who is admitted to the intensive care unit with pneumonia and possibly sepsis. He is intubated. Over the next 2 weeks, he develops acute respiratory distress syndrome (ARDS), *Clostridium difficile* colitis, and a pressure ulcer. His renal function is deteriorating. He remains delirious, and is therefore unable to participate in medical decision-making.

Mr. G's advance directive states that he wouldn't want life-prolonging measures for a terminal condition, and would want care focused on comfort. His wife, whom he designated as his health care agent through a Durable Power of Attorney for Health Care, insists that treatments

with the goal of life prolongation be continued. His nurses are concerned that his prognosis is poor and his preferences aren't being honored. His physicians say he isn't "terminal" yet.

His nurses are concerned that he has pain. They note that he grimaces when they care for him. His wife refuses for him to have pain medications because of a fear that they will prevent him from waking up.

When the doctors, nurses, social workers, and the chaplain have tried to talk to Mrs. G about her husband's goals of care or his pain management, she becomes agitated and threatens lawsuits. Team members are frustrated. Rumors begin to circulate that Mrs. G is demented, in denial, revengeful, and wanting to keep him alive for his pension. Some have commented that scarce resources are being wasted in keeping him alive.

Question: In reading this case, can you sense the distress experienced by those caring for Mr. G?

Moral distress is a term that was coined by Jameton in 1984. Moral distress occurs when someone or something prevents you from doing what you believe is right, forcing you to act contrary to your core values. In other words, you believe you are providing care that is morally wrong. Although moral distress has been studied most in nursing, it has been documented in numerous health care professions, and even in health care leaders. Health care professionals who interact with older patients may personally experience moral distress, and also may interact with colleagues who are experiencing it.

There are numerous root causes of moral distress (see **Table 72-2**). These causes may be due to clinical factors of a case, but also may be due to individual characteristics of members of the care team, or to institutional factors. Multiple members of the care team may experience moral distress about a given case, but may experience this distress from different aspects of the case.

One theory of moral distress holds that one's level of distress does not return to baseline after experiencing an episode of moral distress, but that, with each case, one's "moral residue" increases to a new baseline. Thus, over time, there is a crescendo in a clinician's moral residue. This crescendo of moral residue and "untreated" moral distress can lead to burnout, to individuals leaving their positions, and even to individuals leaving their professions. This can be psychologically costly for individuals and financially costly for institutions **(Figure 72-1)**.

After recognizing that a case is causing moral distress, what can health care professionals do about it? One helpful intervention is for the team to talk about the case and the aspects of the case that are giving each of them moral distress. Building one's moral resilience may be helpful (see Further Reading). Finally, if organizational factors are leading to moral distress in staff, these need to be brought to the attention of the organization's management.

TABLE 72-2 ■ SELECTED ROOT CAUSES OF MORAL DISTRESS

CLINICAL SITUATIONS	
• Providing unnecessary/futile treatment	• Using resources inappropriately
• Prolonging the dying process through aggressive treatment	• Providing care that is not in the best interest of the patient
• Inadequate informed consent	• Providing inadequate pain relief
• Working with caregivers who are not as competent as care requires	• Providing false hope to patients and families
• Lack of consensus regarding treatment plan	• Hastening the dying process
• Lack of continuity of care	• Lack of truth-telling
• Conflicting duties	• Disregard for patient wishes
INTERNAL CONSTRAINTS	
• Perceived powerlessness	• Lack of knowledge of alternative treatment plans
• Inability to identify the ethical issues	• Increased moral sensitivity
• Lack of understanding the full situation	• Lack of assertiveness
• Self-doubt	• Socialization to follow others
EXTERNAL CONSTRAINTS	
• Inadequate communication among team members	• Tolerance of disruptive and abusive behavior
• Differing inter- (eg, RN to MD) or intraprofessional (eg, RN to RN) perspectives	• Compromising care due to pressures to reduce costs
• Inadequate staffing and increased turnover	• Hierarchies within healthcare system
• Lack of administrative support	• Lack of collegial relationships
• Policies and priorities that conflict with care needs	• Nurses not involved in decision-making
• Following family wishes of patient care for fear of litigation	• Compromised care due to insurance pressure or fear of litigation

Reproduced with permission from Hamric AB, Borchers CT, Epstein EG. Development and testing of an instrument to measure moral distress in healthcare professionals. Am J Bioeth Prim Res. 2012;3(2):1–9.

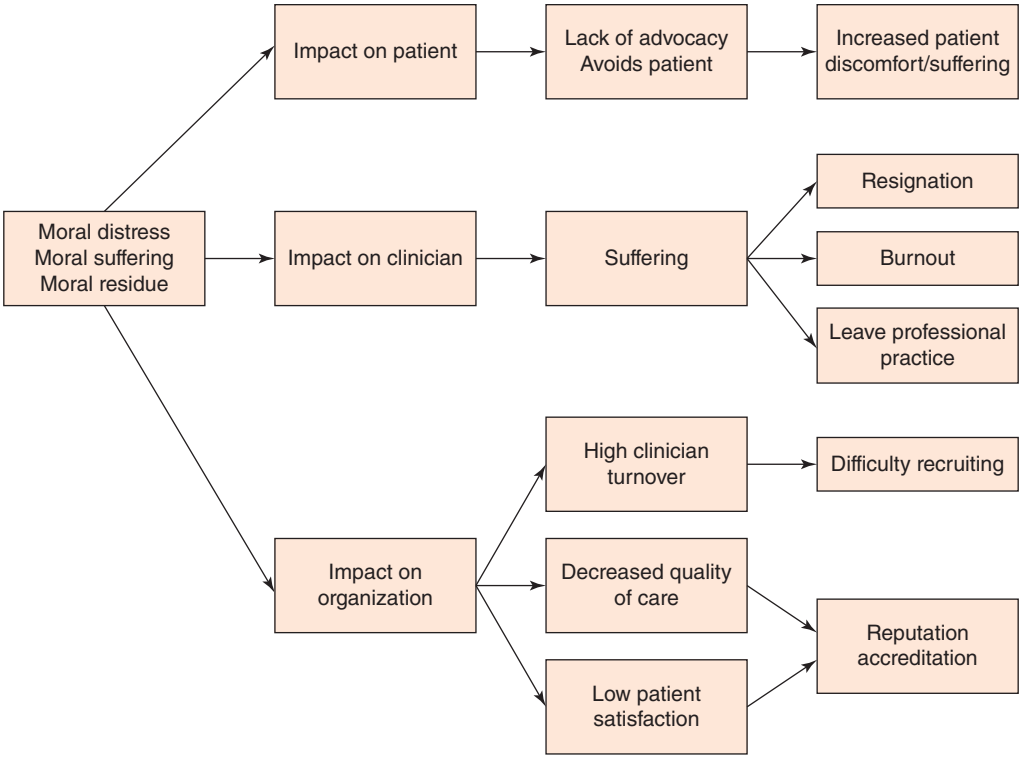

FIGURE 72-1. Impacts of moral distress on the patient, clinician, and organization. (Adapted with permission from Corley MC. Nurse moral distress: a proposed theory and research agenda. *Nurs Ethics*. 2002;9[6]:636–650.)

TRUTHFUL DISCLOSURE

Case

Mr. W is an 83-year-old man who is brought to see his primary care provider by his wife. She confronts the physician outside the examination room and expresses concern that her husband has become more forgetful and no longer is able to pay their bills. She remarks, "If he has Alzheimer's disease, please don't tell him."

Question: What is the ethically justifiable option for the clinician after he examines Mr. W?

In the past, when medicine was more paternalistic, physicians did not routinely inform patients of life-threatening diagnoses. The rationale for not sharing diagnoses, such as cancer, with patients was that this information would cause harm. However, most people want to know their diagnoses. A systematic review of 23 articles with over 9000 participants found that 91% of cognitively intact people and 85% of cognitively impaired people wanted to know if they had dementia. A second study found that there were no increases in rates of depression or anxiety after people learned of a dementia diagnosis.

As medicine has moved from being paternalistic to being patient-centered, the practice of withholding information from patients decreased. In this era of patient-centered care, patients should be offered the opportunity to learn about their medical conditions. Patients also have the right to decline to hear about their conditions. In certain cultures, it is still standard to not share information about serious illness with patients, but with their family members (ie, familismo). In some cultures, there is a belief that talking about bad things will make them happen. Thus if a patient doesn't want information, for any reason, giving this information without permission is not ethically justifiable, and could cause harm.

One method advocated for use when breaking bad news is the six-step SPIKES approach (see Further Reading). The steps of this approach are explained in **Table 72-3**.

In response to the case of Mr. W, it would be appropriate to understand Mrs. W's reasons for not wanting her husband to know his diagnosis. Perhaps he has asked her on multiple occasions not to ever tell him if he develops dementia, because of a previous experience with a family member who suffered from dementia. In this case, it might be appropriate not to share this information with Mr. W. Ultimately, however, he has a right to this information, if he wants it. After hearing her reasons, we could use the SPIKES approach to talk to Mr. W. We would ask about his understanding of his current situation and whether he would want information or not. In the case of dementia, it is appropriate to share information about the diagnosis with a patient with mild dementia who has decisional capacity to make the decision about whether or not to hear the cause of his memory loss. It might not be appropriate to repeat this information over and over to a patient with more advanced

TABLE 72-3 ■ THE SPIKES APPROACH TO BREAKING BAD NEWS

1. **S—Setting** up the interview
   a. Find a private, quiet room with room for all participants to sit down
   b. Encourage family and involved others to participate
   c. Minimize distractions—turn off sound from pagers and phones
   d. Establish rapport with the patient with eye contact and touch
2. **P**—Assessing the patient's **Perception**
   a. What do the patient and family understand about the patient's medical condition?
   b. Correct any misperceptions
3. **I**—Obtaining the patient's **Invitation**
   a. Ask the patient and family how much information and what kind of information they want or don't want
4. **K**—Giving **Knowledge** and information to the patient
   a. Start with a statement that you're about to share bad news
   b. Share the information succinctly in lay language
   c. Allow time for the patient and family to absorb this information
5. **E**—Addressing the patient's **Emotions** with empathic responses
   a. Acknowledge and validate the patient and family's emotional response
6. **S—Strategy and Summary**
   a. Summarize the conversation and answer questions
   b. Develop a clear plan for what comes next

Data from Baile WF, Buckman R, Lenzi R, et al. SPIKES-A six-step protocol for delivering bad news: application to the patient with cancer. Oncologist. 2000;5(4):302–311.

dementia, who might experience the emotional distress of receiving this diagnosis multiple times or might not be able to fully understand what this means.

DECISION-MAKING IN CHRONIC AND ADVANCED ILLNESS

Case

Mrs. S is an 82-year-old woman with mild dementia, coronary artery disease, hypertension, and peripheral vascular disease. She misses a meal in the dining room of her assisted living facility, and is found in her apartment lethargic and confused. She is sent to the hospital, admitted to the ICU, and diagnosed with urosepsis. She has a Physician Orders for Life-Sustaining Treatment (POLST) form, which accompanies her to the hospital, with preferences for Do Not Resuscitate status (DNR), Limited Additional Interventions, antibiotics to prolong life, and a trial of use of a feeding tube.

Despite aggressive care in the ICU, she remains delirious. Over the next 2 weeks in the ICU, she suffers a non-ST segment elevation MI, develops a pneumonia thought secondary to aspiration, and her kidney function worsens. Her adult children, who are her legal decision-makers, are given regular updates, but are overwhelmed by the numerous medical decisions they are asked to make including: whether to insert a temporary feeding tube in her nose, followed later by whether to insert a more permanent tube in her stomach, and whether to start dialysis. Although they had initially told the ICU to do "everything" to make their mother better, short of resuscitating or intubating her, they begin to wonder if they are doing the right thing. They wonder how long their mother would want this level of care continued. During a routine meeting with the family, the ICU team uses the term "futile" and recommends that they change the goal of care to focus on maximizing their mother's comfort.

Question: What should the family and ICU team consider in deciding how to proceed?

In older patients with chronic medical conditions, a multitude of medical decisions may arise during acute illness, and in the course of worsening of the chronic conditions. When patients are unable to make their own decisions, surrogates are called on to make these decisions. Decision-making for others can be stressful for surrogates, while knowledge of a loved one's preferences can reduce the burden of decision-making for surrogates. In the case of Mrs. S, her children have attempted to implement her wishes, but her condition has worsened, and they are no longer sure of the right thing to do. In what follows, several decisions related to this case will be discussed.

We operate under the notion that the patient possesses a fundamental legal and moral right to have his or her care preferences respected. Understanding any prior conversations or specific directives (such as her POLST form), and knowing her overall values, hopefully Mrs. S's family would be able to speak on behalf of Mrs. S, as though they were relaying her explicit wishes, or "standing in her shoes." This concept illustrates the ethical principle of substituted judgment. When such patient preferences are known, it is expected that the patient's surrogate decision-maker would operate under this principle, communicating the patient's wishes, and choosing the option that would best fit her stated care preferences.

Often, the patient's explicit wishes regarding a specific clinical scenario are not known. It can be difficult to anticipate all of the different clinical scenarios and medical decisions that may need to be considered in one's future ahead of time. In this type of situation, surrogate decision-makers often need to rely on the principle of best interest. In deciding what satisfies a person's best interest, the surrogate decision-maker—ideally with guidance from the primary team of health care providers caring for the patient—is able to objectively weigh risks, potential benefits and treatment burdens to make an individualized decision for the patient. However, when evidence of the patient's wishes does exist, making decisions by

substituted judgment should take precedence so that the patient's care preferences are honored.

At the beginning of her hospitalization, Mrs. S's children believed that they were doing what she would want in inserting a temporary nasogastric tube. When she didn't improve despite aggressive interventions and remained unable to swallow safely, her children were less sure if she would want a more permanent tube inserted. Although the literature, position statements from organizations such as the American Geriatrics Society, and the Choosing Wisely campaign have argued against the insertion of feeding tubes in patients with advanced dementia (which is discussed elsewhere), the appropriate use of feeding tubes in other situations is less clear. In Mrs. S's situation, the decision of whether or not to place a permanent feeding tube depends on her overall goals of care. It is not clearly right or wrong to place it.

Another decision that families often will make for their loved ones is whether to treat infections with antibiotics. Antibiotic therapy is thought to be beneficial and the usual standard of care in conditions such as pneumonia and urinary tract infections, which often arise among older adults. In some patients with advanced serious illness, however, antimicrobials can carry substantial risk and treatment burdens that need to be carefully weighed, especially among patients receiving end of life care. For example, patients receiving antimicrobials are at increased risk of acquiring infections with resistant organisms as well *C. difficile* infections. Some of the unintended consequences of antimicrobial therapy not only include burdens to the individual patient, but also extend to other patients, and to society through the possible selection of antimicrobial resistance. Studies are conflicting as to whether antimicrobial treatment of pneumonia may provide symptomatic benefit among patients with advanced dementia. A recent prospective study of patients with advanced dementia and pneumonia demonstrated an association between antimicrobials and increased discomfort. For patients receiving comfort-focused care near the end of life, the treatment of pain, fever, and dyspnea with opioids and antipyretics remains effective to provide symptomatic benefit. Whether a course of antibiotics is warranted will depend on the risks and benefits of the treatment and whether providing that course of treatment is consistent with the patient's goals of care.

In living with chronic conditions, older patients may have acute or chronic worsening of their kidney function. The number of older individuals who are beginning chronic dialysis is increasing. Many of these individuals start dialysis while in the hospital for an acute illness. Studies have shown that older people with poor functional status, multiple chronic conditions, and age over 85 when starting dialysis may not necessarily live longer with dialysis. Conservative management programs for older patients with advanced kidney disease who are forgoing dialysis are prevalent in the United Kingdom, but not elsewhere. In deciding about whether or not to begin dialysis, Mrs. S's children need to be fully informed about the risks and benefits of short- and long-term dialysis. As with other decisions, the decision to start dialysis depends on Mrs. S's goals of care.

After a trial of aggressive therapies aimed at life prolongation, Mrs. S's clinicians begin to talk about her situation being "futile." This term is used by health care professionals, but not by lay people, and therefore should not be used in routine conversations with patients or their loved ones. The term is also commonly used when clinicians don't believe that continuing on the current trajectory is "worth it." Different definitions of futility exist, including quantitative futility (very low probability of survival) and qualitative futility (patient's quality of life is below an acceptable standard). These definitions can be problematic because clinicians cannot accurately predict a patient's chances of survival and may not be able to separate their own views about acceptable quality of life from the patient's perceived quality of life. In the case of Mrs. S, decision-making might be easier if both the medical team and her family were clear about her goals of care; what makes life worth living for her (in keeping with the Institute for Healthcare Improvement and John A. Hartford Foundation's Age-Friendly Health Systems' 4Ms Framework in which eliciting what Matters to the patient is emphasized), and would she consider life to be acceptable if she lived in a nursing home, leaving three times a week for dialysis, and being fed through a tube.

When patients are very sick and care is focused on life prolongation, it can be hard for clinicians and family members to step back and examine the big picture. When the goal is to do "everything," medical teams may want to better understand what this means to a patient and family. They should not feel compelled to offer every intervention that is technically feasible. Instead, they should consider whether the intervention is clinically meaningful, and whether it will help to meet the patient's goals of care.

Mrs. S's medical team has recommended a change in the focus of her care from life prolongation to comfort care. Although this may be a clinically acceptable option, it is important that care recommendations be based on an understanding of the patient's values and preferences. Making recommendations tailored to patient preferences would be consistent with a shared decision-making approach. At the same time, it is important to acknowledge that although withholding and withdrawing care is ethically equivalent, withdrawing life prolonging measures may be more psychologically difficult for decision-makers.

In deciding how to proceed with Mrs. S's care, consulting a palliative care team may be helpful. The palliative care team could not only help the medical team develop a richer understanding of Mrs. S's values, care preferences, and goals of care by engaging in in-depth conversations with her children about her life, but also support her family during a stressful time.

MANAGING REFRACTORY SYMPTOMS CAUSING INTRACTABLE SUFFERING NEAR THE END OF LIFE

Case

Mr. P is a 90-year-old man with widely metastatic prostate cancer receiving hospice services in a nursing home. He continues to have severe bone pain from his metastatic disease despite the use of opioids, steroids, zolendronic acid injections, and a course of palliative radiation. As his opioid doses are increased, he develops myoclonus and confusion. Rotation to two different opioids over the next two weeks is minimally effective. His confusion worsens, becoming hyperactive delirium with bouts of yelling and attempting to climb out of bed. Addition of antipsychotic medications is only slightly effective. His family and the nursing staff are distressed about his situation and their inability to ensure his comfort and safety.

Question: What are the ethically and legally appropriate options to reduce Mr. P's suffering?

In terminally ill patients such as Mr. P, questions may arise about ways to reduce his intractable suffering and promote his comfort at the very end of life, to ensure a peaceful death. His family may ask if there is something that could be done to end his suffering, such as euthanasia. Those taking care of him may wonder about the use of palliative sedation or physician aid in dying (PAD; also known as physician-assisted suicide) and whether alternative options exist. The differences between these entities can be confusing to families and clinicians, and are defined in **Table 72-4**.

TABLE 72-4 ■ KEY TERMS, DEFINITIONS, AND LEGAL STATUS OF INTERVENTIONS TO MANAGE INTRACTABLE SUFFERING AT THE END OF LIFE

TERM	DEFINITION	STATUS	SOURCE
Palliative sedation Also called terminal sedation, end-of-life sedation, total sedation, and proportional palliative sedation	Lowering of patient consciousness using medications for the express purpose of limiting patient awareness of suffering that is intractable and intolerable. Proportional palliative sedation is the process of reducing consciousness only as much as needed to reduce patient suffering. Patients may only be mildly sedated, not necessarily sedated to unconsciousness. Deep continuous sedation is used to describe the last step of a gradual process of palliative sedation in which the patient is sedated to deep unconsciousness.	Legal in the United States	Kirk et al. NHPCO Consensus JPSM 2010 AMA Code of Medical Ethics https://www.ama-assn.org/system/files/2019-06/code-of-medical-ethics-chapter-5.pdf AAHPM position statement 2014
Voluntary stopping of eating and drinking (VSED)	"An action of a competent, capacitated person who voluntarily and deliberately chooses to stop eating and drinking with the primary intention to hasten death because unacceptable suffering persists."	Legal in the United States	Wax JW, An AW, Kosier N, Quill TE. Voluntary stopping eating and drinking. *J Am Geriatr Soc*. 2018;66(3):441–445.
Physician aid in dying (PAD) Also called physician-assisted suicide (PAS) or physician-assisted death	A practice that "occurs when a physician facilitates a patient's death by providing the necessary means and/or information to enable the patient to perform the life-ending act." Medications are self-administered by the patient, thereby distinguishing it from euthanasia. Physician-assisted suicide is a term being used less often because of the impression that the patient commits suicide.	Legal in some US states, and in Canada, Belgium, the Netherlands, and Luxembourg	AMA Code of Medical Ethics https://www.ama-assn.org/system/files/2019-06/code-of-medical-ethics-chapter-5.pdf
Euthanasia	The administration of a lethal agent by another person, such as a clinician, to a patient for the purpose of relieving the patient's intolerable and incurable suffering by ending the patient's life. Intentionally ending the life of another person, usually with the goal of alleviating or avoiding suffering.	Not legal in the United States Legal in Belgium, the Netherlands, and Luxembourg under specific circumstances with procedures to guide use	AMA Code of Medical Ethics. https://www.ama-assn.org/system/files/2019-06/code-of-medical-ethics-chapter-5.pdf

Palliative sedation is legal in the United States. It is an intervention provided to relieve intractable and intolerable symptoms at the end of life, and is accomplished by intentionally lowering the patient's level of consciousness using sedating medications. Palliative sedation should be used as an intervention of last resort in the setting of a palliative care interdisciplinary team strategy to achieve symptom control within the spectrum of palliative care interventions. The American Medical Association, the American Academy of Hospice and Palliative Medicine, and guidelines from Europe, Canada, and Japan consider palliative sedation to unconsciousness to be appropriate as a last resort in terminally ill patients with goals for comfort and severe refractory distressing symptoms including agitated delirium, dyspnea, pain, nausea and vomiting, shortness of breath, urinary retention due to clot formation, gastrointestinal pain, uncontrolled bleeding, and myoclonus. Palliative sedation for suffering that does not result from physical symptoms, but from existential, psychological, or spiritual distress remains controversial.

The recommendations on palliative sedation from the American Academy of Hospice and Palliative Care in 2014 are summarized as follows: (1) it is ethically defensible if used after careful interdisciplinary evaluation and treatment of the patient; (2) it should be used when palliative treatments that are not intended to affect consciousness have failed or, in the judgment of the clinician, are very likely to fail; (3) it should be used where its use is not expected to shorten the patient's time to death; and (4) it should be used only for the actual or expected duration of symptoms. Ideally, the patient, family, and participating interdisciplinary health care professionals should be in agreement that this intervention is consistent with patient goals and is the right thing to do. Health care professionals who believe that participating in palliative sedation is morally wrong should not be forced to participate in the care of patients undergoing this intervention. Treatment of existential suffering should include the full interdisciplinary team including mental health and spiritual care experts. Respite sedation, or sedation for a limited time with subsequent removal of sedation to reassess the patient, may be an alternative approach.

An alternative to palliative sedation is the voluntary cessation of eating and drinking (also called voluntary stopping of eating and drinking or VSED). This practice is legal if chosen because of intractable suffering by a competent person with decisional capacity who is approaching the end of life. Many individuals who are approaching the end of life do not have an appetite, and thus do not experience discomfort from VSED. For others, symptoms resulting from voluntary cessation of eating and drinking may require intervention by a palliative care team. Individuals are encouraged to let their medical teams and families know if they decide to pursue VSED. However, there are cases in which older individuals who had decided to pursue this were evicted from long-term care facilities due to legal concerns.

VSED is a patient-initiated, voluntary process, which necessitates that the patient remains competent at least at the start of the process. The health care team may need to palliate symptoms of thirst and delirium, and sustain the continued withholding of food and fluids from some patients who pursue VSED. Due to the loss of decisional capacity inherent in the progression of dementia, VSED in the setting of dementia deserves further thoughtful consideration. While some dementia-specific directives allow for delineation of potential withholding or withdrawing of food and fluids (ie, stopping eating and drinking by advance directive, or SED by AD) when one's dementia progression reaches a certain point, implementation of these directives across varying state jurisdictions will need to be understood and studied further. While the concept of SED by AD has gained traction among some advocates for advance care planning, these directives have also been criticized for utilizing a potentially problematic ethical concept called "precedent autonomy" in which a fully cognizant and healthy patient directs their care preferences to be carried out after loss of decisional capacity, should their degree of functional incapacity and circumstance meet preset conditions, regardless of the clinical situation and social setting in which the patient might land. SED by AD is a process in which a surrogate decision-maker directs the withholding of food and fluids when a certain degree of cognitive and functional impairment is reached due to dementia progression. Advocates have focused on the ethical principle of patient autonomy to direct preferences for SED by AD, as is the case with more general advance directives. However, others, including a recent position statement put forth by the Ethics Subcommittee of AMDA—The Society for Post-Acute and Long-Term Care Medicine, argue that the principle of justice (the obligation to treat all individuals equally regardless of race, gender, cognitive or physical ability needs to be the overarching principle to guide these decisions), as well as the physician's obligations to beneficence and nonmaleficence, thereby advocating that comfort feeding should be implemented for those with advanced dementia. Some of the more detailed directives, that include specific preferences for SED by AD, are difficult to implement in care settings such as nursing homes and inpatient hospices. Specific instructions to discontinue hand feeding in advanced dementia or to instruct surrogates to not allow hand feeding if one loses the ability to feed oneself are potentially difficult to implement in care settings where the provision of oral food and fluids are considered basic and comfort-oriented care, especially if the patient appears to derive enjoyment comfort out of eating or tasting food. A policy of comfort feeding in advanced dementia, on the other hand, allows for continued assisted feeding for those who are accepting food and fluids as tolerated until their feeding behaviors indicate refusal or show signs of any distress with eating.

PAD, also called physician-assisted suicide or death with dignity, is legal in some US states and the

District of Columbia. The patient pursuing PAD intentionally self-administers lethal doses of medications with the intention of ending their suffering. The Death with Dignity laws in Oregon and Washington and similar laws in other states in the United States stipulate that only terminally ill patients who are decisionally intact and voluntarily making this request are eligible to use the law. Interested individuals are required to make two verbal and one written request, and must be adult residents of their state. Two attending physicians must verify that the patient is terminally ill with intact decisional capacity. The majority of people requesting this are already receiving hospice services, and often make the request because of concerns about loss of autonomy and control. Approximately one-third of individuals who obtain medications do not use them. Finally, patients may raise the subject of PAD with their clinicians as a way to begin a conversation about death and dying. Even if clinicians cannot or will not participate, they should still view this as an opportunity to talk about the patient's sources of unbearable suffering, and fears about dying.

Euthanasia is a process by which an individual, such as a clinician, intentionally administers a lethal dose of medication to a terminally ill patient, which directly ends their life. In contrast to PAD, euthanasia is not legal in the United States. Internationally, it is legal in Belgium and the Netherlands, which have widespread palliative care policies and availability of palliative care professionals. In these countries, the practice of euthanasia is considered one option to manage intractable suffering at the end of life, and is provided in conjunction with palliative care. Of note, a qualitative study characterizing the Dutch experience with euthanasia or physician-assisted suicide requests among older adults without terminal illness and with multiple geriatric conditions between 2013 and 2019 was recently published. These older adults attributed unbearable suffering to a combination of medical, social, and existential issues that include loss of mobility, fear of decline, social isolation, and loss of meaning. This study found that patients whose suffering was attributed to multiple geriatric conditions accounted for 4% of all cases of euthanasia or physician-assisted suicide in the Netherlands during this period, calling attention to the potential for a "slippery slope" of allowing the use of euthanasia or physician-assisted suicide for suffering due to nonterminal conditions that could instead be addressed with enhanced geriatric or palliative supportive efforts to enhance quality of life.

The ethical distinction between palliative sedation and euthanasia has been debated, hinging on three areas: intention, monitoring, and methods. Palliative sedation does not have the intention to hasten death, the patient is monitored to provide the minimum level of sedation required to provide symptom relief, and the medication is carefully titrated to achieve this minimum level of sedation required to provide symptom relief. The American Medical Association (AMA) asserts that the distinction between palliative sedation and euthanasia may be evident on examination of the medical record, as repeated doses and continuous infusions suggest palliative sedation, while one large dose or rapidly accelerating doses out of proportion to the level of immediate patient suffering may reflect an intention to hasten death. Other indications of inappropriate palliative sedation include inadequate communication and support between patients, relatives, and staff; inadequate evaluation of psychological distress and existential suffering; and inadequate monitoring. There are multiple published guidelines on appropriate indications, monitoring, and methods for palliative sedation, and most guidelines include the recommendation that cases in which this procedure is being considered should include a physician with expertise in palliative care.

BALANCING AUTONOMY AND SELF-DETERMINATION WITH INDIVIDUAL AND SOCIETAL SAFETY DRIVING

Case

Mr. J is an 89-year-old man living temporarily in a skilled nursing facility where he is receiving physical and occupational therapy after he fell and sustained a patellar fracture. He is doing well, and the interdisciplinary team informs you that he will be ready to discharge home soon. He lives in a one-story home with his wife. He drives to the neighborhood shopping center to meet his friends for lunch, buy groceries, and do other chores. During your assessment, you note that his memory is mildly impaired, with only one out of three items recalled after 5 minutes, and he has difficulty with attention. His clock-drawing test is markedly abnormal, with impaired spacing and numbering.

Questions for the interdisciplinary team: Should he still be driving? What do we do about it?

Decisions regarding older adults' driving ability can lead to ethical dilemmas for health care professionals, especially when the patient is upset at the loss of the independence and autonomy that driving provides. Common reasons for recommending driving cessation are epilepsy, stroke, dementia, visual impairment, and medication adverse effects. Strategies to employ when addressing driving decisions with patients are summarized in **Table 72-5**. Some key ethical considerations with regard to driving include protecting the patient's safety, the public's safety, and patient confidentiality. Protecting Mr. J would include counseling him about conditions or medications that may impair his ability to drive safely. Some states have mandatory reporting requirements for physicians regarding driving ability, and some states have found physicians liable for harm to others because their patients were not warned about conditions or medication side effects that could impair driving performance. Physicians must weigh the responsibility for ensuring patient confidentiality against the responsibility for the patient's safety and public safety. Involving patients and family, and carefully counseling patients,

TABLE 72-5 ■ STRATEGIES TO EMPLOY WHEN WORKING WITH OLDER ADULTS REGARDING DRIVING CESSATION

SITUATION	PROPOSED STRATEGIES
Patient threatens to sue physician if reported to Department of Motor Vehicles	• Clearly document how the patient was determined to be an unsafe driver (driving fitness evaluation, assessing driving-related skills [ADReS] battery, driving rehabilitation referral, medications, etc). • Advise patient that the state has the ultimate responsibility to determine licensing. • If physician is mandated to report in that state, physician must report. The physician should also advise the patient of this. • Check if the state has legislation protecting health professionals against liability for reporting unsafe drivers in good faith. • Consult malpractice insurance or legal counsel to determine degree of risk (physicians generally run little risk of liability for following mandatory reporting).
Physician identifies patient as an unsafe driver, state does not have mandatory reporting laws	• Attempt to counsel patient to stop driving voluntarily without revocation of license. • Explore ways to ensure that patient no longer drives. • If patient refuses to stop driving, physician must weigh breach of confidentiality vs potential injury to patient and others from a crash.
Patient had license suspended, but continues to drive	• Talk to the patient about why he or she continues to drive. • With the patient's permission, involve the patient's family or caregivers to identify transportation resources and alternative transportation. • Counsel patient that he or she is breaking the law, cite concerns about patient's safety, and discuss consequences of a crash (being injured, injuring others, financial and legal implications). • If the patient is cognitively impaired and lacks insight, counsel surrogate decision-makers regarding responsibility to prevent patient from driving. • Report patient as an unsafe driver again.
Patient threatens to find a new doctor if physician reports to Department of Motor Vehicles	• Diffuse the situation by reiterating how unsafe driving determination was made. • Convey concern for patient's safety, and safety of others who would be affected if patient crashed. • Remind patient that driving safety is part of good health care as much as other physician responsibilities. • Encourage patient to be evaluated by a driving rehabilitation specialist. • Emphasize that the state Department of Motor Vehicles makes the final determination of driving safety. • Maintain professional behavior by not expressing hostility even if he or she seeks a new physician.

Data from Clinician's Guide to Assessing and Counseling Older Drivers, 4th ed. New York, NY: American Geriatrics Society; 2019.

including providing a written letter of recommendations regarding driving cessation, can be helpful to navigate these challenging situations. Careful documentation of a patient's risks regarding driving and counseling provided, as well as additional referrals made to social workers, driving rehabilitation or other professionals, may reduce clinician liability. In addition, physicians must make sure they are in compliance with their own state reporting laws for physicians, available from each state's Department of Motor Vehicles (DMV). Six states have mandatory reporting requirements (California, Delaware, Nevada, New Jersey, Oregon, and Pennsylvania). In states without mandatory reporting laws, providers need to obtain release of information to report patients to the DMV, or they could be held liable for breach of confidentiality. Some states provide civil immunity if health care professionals report in good faith.

FIREARMS

The possession of firearms by people with dementia is another area in which the autonomy and self-determination of the individual needs to be weighed against the safety of the individual and of society. Of people with dementia, 60% have firearms in their homes and only 17% of families report storing these weapons unloaded and locked. Having firearms in the home increases the likelihood of suicide completion. Ten states and the District of Columbia require a permit to buy a firearm. Hawaii restricts persons with organic brain syndromes from possessing firearms, while Texas prohibits persons with dementia from obtaining a license to carry a handgun in public. There are "red flag laws" called extreme risk protection order (ERPO) laws in 17 states, which provide a means for court-ordered firearm removal based on reasonable or substantial threat. Strategies for assessing and counseling firearm safety in patients with dementia and their families are shown in **Table 72-6**. As dementia progresses, the recommendations become more focused on counseling families to restrict access to the firearm and remove the firearm from the home environment. A website has been developed to assist families and caregivers in discussing changes regarding safety in dementia, including firearms, driving and home safety (https://safetyindementia.org/).

TABLE 72-6 ■ RECOMMENDED STRATEGIES FOR COUNSELING PERSONS WITH DEMENTIA AND THEIR FAMILIES REGARDING FIREARMS

5 Ls (Loaded, Locked, Little Children, Low, and Learned)

Is it Loaded?

Is it Locked? (gun safe, trigger lock or cable lock in place, ammunition locked separate)

Are Little children present? (grandchildren at risk for injury)

Is the person feeling Low? (risk of suicide)

Is the person Learned? (knows how to use the weapon, or is the weapon inherited or not intentionally purchased, presence of dementia)

Firearm Safety Counseling Protocol Questions to Assess Access, Safety Profile, and Capacity

- Assess dementia severity:
  - Include person with dementia in assessment and counseling when dementia is mild.
  - As dementia severity increases, the protocol involves family members only.
- Risk assessment:
  - Firearm: type, number, status (locked, unloaded, with ammunition)
  - Patient: vision, neurological changes, PHQ-2
  - Environment: small children, teenagers
  - Behaviors: last time handled/disarmed firearm
- Counseling based on risk profile:
  - Safe storage (locked, unloaded, stored separate from ammunition, safety device)
  - Access for suicide and injury-related fatalities (Lethal Means Access)
  - Firearm sunset: ways to remove firearm from home, remove ammunition/ disable firearm (remove firing pins), red flag laws if family feels unsafe.

Data from Pinholt EM, Mitchell JD, Butler JH, et al. "Is there a gun in the home?" Assessing the risks of gun ownership in older adults. J Am Geriatr Soc. 2014;62(6):1142–1146 and Doucette ML, Dayton H, Lapidus G, et al. Firearms, dementia, and the clinician: development of a safety counseling protocol. J Am Geriatr Soc. 2020;68(9):2128–2133.

MANAGING SEXUAL INTIMACY AND OTHER ETHICAL ISSUES IN NURSING HOMES

Case

Mrs. N, an ambulatory 87-year-old nursing home resident with moderate dementia, is found by nursing staff in the room of Mr. O, an 85-year-old nursing home resident with COPD, Parkinson disease, and dementia. They are seated on Mr. O's bed, hugging. Mr. O is wearing his incontinence briefs, but no pants, and is audibly wheezing.

Question: What is the appropriate way to approach sexual intimacy in the long-term care setting?

Common ethical issues encountered in long-term care are summarized in **Table 72-7**. In general, ethical issues in the nursing home often involve tension between honoring resident autonomy and protecting their health and safety. During the COVID-19 pandemic, for example, many nursing homes prioritized beneficence and safety over resident autonomy. This has resulted in isolation and cognitive decline.

TABLE 72-7 ■ ETHICAL ISSUES IN LONG-TERM CARE FACILITIES

ETHICAL ISSUE	KEY CONSIDERATIONS
Conflict between resident autonomy and beneficence	Managing resident refusal of care, medications, bathing, or wanting to leave the facility. The facility staff and physician must communicate standards of care clearly with the patient and the surrogate decision-makers at the time of admission and at regular intervals.
Decision-making at the end of life	A facility ethics committee can assist with decisions regarding withholding, withdrawing, or otherwise limiting aggressive medical care.
Ethical issues in the care of LBGT persons in the long-term care setting	Adoption of congruent and appropriate wording on admission and procedure forms; nonrestriction of visitation and inclusion of partner in family meetings; create programs and staff education to facilitate respect; cultural competency and genuine inclusion; reducing isolation.
Role of ethics committees in nursing homes	Each nursing facility should have a mechanism to assist the facility, residents, physicians, leadership, and staff in managing ethical dilemmas, including policy development and review, quality assurance, education, monitoring judicial decisions and legislative action and consultation, and review of case-specific dilemmas.
Long-term care research	Autonomy and informed consent—residents may lack decision-making capacity necessitating consent by surrogate.

Sexual intimacy in the long-term care setting requires a specific approach that is different from usual medical decision-making capacity evaluations. One recommended approach is summarized in **Table 72-8**.

In the case, the physician caring for Mrs. N and Mr. O must assess their capacity for sexual consent and the long-term care facility must develop a plan for sexual intimacy within the facility. Involving the residents' surrogate decision-makers is recommended for open communication and a coordinated approach that balances patient preference (if the patient is determined to have capacity for sexual consent) with safety and privacy.

RESOURCE ALLOCATION AND AGEISM

Case

A hospital system is approaching 100% ICU capacity due to the COVID-19 pandemic. This health system's crisis standards of care specify that adults older than age 85 should not enter the resource allocation algorithm, meaning that

TABLE 72-8 ■ COMPONENTS OF SEXUAL CONSENT IN THE NURSING HOME

COMPONENT OF DETERMINATION OF SEXUAL CONSENT	KEY CONSIDERATIONS
Is the person with dementia (PWD)...	
• Familiar with the sexual partner and the nature of that person's relationship to the PWD?	Can the PWD name the person who would be in sexual contact with them? Can the PWD accurately explain their relationship (ie, does the PWD mistake someone else for their spouse)?
• Able to describe or name the type of intimate sexual activity they are comfortable with?	Can the PWD state what kind of intimate sexual activity they would like to have with this person? (Note that the standard for capacity might be different, for example, for kissing their spouse versus sexual penetration.)
• Able to decline unwanted sexual activity?	Can the PWD say "no" to this person or to this activity? Can the PWD physically stop the person or the activity?
• Able to articulate what their reaction will be if sexual activity ends?	Can the PWD say how this activity will make them feel, react after it is over, or after the relationship ends?
Is this behavior consistent with previously expressed beliefs and preferences?	Observe the PWD in the long-term care setting: Are they acting in a manner that is consistent with how they would have acted prior to dementia?
Can the facility...	
• Enlist the PWD's proposed sexual partner in the assessment?	Observe partner for signs of cognitive impairment, awareness of the PWD's wishes, ability to assure their safety and privacy, and potential conflict of interest, coercion, or undue influence. Involve additional surrogate decision-makers if there are concerns.
• Address concerns of family or surrogate decision-makers?	Consider family members' personal, generational, and cultural background and acknowledge sensitive nature of the topic when discussing with family.
• Address concerns of staff members?	Elicit concerns from staff in a safe forum when developing plan regarding sexual intimacy and how to protect privacy and dignity of patients during sanctioned sexual activity.
• Address safety concerns?	Discuss with PWD, partner, and surrogate decision-maker the nature of the sexual activity and physical and other risks, including risk of falling, infection, or cardiovascular event. Develop plan for staff monitoring, if necessary.

Data from Metzger 2017 and American Medical Directors Association, 2008, White paper.

this group would be denied intensive care should crisis care standards need to be invoked.

Question: Which ethical principles should guide the fair allocation of health care resources under conditions of resources scarcity, especially when considering age?

The specter of denying health care resources to patients under conditions of resource scarcity is perhaps one of the starkest examples in which health care professionals rely on sound and transparent ethical principles for guidance. In usual conditions in Western societies, the principle of autonomy dominates health care decision-making, but under conditions of resource scarcity, the principle of justice has greater weight. During the 2009 H1N1 pandemic, many states began adopting crisis standards of care documents designed to provide guidance to hospitals faced with conditions of resource scarcity. More recently, the COVID-19 pandemic in 2020 brought renewed attention to these crisis standards of care as ICU capacity in terms of "space" (ie, ICU beds), "staff" (eg, intensivists, nurses, respiratory therapists), and "stuff" (eg, ventilators, remdesivir) was pushed to the brink, and many hospitals developed scarce resource allocation triage teams and revised relevant protocols.

A key ethical consideration with respect to allocating health care resources under conditions of resource scarcity is that clinicians engaged in patient care on the "front line" should not be making decisions regarding resource allocation. Leaving resource allocation decisions to individual clinicians increases the chance that resources will be allocated in an ad hoc and not a systematic fashion, increases the risk of bias including the potential for decisions to disadvantage underrepresented groups (eg, a first-come, first-serve approach), and also burdens the clinician with moral distress. Each hospital should have an interdisciplinary triage committee that is removed from clinical care and that applies their state's crisis standards of care guidelines, with outcomes of these deliberations communicated to clinicians who apply the triage committee's recommendation. It is critically important that triage committees apply crisis standards of care fairly and that their decisions are subject to retrospective review. Otherwise, public confidence in the process by which limited resources are allocated may be undermined.

In early 2020, reports surfaced in Italy of rationing limited health care resources based on age. In the United States, some states' crisis standards of care included age-based cutoffs beyond which patients would not be eligible to receive limited health care resources. Additional age-related provisions include "tiebreaker" provisions in which older age counts against a patient if two patients

have identical scores on measures of physiological function such as the MSOFA (Modified Sequential Organ Failure Assessment) score. Although the MSOFA is not based on age, other assessments used to distinguish between patients during times of crisis, such as the 4C mortality score, weight age heavily. While it is the case that older adults with COVID-19 have a higher overall mortality rate than younger adults with COVID-19, the use of age per se to ration health care resources has been criticized because the use of age alone does not consider individual differences that contribute to short-term prognosis. Other arguments to ration health care resources based on age include long-term predicted life expectancy and utilitarian frameworks such as maximizing "life-years saved." Long-term predicted life expectancy has been criticized in this context due to difficulty with prognosticating life expectancy and because, among underrepresented patients, it overlooks the fact that life expectancy may be shorter due to social determinants of health outside of patients' control. The "life-years saved" approach does not consider individual differences among patients and instead withholds resources based on membership in a class (ie, one's age group). Finally, the "fair innings" argument contends that older adults have a lesser claim on health care resources than younger adults since they have lived through more "innings." However, opponents of this argument argue that value judgments about which "innings" of life are more important than others cannot be made. Although situations in which patients are truly tied after an exhaustive consideration of individual characteristics such as physiological parameters should be rare, promising approaches to settling these ties using assessments that are not based on age, such as the Clinical Frailty Scale, are gaining acceptance.

Although the principle of justice outweighs the principle of autonomy under conditions of resource scarcity, autonomy should still be respected. It is critically important to consider the role of advance care planning in ensuring that older adults' preferences are honored even under these difficult conditions. Clinicians should engage in advance care planning conversations, but should be careful to avoid pressuring patients to make decisions for the sole purpose of conserving resources, regardless of whether the patient is healthy or ill at the time the advance care planning conversation occurs. Some have argued that adults should voluntarily relinquish their claim on health care resources to benefit younger generations, but this position is controversial and should not be assumed or imposed on older adults.

FURTHER READING

American Medical Directors Association, 2008, White paper, The role of a facility ethics committee in decision-making at the end of life. Available at https://paltc.org/amda-white-papers-and-resolution-position-statements/role-facility-ethics-committee-decision-making. Accessed February 22, 2021.

American Medical Association Code of Medical Ethics. Chapter 5: Opinions on caring for patients at the end of life. Available at https://www.ama-assn.org/system/files/2019-06/code-of-medical-ethics-chapter-5.pdf. Accessed February 7, 2021.

American Medical Directors Association. 2016, Capacity for sexual consent in dementia in long-term care. Available at https://paltc.org/amda-white-papers-and-resolution-position-statements/capacity-sexual-consent-dementia-long-term-care. Accessed January 10, 2021.

Baile W, Buckman R, Lenzi R, Glober G, Beale EA, Kudelka AP. SPIKES—a six-step protocol for delivering bad news: application to the patient with cancer. *Oncologist*. 2000;5:302–311.

Betz ME, McCourt AD, Vernick JS, Ranney ML, Maust DT, Wintemute GJ. Firearms and dementia: clinical considerations. *Ann Intern Med*. 2018;169(1):47–49.

Clinician's Guide to Assessing and Counseling Older Drivers. Fourth edition. American Geriatrics Society 2019. Available at https://geriatricscareonline.org/ProductAbstract/clinicians-guide-to-assessing-and-counseling-older-drivers-4th-edition/B047. Accessed February 22, 2021.

Epstein, EG, Hamric AB. Moral distress, moral residue, and the crescendo effect. *J Clin Ethics*. 2009;20(4):330–342.

Farrell TW, Widera E, Rosenberg L, et al. AGS position statement: making medical treatment decisions for unbefriended older adults. *J Am Geriatr Soc*. 2017;65(1):14–15.

Farrell TW, Francis L, Brown T, et al. Rationing limited healthcare resources in the COVID-19 era and beyond: ethical considerations regarding older adults. *J Am Geriatr Soc*. 2020;68(6):1143–1149.

Fox E, Berkowitz KA, Chanko BL, Powell T. *Ethics Consultation: Responding to Ethics Questions in Health Care*. Second edition. Available at http://vaww.ethics.va.gov/docs/integratedethics/ec_primer.pdf. Accessed February 22, 2021.

Gruenewald DA. Voluntary stopping eating and drinking: a practical approach for long-term care facilities. *J Palliat Med*. 2018;21(9):1214–1220.

Metzer E. Ethics and intimate sexual activity in long-term care. *AMA J Ethics*. 2017;19(7):640–648.

Preshaw DHI, Brazil K, McLaughlin D, Frolic A. Ethical issues experienced by healthcare workers in nursing homes: literature review. *Nurs Ethics*. 2016;23(5):490–506.

Quill T, Arnold RM. Evaluating requests for hastened death #156. *J Palliat Med*. 2008;11:1151–1152.

Rushton, CH. Cultivating moral resilience. *Am J Nurs*. 2017;117(2 Suppl 1):S11–S15.

Wong SP, Sharda N, Zietlow KE, Heflin MT. Planning for a safe discharge: more than a capacity evaluation. *J Am Geriatr Soc*. 2020;68(4):859–866.

van den Berg V, van Thiel G, Zomers M, et al. Euthanasia and physician-assisted suicide in patients with multiple geriatric syndromes. *JAMA Intern Med*. 2021;181(2):245–250.

VHA National Center for Ethics in Health Care. Ethics guidance for pandemics 2020. Available at http://vaww.ethics.va.gov/ETHICS/activities/pandemic/Ethics_Guidance_for_Pandemics_2020.pdf. Accessed February 22, 2021.

Wax JW, An AW, Kosier N, Quill TE. Voluntary stopping eating and drinking. *J Am Geriatr Soc*. 2018;66(3):441–445.

Wright JL, Jaggard PM, Holahan T. Stopping eating and drinking by advance directives (SED by AD) in assisted living and nursing homes. *J Am Med Dir Assoc*. 2019;20(11):1362–1366.